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Systems of Linear Equations

- 1.1 Introduction to Systems of Linear Equations
- 1.2 Gaussian Elimination and Gauss-Jordan Elimination
- 1.3 Applications of Systems of Linear Equations

CHAPTER OBJECTIVES

- Recognize, graph, and solve a system of linear equations in n variables.
- Use back-substitution to solve a system of linear equations.
- Determine whether a system of linear equations is consistent or inconsistent.
- Determine if a matrix is in row-echelon form or reduced row-echelon form.
- Use elementary row operations with back-substitution to solve a system in row-echelon form.
- Use elimination to rewrite a system in row-echelon form.
- Write an augmented or coefficient matrix from a system of linear equations, or translate a matrix into a system of linear equations.
- Solve a system of linear equations using Gaussian elimination and Gaussian elimination with back-substitution.
- Solve a homogeneous system of linear equations.
- Set up and solve a system of equations to fit a polynomial function to a set of data points, as well as to represent a network.

HISTORICAL NOTE

**Carl Friedrich Gauss
(1777–1855)**

is often ranked—along with Archimedes and Newton—as one of the greatest mathematicians in history. To read about his contributions to linear algebra, visit college.hmco.com/pic/larsonELA6e.

1.1 Introduction to Systems of Linear Equations

Linear algebra is a branch of mathematics rich in theory and applications. This text strikes a balance between the theoretical and the practical. Because linear algebra arose from the study of systems of linear equations, you shall begin with linear equations. Although some material in this first chapter will be familiar to you, it is suggested that you carefully study the methods presented here. Doing so will cultivate and clarify your intuition for the more abstract material that follows.

The study of linear algebra demands familiarity with algebra, analytic geometry, and trigonometry. Occasionally you will find examples and exercises requiring a knowledge of calculus; these are clearly marked in the text.

Early in your study of linear algebra you will discover that many of the solution methods involve dozens of arithmetic steps, so it is essential to strive to avoid careless errors. A computer or calculator can be very useful in checking your work, as well as in performing many of the routine computations in linear algebra.

Linear Equations in n Variables

Recall from analytic geometry that the equation of a line in two-dimensional space has the form

$$a_1x + a_2y = b, \quad a_1, a_2, \text{ and } b \text{ are constants.}$$

This is a **linear equation in two variables** x and y . Similarly, the equation of a plane in three-dimensional space has the form

$$a_1x + a_2y + a_3z = b, \quad a_1, a_2, a_3, \text{ and } b \text{ are constants.}$$

Such an equation is called a **linear equation in three variables** x , y , and z . In general, a linear equation in n variables is defined as follows.

Definition of a Linear Equation in n Variables

A **linear equation in n variables** $x_1, x_2, x_3, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = b.$$

The **coefficients** $a_1, a_2, a_3, \dots, a_n$ are real numbers, and the **constant term** b is a real number. The number a_1 is the **leading coefficient**, and x_1 is the **leading variable**.

REMARK: Letters that occur early in the alphabet are used to represent constants, and letters that occur late in the alphabet are used to represent variables.

Linear equations have no products or roots of variables and no variables involved in trigonometric, exponential, or logarithmic functions. Variables appear only to the first power. Example 1 lists some equations that are linear and some that are not linear.

EXAMPLE 1

Examples of Linear Equations and Nonlinear Equations

Each equation is linear.

(a) $3x + 2y = 7$

(b) $\frac{1}{2}x + y - \pi z = \sqrt{2}$

(c) $x_1 - 2x_2 + 10x_3 + x_4 = 0$

(d) $\left(\sin \frac{\pi}{2}\right)x_1 - 4x_2 = e^2$

Each equation is not linear.

(a) $xy + z = 2$

(b) $e^x - 2y = 4$

(c) $\sin x_1 + 2x_2 - 3x_3 = 0$

(d) $\frac{1}{x} + \frac{1}{y} = 4$

A **solution** of a linear equation in n variables is a sequence of n real numbers $s_1, s_2, s_3, \dots, s_n$ arranged so the equation is satisfied when the values

$$x_1 = s_1, \quad x_2 = s_2, \quad x_3 = s_3, \quad \dots, \quad x_n = s_n$$

are substituted into the equation. For example, the equation

$$x_1 + 2x_2 = 4$$

is satisfied when $x_1 = 2$ and $x_2 = 1$. Some other solutions are $x_1 = -4$ and $x_2 = 4$, $x_1 = 0$ and $x_2 = 2$, and $x_1 = -2$ and $x_2 = 3$.

The set of *all* solutions of a linear equation is called its **solution set**, and when this set is found, the equation is said to have been **solved**. To describe the entire solution set of a linear equation, a **parametric representation** is often used, as illustrated in Examples 2 and 3.

EXAMPLE 2**Parametric Representation of a Solution Set**

Solve the linear equation $x_1 + 2x_2 = 4$.

SOLUTION

To find the solution set of an equation involving two variables, solve for one of the variables in terms of the other variable. If you solve for x_1 in terms of x_2 , you obtain

$$x_1 = 4 - 2x_2.$$

In this form, the variable x_2 is **free**, which means that it can take on any real value. The variable x_1 is not free because its value depends on the value assigned to x_2 . To represent the infinite number of solutions of this equation, it is convenient to introduce a third variable t called a **parameter**. By letting $x_2 = t$, you can represent the solution set as

$$x_1 = 4 - 2t, \quad x_2 = t, \quad t \text{ is any real number.}$$

Particular solutions can be obtained by assigning values to the parameter t . For instance, $t = 1$ yields the solution $x_1 = 2$ and $x_2 = 1$, and $t = 4$ yields the solution $x_1 = -4$ and $x_2 = 4$.

The solution set of a linear equation can be represented parametrically in more than one way. In Example 2 you could have chosen x_1 to be the free variable. The parametric representation of the solution set would then have taken the form

$$x_1 = s, \quad x_2 = 2 - \frac{1}{2}s, \quad s \text{ is any real number.}$$

For convenience, choose the variables that occur last in a given equation to be free variables.

EXAMPLE 3**Parametric Representation of a Solution Set**

Solve the linear equation $3x + 2y - z = 3$.

SOLUTION

Choosing y and z to be the free variables, begin by solving for x to obtain

$$\begin{aligned} 3x &= 3 - 2y + z \\ x &= 1 - \frac{2}{3}y + \frac{1}{3}z. \end{aligned}$$

Letting $y = s$ and $z = t$, you obtain the parametric representation

$$x = 1 - \frac{2}{3}s + \frac{1}{3}t, \quad y = s, \quad z = t$$

where s and t are any real numbers. Two particular solutions are

$$x = 1, y = 0, z = 0 \quad \text{and} \quad x = 1, y = 1, z = 2.$$

Systems of Linear Equations

A **system of m linear equations in n variables** is a set of m equations, each of which is linear in the same n variables:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n &= b_3 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= b_m. \end{aligned}$$

REMARK: The double-subscript notation indicates a_{ij} is the coefficient of x_j in the i th equation.

A **solution** of a system of linear equations is a sequence of numbers $s_1, s_2, s_3, \dots, s_n$ that is a solution of each of the linear equations in the system. For example, the system

$$\begin{aligned} 3x_1 + 2x_2 &= 3 \\ -x_1 + x_2 &= 4 \end{aligned}$$

has $x_1 = -1$ and $x_2 = 3$ as a solution because *both* equations are satisfied when $x_1 = -1$ and $x_2 = 3$. On the other hand, $x_1 = 1$ and $x_2 = 0$ is not a solution of the system because these values satisfy only the first equation in the system.

Discovery

Graph the two lines

$$3x - y = 1$$

$$2x - y = 0$$

in the xy -plane. Where do they intersect? How many solutions does this system of linear equations have?

Repeat this analysis for the pairs of lines

$$3x - y = 1 \quad 3x - y = 1$$

$$3x - y = 0 \quad 6x - 2y = 2.$$

In general, what basic types of solution sets are possible for a system of two equations in two unknowns?

It is possible for a system of linear equations to have exactly one solution, an infinite number of solutions, or no solution. A system of linear equations is called **consistent** if it has at least one solution and **inconsistent** if it has no solution.

EXAMPLE 4 Systems of Two Equations in Two Variables

Solve each system of linear equations, and graph each system as a pair of straight lines.

$$\begin{array}{lll} \text{(a)} & x + y = 3 & \text{(b)} \quad x + y = 3 \\ & x - y = -1 & 2x + 2y = 6 \\ & & \text{(c)} \quad x + y = 3 \\ & & x + y = 1 \end{array}$$

- SOLUTION**
- (a) This system has exactly one solution, $x = 1$ and $y = 2$. The solution can be obtained by adding the two equations to give $2x = 2$, which implies $x = 1$ and so $y = 2$. The graph of this system is represented by two *intersecting* lines, as shown in Figure 1.1(a).
- (b) This system has an infinite number of solutions because the second equation is the result of multiplying both sides of the first equation by 2. A parametric representation of the solution set is shown as

$$x = 3 - t, \quad y = t, \quad t \text{ is any real number.}$$

The graph of this system is represented by two *coincident* lines, as shown in Figure 1.1(b).

- (c) This system has no solution because it is impossible for the sum of two numbers to be 3 and 1 simultaneously. The graph of this system is represented by two *parallel* lines, as shown in Figure 1.1(c).

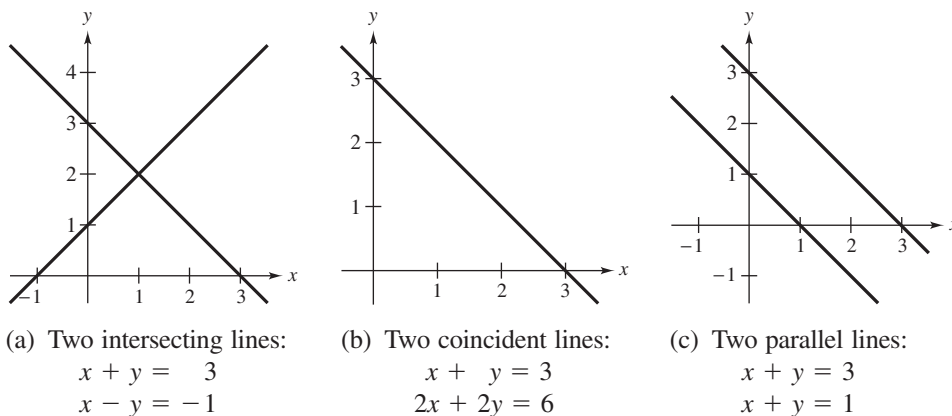


Figure 1.1

Example 4 illustrates the three basic types of solution sets that are possible for a system of linear equations. This result is stated here without proof. (The proof is provided later in Theorem 2.5.)

Number of Solutions of a System of Linear Equations

For a system of linear equations in n variables, precisely one of the following is true.

1. The system has exactly one solution (consistent system).
2. The system has an infinite number of solutions (consistent system).
3. The system has no solution (inconsistent system).

Solving a System of Linear Equations

Which system is easier to solve algebraically?

$$\begin{array}{rcl} x - 2y + 3z & = & 9 \\ -x + 3y & = & -4 \\ 2x - 5y + 5z & = & 17 \end{array} \qquad \begin{array}{rcl} x - 2y + 3z & = & 9 \\ y + 3z & = & 5 \\ z & = & 2 \end{array}$$

The system on the right is clearly easier to solve. This system is in **row-echelon form**, which means that it follows a stair-step pattern and has leading coefficients of 1. To solve such a system, use a procedure called **back-substitution**.

EXAMPLE 5 Using Back-Substitution to Solve a System in Row-Echelon Form

Use back-substitution to solve the system.

$$\begin{array}{rcl} x - 2y & = & 5 \\ y & = & -2 \end{array} \qquad \begin{array}{l} \text{Equation 1} \\ \text{Equation 2} \end{array}$$

SOLUTION From Equation 2 you know that $y = -2$. By substituting this value of y into Equation 1, you obtain

$$\begin{array}{rcl} x - 2(-2) & = & 5 \\ x & = & 1. \end{array} \qquad \begin{array}{l} \text{Substitute } y = -2. \\ \text{Solve for } x. \end{array}$$

The system has exactly one solution: $x = 1$ and $y = -2$.

The term “back-substitution” implies that you work *backward*. For instance, in Example 5, the second equation gave you the value of y . Then you substituted that value into the first equation to solve for x . Example 6 further demonstrates this procedure.

EXAMPLE 6 Using Back-Substitution to Solve a System in Row-Echelon Form

Solve the system.

$$\begin{array}{rcl} x - 2y + 3z & = & 9 \\ y + 3z & = & 5 \\ z & = & 2 \end{array} \qquad \begin{array}{l} \text{Equation 1} \\ \text{Equation 2} \\ \text{Equation 3} \end{array}$$

SOLUTION From Equation 3 you already know the value of z . To solve for y , substitute $z = 2$ into Equation 2 to obtain

$$\begin{aligned} y + 3(2) &= 5 && \text{Substitute } z = 2. \\ y &= -1. && \text{Solve for } y. \end{aligned}$$

Finally, substitute $y = -1$ and $z = 2$ in Equation 1 to obtain

$$\begin{aligned} x - 2(-1) + 3(2) &= 9 && \text{Substitute } y = -1, z = 2. \\ x &= 1. && \text{Solve for } x. \end{aligned}$$

The solution is $x = 1$, $y = -1$, and $z = 2$.

Two systems of linear equations are called **equivalent** if they have precisely the same solution set. To solve a system that is not in row-echelon form, first change it to an *equivalent* system that is in row-echelon form by using the operations listed below.

Operations That Lead to Equivalent Systems of Equations

Each of the following operations on a system of linear equations produces an *equivalent* system.

1. Interchange two equations.
2. Multiply an equation by a nonzero constant.
3. Add a multiple of an equation to another equation.

Rewriting a system of linear equations in row-echelon form usually involves a *chain* of equivalent systems, each of which is obtained by using one of the three basic operations. This process is called **Gaussian elimination**, after the German mathematician Carl Friedrich Gauss (1777–1855).

EXAMPLE 7

Using Elimination to Rewrite a System in Row-Echelon Form

Solve the system.

$$\begin{aligned} x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17 \end{aligned}$$

SOLUTION Although there are several ways to begin, you want to use a systematic procedure that can be applied easily to large systems. Work from the upper left corner of the system, saving the x in the upper left position and eliminating the other x 's from the first column.

$$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 && \leftarrow \text{Adding the first equation to the second equation produces a new second equation.} \\ 2x - 5y + 5z &= 17 \\ x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ -y - z &= -1 && \leftarrow \text{Adding } -2 \text{ times the first equation to the third equation produces a new third equation.} \end{aligned}$$

Now that everything but the first x has been eliminated from the first column, work on the second column.

$$x - 2y + 3z = 9$$

$$y + 3z = 5$$

$$2z = 4$$

Adding the second equation to the third equation produces a new third equation.

$$x - 2y + 3z = 9$$

$$y + 3z = 5$$

$$z = 2$$

Multiplying the third equation by $\frac{1}{2}$ produces a new third equation.

This is the same system you solved in Example 6, and, as in that example, the solution is

$$x = 1, \quad y = -1, \quad z = 2.$$

Each of the three equations in Example 7 is represented in a three-dimensional coordinate system by a plane. Because the unique solution of the system is the point

$$(x, y, z) = (1, -1, 2),$$

the three planes intersect at the point represented by these coordinates, as shown in Figure 1.2.

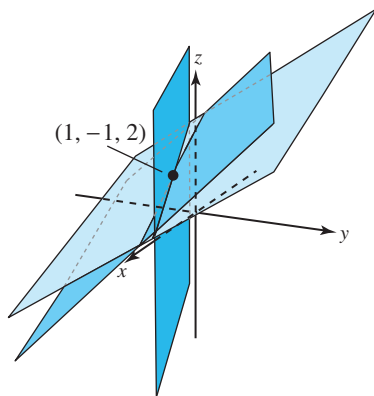


Figure 1.2

Technology Note

Many graphing utilities and computer software programs can solve a system of m linear equations in n variables. Try solving the system in Example 7 using the simultaneous equation solver feature of your graphing utility or computer software program. Keystrokes and programming syntax for these utilities/programs applicable to Example 7 are provided in the **Online Technology Guide**, available at college.hmco.com/pic/larsonELA6e.

Because many steps are required to solve a system of linear equations, it is very easy to make errors in arithmetic. It is suggested that you develop the habit of *checking your solution by substituting it into each equation in the original system*. For instance, in Example 7, you can check the solution $x = 1$, $y = -1$, and $z = 2$ as follows.

$$\begin{array}{ll} \text{Equation 1:} & (1) - 2(-1) + 3(2) = 9 \\ \text{Equation 2:} & -(1) + 3(-1) = -4 \\ \text{Equation 3:} & 2(1) - 5(-1) + 5(2) = 17 \end{array} \quad \begin{array}{l} \text{Substitute solution in} \\ \text{each equation of the} \\ \text{original system.} \end{array}$$

Each of the systems in Examples 5, 6, and 7 has exactly one solution. You will now look at an inconsistent system—one that has no solution. The key to recognizing an inconsistent system is reaching a false statement such as $0 = 7$ at some stage of the elimination process. This is demonstrated in Example 8.

EXAMPLE 8 An Inconsistent System

Solve the system.

$$\begin{array}{rcl} x_1 - 3x_2 + x_3 & = & 1 \\ 2x_1 - x_2 - 2x_3 & = & 2 \\ x_1 + 2x_2 - 3x_3 & = & -1 \end{array}$$

SOLUTION

$$\begin{array}{rcl} x_1 - 3x_2 + x_3 & = & 1 \\ 5x_2 - 4x_3 & = & 0 \\ x_1 + 2x_2 - 3x_3 & = & -1 \\ x_1 - 3x_2 + x_3 & = & 1 \\ 5x_2 - 4x_3 & = & 0 \\ 5x_2 - 4x_3 & = & -2 \end{array} \quad \begin{array}{l} \leftarrow \text{Adding } -2 \text{ times the first} \\ \text{equation to the second equation} \\ \text{produces a new second equation.} \\ \\ \leftarrow \text{Adding } -1 \text{ times the first} \\ \text{equation to the third equation} \\ \text{produces a new third equation.} \end{array}$$

(Another way of describing this operation is to say that you *subtracted* the first equation from the third equation to produce a new third equation.) Now, continuing the elimination process, add -1 times the second equation to the third equation to produce a new third equation.

$$\begin{array}{rcl} x_1 - 3x_2 + x_3 & = & 1 \\ 5x_2 - 4x_3 & = & 0 \\ 0 & = & -2 \end{array} \quad \begin{array}{l} \leftarrow \text{Adding } -1 \text{ times the second} \\ \text{equation to the third equation} \\ \text{produces a new third equation.} \end{array}$$

Because the third “equation” is a false statement, this system has no solution. Moreover, because this system is equivalent to the original system, you can conclude that the original system also has no solution.

As in Example 7, the three equations in Example 8 represent planes in a three-dimensional coordinate system. In this example, however, the system is inconsistent. So, the planes do not have a point in common, as shown in Figure 1.3 on the next page.

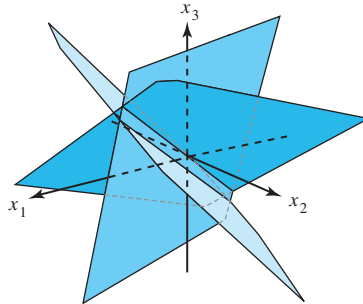


Figure 1.3

This section ends with an example of a system of linear equations that has an infinite number of solutions. You can represent the solution set for such a system in parametric form, as you did in Examples 2 and 3.

EXAMPLE 9 A System with an Infinite Number of Solutions

Solve the system.

$$\begin{array}{rcl} x_2 - x_3 & = & 0 \\ x_1 & - & 3x_3 = -1 \\ -x_1 + 3x_2 & = & 1 \end{array}$$

SOLUTION Begin by rewriting the system in row-echelon form as follows.

$$\begin{array}{rcl} x_1 & - & 3x_3 = -1 \\ x_2 - x_3 & = & 0 \\ -x_1 + 3x_2 & = & 1 \end{array} \quad \begin{array}{l} \leftarrow \\ \leftarrow \end{array} \quad \begin{array}{l} \text{The first two equations} \\ \text{are interchanged.} \end{array}$$

$$\begin{array}{rcl} x_1 & - & 3x_3 = -1 \\ x_2 - x_3 & = & 0 \\ 3x_2 - 3x_3 & = & 0 \end{array} \quad \leftarrow \quad \begin{array}{l} \text{Adding the first equation to} \\ \text{the third equation produces} \\ \text{a new third equation.} \end{array}$$

$$\begin{array}{rcl} x_1 & - & 3x_3 = -1 \\ x_2 - x_3 & = & 0 \\ 0 & = & 0 \end{array} \quad \leftarrow \quad \begin{array}{l} \text{Adding } -3 \text{ times the second} \\ \text{equation to the third equation} \\ \text{eliminates the third equation.} \end{array}$$

Because the third equation is unnecessary, omit it to obtain the system shown below.

$$\begin{array}{rcl} x_1 & - & 3x_3 = -1 \\ x_2 - x_3 & = & 0 \end{array}$$

To represent the solutions, choose x_3 to be the free variable and represent it by the parameter t . Because $x_2 = x_3$ and $x_1 = 3x_3 - 1$, you can describe the solution set as

$$x_1 = 3t - 1, \quad x_2 = t, \quad x_3 = t, \quad t \text{ is any real number.}$$

Discovery

Graph the two lines represented by the system of equations.

$$\begin{aligned}x - 2y &= 1 \\ -2x + 3y &= -3\end{aligned}$$

You can use Gaussian elimination to solve this system as follows.

$$\begin{array}{rcl}x - 2y &= & 1 \\ -1y &= & -1\end{array} \quad \begin{array}{rcl}x - 2y &= & 1 \\ y &= & 1\end{array} \quad \begin{array}{rcl}x &= & 3 \\ y &= & 1\end{array}$$

Graph the system of equations you obtain at each step of this process. What do you observe about the lines? You are asked to repeat this graphical analysis for other systems in Exercises 91 and 92.

SECTION 1.1 Exercises

In Exercises 1–6, determine whether the equation is linear in the variables x and y .

1. $2x - 3y = 4$
2. $3x - 4xy = 0$
3. $\frac{3}{y} + \frac{2}{x} - 1 = 0$
4. $x^2 + y^2 = 4$
5. $2 \sin x - y = 14$
6. $(\sin 2)x - y = 14$

In Exercises 7–10, find a parametric representation of the solution set of the linear equation.

7. $2x - 4y = 0$
8. $3x - \frac{1}{2}y = 9$
9. $x + y + z = 1$
10. $13x_1 - 26x_2 + 39x_3 = 13$

In Exercises 11–16, use back-substitution to solve the system.

11. $\begin{aligned}x_1 - x_2 &= 2 \\ x_2 &= 3\end{aligned}$
12. $\begin{aligned}2x_1 - 4x_2 &= 6 \\ 3x_2 &= 9\end{aligned}$
13. $\begin{aligned}-x + y - z &= 0 \\ 2y + z &= 3 \\ \frac{1}{2}z &= 0\end{aligned}$
14. $\begin{aligned}x - y &= 4 \\ 2y + z &= 6 \\ 3z &= 6\end{aligned}$
15. $\begin{aligned}5x_1 + 2x_2 + x_3 &= 0 \\ 2x_1 + x_2 &= 0\end{aligned}$
16. $\begin{aligned}x_1 + x_2 + x_3 &= 0 \\ x_2 &= 0\end{aligned}$

In Exercises 17–30, graph each system of equations as a pair of lines in the xy -plane. Solve each system and interpret your answer.

17. $\begin{aligned}2x + y &= 4 \\ x - y &= 2\end{aligned}$
18. $\begin{aligned}x + 3y &= 2 \\ -x + 2y &= 3\end{aligned}$
19. $\begin{aligned}x - y &= 1 \\ -2x + 2y &= 5\end{aligned}$
20. $\begin{aligned}\frac{1}{2}x - \frac{1}{3}y &= 1 \\ -2x + \frac{4}{3}y &= -4\end{aligned}$

21. $\begin{aligned}3x - 5y &= 7 \\ 2x + y &= 9\end{aligned}$
23. $\begin{aligned}2x - y &= 5 \\ 5x - y &= 11\end{aligned}$
25. $\begin{aligned}\frac{x+3}{4} + \frac{y-1}{3} &= 1 \\ 2x - y &= 12\end{aligned}$
27. $\begin{aligned}0.05x - 0.03y &= 0.07 \\ 0.07x + 0.02y &= 0.16\end{aligned}$
29. $\begin{aligned}\frac{x}{4} + \frac{y}{6} &= 1 \\ x - y &= 3\end{aligned}$
22. $\begin{aligned}-x + 3y &= 17 \\ 4x + 3y &= 7\end{aligned}$
24. $\begin{aligned}x - 5y &= 21 \\ 6x + 5y &= 21\end{aligned}$
26. $\begin{aligned}\frac{x-1}{2} + \frac{y+2}{3} &= 4 \\ x - 2y &= 5\end{aligned}$
28. $\begin{aligned}0.2x - 0.5y &= -27.8 \\ 0.3x + 0.4y &= 68.7\end{aligned}$
30. $\begin{aligned}\frac{2}{3}x + \frac{1}{6}y &= \frac{2}{3} \\ 4x + y &= 4\end{aligned}$

 In Exercises 31–36, complete the following set of tasks for each system of equations.

- (a) Use a graphing utility to graph the equations in the system.
- (b) Use the graphs to determine whether the system is consistent or inconsistent.
- (c) If the system is consistent, approximate the solution.
- (d) Solve the system algebraically.
- (e) Compare the solution in part (d) with the approximation in part (c). What can you conclude?

31. $\begin{aligned}-3x - y &= 3 \\ 6x + 2y &= 1\end{aligned}$
33. $\begin{aligned}2x - 8y &= 3 \\ \frac{1}{2}x + y &= 0\end{aligned}$
35. $\begin{aligned}4x - 8y &= 9 \\ 0.8x - 1.6y &= 1.8\end{aligned}$
32. $\begin{aligned}4x - 5y &= 3 \\ -8x + 10y &= 14\end{aligned}$
34. $\begin{aligned}9x - 4y &= 5 \\ \frac{1}{2}x + \frac{1}{3}y &= 0\end{aligned}$
36. $\begin{aligned}-5.3x + 2.1y &= 1.25 \\ 15.9x - 6.3y &= -3.75\end{aligned}$

The symbol  indicates an exercise in which you are instructed to use a graphing utility or a symbolic computer software program.

In Exercises 37–56, solve the system of linear equations.

$$37. \begin{cases} x_1 - x_2 = 0 \\ 3x_1 - 2x_2 = -1 \end{cases}$$

$$39. \begin{cases} 2u + v = 120 \\ u + 2v = 120 \end{cases}$$

$$41. \begin{cases} 9x - 3y = -1 \\ \frac{1}{5}x + \frac{2}{5}y = -\frac{1}{3} \end{cases}$$

$$43. \begin{cases} \frac{x-1}{2} + \frac{y+2}{3} = 4 \\ x - 2y = 5 \end{cases}$$

$$45. \begin{cases} 0.02x_1 - 0.05x_2 = -0.19 \\ 0.03x_1 + 0.04x_2 = 0.52 \end{cases}$$

$$47. \begin{cases} x + y + z = 6 \\ 2x - y + z = 3 \\ 3x - z = 0 \end{cases}$$

$$49. \begin{cases} 3x_1 - 2x_2 + 4x_3 = 1 \\ x_1 + x_2 - 2x_3 = 3 \\ 2x_1 - 3x_2 + 6x_3 = 8 \end{cases}$$

$$51. \begin{cases} 2x_1 + x_2 - 3x_3 = 4 \\ 4x_1 + 2x_3 = 10 \\ -2x_1 + 3x_2 - 13x_3 = -8 \end{cases}$$

$$53. \begin{cases} x - 3y + 2z = 18 \\ 5x - 15y + 10z = 18 \end{cases}$$

$$55. \begin{cases} x + y + z + w = 6 \\ 2x + 3y - w = 0 \\ -3x + 4y + z + 2w = 4 \\ x + 2y - z + w = 0 \end{cases}$$

$$56. \begin{cases} x_1 + 3x_4 = 4 \\ 2x_2 - x_3 - x_4 = 0 \\ 3x_2 - 2x_4 = 1 \\ 2x_1 - x_2 + 4x_3 = 5 \end{cases}$$

$$38. \begin{cases} 3x + 2y = 2 \\ 6x + 4y = 14 \end{cases}$$

$$40. \begin{cases} x_1 - 2x_2 = 0 \\ 6x_1 + 2x_2 = 0 \end{cases}$$

$$42. \begin{cases} \frac{2}{3}x_1 + \frac{1}{6}x_2 = 0 \\ 4x_1 + x_2 = 0 \end{cases}$$

$$44. \begin{cases} \frac{x_1+3}{4} + \frac{x_2-1}{3} = 1 \\ 2x_1 - x_2 = 12 \end{cases}$$

$$46. \begin{cases} 0.05x_1 - 0.03x_2 = 0.21 \\ 0.07x_1 + 0.02x_2 = 0.17 \end{cases}$$

$$48. \begin{cases} x + y + z = 2 \\ -x + 3y + 2z = 8 \\ 4x + y = 4 \end{cases}$$

$$50. \begin{cases} 5x_1 - 3x_2 + 2x_3 = 3 \\ 2x_1 + 4x_2 - x_3 = 7 \\ x_1 - 11x_2 + 4x_3 = 3 \end{cases}$$

$$52. \begin{cases} x_1 + 4x_3 = 13 \\ 4x_1 - 2x_2 + x_3 = 7 \\ 2x_1 - 2x_2 - 7x_3 = -19 \end{cases}$$

$$54. \begin{cases} x_1 - 2x_2 + 5x_3 = 2 \\ 3x_1 + 2x_2 - x_3 = -2 \end{cases}$$

$$58. 0.1x - 2.5y + 1.2z - 0.75w = 108$$

$$2.4x + 1.5y - 1.8z + 0.25w = -81$$

$$0.4x - 3.2y + 1.6z - 1.4w = 148.8$$

$$1.6x + 1.2y - 3.2z + 0.6w = -143.2$$

$$59. 123.5x + 61.3y - 32.4z = -262.74$$

$$54.7x - 45.6y + 98.2z = 197.4$$

$$42.4x - 89.3y + 12.9z = 33.66$$

$$60. 120.2x + 62.4y - 36.5z = 258.64$$

$$56.8x - 42.8y + 27.3z = -71.44$$

$$88.1x + 72.5y - 28.5z = 225.88$$

$$61. \frac{1}{2}x_1 - \frac{3}{7}x_2 + \frac{2}{9}x_3 = \frac{349}{630}$$

$$\frac{2}{3}x_1 + \frac{4}{9}x_2 - \frac{2}{5}x_3 = -\frac{19}{45}$$

$$\frac{4}{5}x_1 - \frac{1}{8}x_2 + \frac{4}{3}x_3 = \frac{139}{150}$$

$$62. \frac{1}{4}x_1 - \frac{3}{5}x_2 + \frac{1}{3}x_3 = \frac{43}{60}$$

$$\frac{2}{5}x_1 + \frac{1}{4}x_2 - \frac{5}{6}x_3 = -\frac{331}{600}$$

$$\frac{3}{4}x_1 - \frac{2}{5}x_2 + \frac{1}{5}x_3 = \frac{81}{100}$$

$$63. \frac{1}{8}x - \frac{1}{7}y + \frac{1}{6}z - \frac{1}{5}w = 1$$

$$\frac{1}{7}x + \frac{1}{6}y - \frac{1}{5}z + \frac{1}{4}w = 1$$

$$\frac{1}{6}x - \frac{1}{5}y + \frac{1}{4}z - \frac{1}{3}w = 1$$

$$\frac{1}{5}x + \frac{1}{4}y - \frac{1}{3}z + \frac{1}{2}w = 1$$

$$64. \frac{1}{8}x + \frac{1}{7}y - \frac{1}{6}z + \frac{1}{5}w = 1$$

$$\frac{1}{7}x - \frac{1}{6}y + \frac{1}{5}z - \frac{1}{4}w = 1$$

$$\frac{1}{6}x + \frac{1}{5}y - \frac{1}{4}z + \frac{1}{3}w = 1$$

$$\frac{1}{5}x - \frac{1}{4}y + \frac{1}{3}z - \frac{1}{2}w = 1$$

In Exercises 65–68, state why each system of equations must have at least one solution. Then solve the system and determine if it has exactly one solution or an infinite number of solutions.

$$65. \begin{cases} 4x + 3y + 17z = 0 \\ 5x + 4y + 22z = 0 \\ 4x + 2y + 19z = 0 \end{cases}$$

$$66. \begin{cases} 2x + 3y = 0 \\ 4x + 3y - z = 0 \\ 8x + 3y + 3z = 0 \end{cases}$$

$$67. \begin{cases} 5x + 5y - z = 0 \\ 10x + 5y + 2z = 0 \\ 5x + 15y - 9z = 0 \end{cases}$$

$$68. \begin{cases} 12x + 5y + z = 0 \\ 12x + 4y - z = 0 \end{cases}$$

 In Exercises 57–64, use a computer software program or graphing utility to solve the system of linear equations.

$$57. \begin{cases} x_1 + 0.5x_2 + 0.33x_3 + 0.25x_4 = 1.1 \\ 0.5x_1 + 0.33x_2 + 0.25x_3 + 0.21x_4 = 1.2 \\ 0.33x_1 + 0.25x_2 + 0.2x_3 + 0.17x_4 = 1.3 \\ 0.25x_1 + 0.2x_2 + 0.17x_3 + 0.14x_4 = 1.4 \end{cases}$$

The symbol  indicates that electronic data sets for these exercises are available at college.hmco.com/pic/larsonELA6e. These data sets are compatible with each of the following technologies: MATLAB, *Mathematica*, Maple, Derive, TI-83/TI-83 Plus, TI-84/TI-84 Plus, TI-86, TI-89, TI-92, and TI-92 Plus.

True or False? In Exercises 69 and 70, determine whether each statement is true or false. If a statement is true, give a reason or cite an appropriate statement from the text. If a statement is false, provide an example that shows the statement is not true in all cases or cite an appropriate statement from the text.

69. (a) A system of one linear equation in two variables is always consistent.
 (b) A system of two linear equations in three variables is always consistent.
 (c) If a linear system is consistent, then it has an infinite number of solutions.

70. (a) A system of linear equations can have exactly two solutions.
 (b) Two systems of linear equations are equivalent if they have the same solution set.
 (c) A system of three linear equations in two variables is always inconsistent.
71. Find a system of two equations in two variables, x_1 and x_2 , that has the solution set given by the parametric representation $x_1 = t$ and $x_2 = 3t - 4$, where t is any real number. Then show that the solutions to your system can also be written as $x_1 = \frac{4}{3} + \frac{t}{3}$ and $x_2 = t$.
72. Find a system of two equations in three variables, x_1 , x_2 , and x_3 , that has the solution set given by the parametric representation $x_1 = t$, $x_2 = s$, and $x_3 = 3 + s - t$, where s and t are any real numbers. Then show that the solutions to your system can also be written as $x_1 = 3 + s - t$, $x_2 = s$, and $x_3 = t$.

In Exercises 73–76, solve the system of equations by letting $A = 1/x$, $B = 1/y$, and $C = 1/z$.

73. $\frac{12}{x} - \frac{12}{y} = 7$
 $\frac{3}{x} + \frac{4}{y} = 0$
74. $\frac{2}{x} + \frac{3}{y} = 0$
 $\frac{3}{x} - \frac{4}{y} = -\frac{25}{6}$
75. $\frac{2}{x} + \frac{1}{y} - \frac{3}{z} = 4$
 $\frac{4}{x} + \frac{2}{z} = 10$
 $-\frac{2}{x} + \frac{3}{y} - \frac{13}{z} = -8$
76. $\frac{2}{x} + \frac{1}{y} - \frac{2}{z} = 5$
 $\frac{3}{x} - \frac{4}{y} = -1$
 $\frac{2}{x} + \frac{1}{y} + \frac{3}{z} = 0$

In Exercises 77 and 78, solve the system of linear equations for x and y .

77. $(\cos \theta)x + (\sin \theta)y = 1$
 $(-\sin \theta)x + (\cos \theta)y = 0$
78. $(\cos \theta)x + (\sin \theta)y = 1$
 $(-\sin \theta)x + (\cos \theta)y = 1$

In Exercises 79–84, determine the value(s) of k such that the system of linear equations has the indicated number of solutions.

79. An infinite number of solutions
 $4x + ky = 6$
 $kx + y = -3$
80. An infinite number of solutions
 $kx + y = 4$
 $2x - 3y = -12$

81. Exactly one solution

$$\begin{aligned} x + ky &= 0 \\ kx + y &= 0 \end{aligned}$$

83. No solution

$$\begin{aligned} x + 2y + kz &= 6 \\ 3x + 6y + 8z &= 4 \end{aligned}$$

82. No solution

$$\begin{aligned} x + ky &= 2 \\ kx + y &= 4 \end{aligned}$$

84. Exactly one solution

$$\begin{aligned} kx + 2ky + 3kz &= 4k \\ x + y + z &= 0 \\ 2x - y + z &= 1 \end{aligned}$$

85. Determine the values of k such that the system of linear equations does not have a unique solution.

$$\begin{aligned} x + y + kz &= 3 \\ x + ky + z &= 2 \\ kx + y + z &= 1 \end{aligned}$$

86. Find values of a , b , and c such that the system of linear equations has (a) exactly one solution, (b) an infinite number of solutions, and (c) no solution.

$$\begin{aligned} x + 5y + z &= 0 \\ x + 6y - z &= 0 \\ 2x + ay + bz &= c \end{aligned}$$

87. **Writing** Consider the system of linear equations in x and y .

$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \\ a_3x + b_3y &= c_3 \end{aligned}$$

Describe the graphs of these three equations in the xy -plane when the system has (a) exactly one solution, (b) an infinite number of solutions, and (c) no solution.

88. **Writing** Explain why the system of linear equations in Exercise 87 must be consistent if the constant terms c_1 , c_2 , and c_3 are all zero.

89. Show that if $ax^2 + bx + c = 0$ for all x , then $a = b = c = 0$.

90. Consider the system of linear equations in x and y .

$$\begin{aligned} ax + by &= e \\ cx + dy &= f \end{aligned}$$

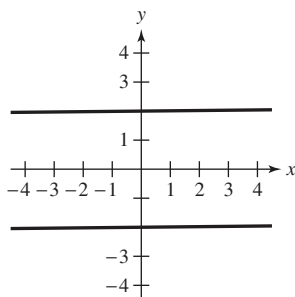
Under what conditions will the system have exactly one solution?

In Exercises 91 and 92, sketch the lines determined by the system of linear equations. Then use Gaussian elimination to solve the system. At each step of the elimination process, sketch the corresponding lines. What do you observe about these lines?

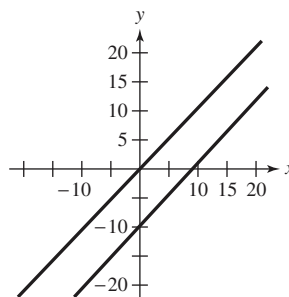
91. $x - 4y = -3$
 $5x - 6y = 13$
92. $2x - 3y = 7$
 $-4x + 6y = -14$

Writing In Exercises 93 and 94, the graphs of two equations are shown and appear to be parallel. Solve the system of equations algebraically. Explain why the graphs are misleading.

93. $100y - x = 200$
 $99y - x = -198$



94. $21x - 20y = 0$
 $13x - 12y = 120$



1.2 Gaussian Elimination and Gauss-Jordan Elimination

In Section 1.1, Gaussian elimination was introduced as a procedure for solving a system of linear equations. In this section you will study this procedure more thoroughly, beginning with some definitions. The first is the definition of a **matrix**.

Definition of a Matrix

If m and n are positive integers, then an $m \times n$ **matrix** is a rectangular array

$$\left[\begin{array}{ccccc} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{array} \right] \quad \left. \begin{array}{l} \\ \\ \\ \\ \end{array} \right\} \begin{array}{l} \\ \\ \\ \\ m \text{ rows} \end{array}$$

$\underbrace{\hspace{10em}}_{n \text{ columns}}$

in which each **entry**, a_{ij} , of the matrix is a number. An $m \times n$ matrix (read “ m by n ”) has m **rows** (horizontal lines) and n **columns** (vertical lines).

REMARK: The plural of matrix is *matrices*. If each entry of a matrix is a *real* number, then the matrix is called a **real matrix**. Unless stated otherwise, all matrices in this text are assumed to be real matrices.

The entry a_{ij} is located in the i th row and the j th column. The index i is called the **row subscript** because it identifies the row in which the entry lies, and the index j is called the **column subscript** because it identifies the column in which the entry lies.

A matrix with m rows and n columns (an $m \times n$ matrix) is said to be of **size** $m \times n$. If $m = n$, the matrix is called **square** of **order** n . For a square matrix, the entries a_{11} , a_{22} , a_{33} , $\dots$ are called the **main diagonal** entries.

EXAMPLE 1 Examples of Matrices

Each matrix has the indicated size.

(a) Size: 1×1

$$[2]$$

(b) Size: 2×2

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

(c) Size: 1×4

$$\left[1 \quad -3 \quad 0 \quad \frac{1}{2} \right]$$

(d) Size: 3×2

$$\begin{bmatrix} e & \pi \\ 2 & \sqrt{2} \\ -7 & 4 \end{bmatrix}$$

One very common use of matrices is to represent systems of linear equations. The matrix derived from the coefficients and constant terms of a system of linear equations is called the **augmented matrix** of the system. The matrix containing only the coefficients of the system is called the **coefficient matrix** of the system. Here is an example.

System	Augmented Matrix	Coefficient Matrix
$\begin{array}{rcl} x - 4y + 3z & = & 5 \\ -x + 3y - z & = & -3 \\ 2x & - & 4z = 6 \end{array}$	$\begin{bmatrix} 1 & -4 & 3 & 5 \\ -1 & 3 & -1 & -3 \\ 2 & 0 & -4 & 6 \end{bmatrix}$	$\begin{bmatrix} 1 & -4 & 3 \\ -1 & 3 & -1 \\ 2 & 0 & -4 \end{bmatrix}$

REMARK: Use 0 to indicate coefficients of zero. The coefficient of y in the third equation is zero, so a 0 takes its place in the matrix. Also note the fourth column of constant terms in the augmented matrix.

When forming either the coefficient matrix or the augmented matrix of a system, you should begin by aligning the variables in the equations vertically.

Given System	Align Variables	Augmented Matrix
$\begin{array}{rcl} x_1 + 3x_2 & = & 9 \\ -x_2 + 4x_3 & = & -2 \\ x_1 - 5x_3 & = & 0 \end{array}$	$\begin{array}{rcl} x_1 + 3x_2 & = & 9 \\ & -x_2 + 4x_3 & = -2 \\ x_1 & - & 5x_3 = 0 \end{array}$	$\begin{bmatrix} 1 & 3 & 0 & 9 \\ 0 & -1 & 4 & -2 \\ 1 & 0 & -5 & 0 \end{bmatrix}$

Elementary Row Operations

In the previous section you studied three operations that can be used on a system of linear equations to produce equivalent systems.

1. Interchange two equations.
2. Multiply an equation by a nonzero constant.
3. Add a multiple of an equation to another equation.

In matrix terminology these three operations correspond to **elementary row operations**. An elementary row operation on an augmented matrix produces a new augmented matrix corresponding to a new (but equivalent) system of linear equations. Two matrices are said to be **row-equivalent** if one can be obtained from the other by a finite sequence of elementary row operations.

Elementary Row Operations

1. Interchange two rows.
2. Multiply a row by a nonzero constant.
3. Add a multiple of a row to another row.

Although elementary row operations are simple to perform, they involve a lot of arithmetic. Because it is easy to make a mistake, you should get in the habit of noting the elementary row operation performed in each step so that it is easier to check your work.

Because solving some systems involves several steps, it is helpful to use a shorthand method of notation to keep track of each elementary row operation you perform. This notation is introduced in the next example.

EXAMPLE 2 Elementary Row Operations

- (a) Interchange the first and second rows.

Original Matrix	New Row-Equivalent Matrix	Notation
$\begin{bmatrix} 0 & 1 & 3 & 4 \\ -1 & 2 & 0 & 3 \\ 2 & -3 & 4 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 2 & 0 & 3 \\ 0 & 1 & 3 & 4 \\ 2 & -3 & 4 & 1 \end{bmatrix}$	$R_1 \leftrightarrow R_2$

- (b) Multiply the first row by $\frac{1}{2}$ to produce a new first row.

Original Matrix	New Row-Equivalent Matrix	Notation
$\begin{bmatrix} 2 & -4 & 6 & -2 \\ 1 & 3 & -3 & 0 \\ 5 & -2 & 1 & 2 \end{bmatrix}$	$\begin{bmatrix} 1 & -2 & 3 & -1 \\ 1 & 3 & -3 & 0 \\ 5 & -2 & 1 & 2 \end{bmatrix}$	$(\frac{1}{2})R_1 \rightarrow R_1$

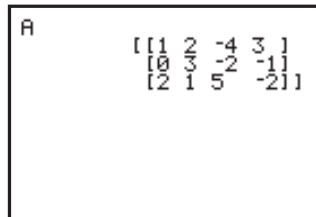
- (c) Add -2 times the first row to the third row to produce a new third row.

Original Matrix	New Row-Equivalent Matrix	Notation
$\begin{bmatrix} 1 & 2 & -4 & 3 \\ 0 & 3 & -2 & -1 \\ 2 & 1 & 5 & -2 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & -4 & 3 \\ 0 & 3 & -2 & -1 \\ 0 & -3 & 13 & -8 \end{bmatrix}$	$R_3 + (-2)R_1 \rightarrow R_3$

REMARK: Notice in Example 2(c) that adding -2 times row 1 to row 3 does not change row 1.

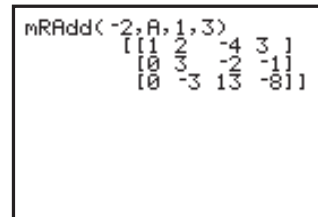
Technology Note

Many graphing utilities and computer software programs can perform elementary row operations on matrices. If you are using a graphing utility, your screens for Example 2(c) may look like those shown below. Keystrokes and programming syntax for these utilities/programs applicable to Example 2(c) are provided in the **Online Technology Guide**, available at college.hmco.com/pic/larsonELA6e.



A

$$\begin{bmatrix} 1 & 2 & -4 & 3 \\ 0 & 3 & -2 & -1 \\ 2 & 1 & 5 & -2 \end{bmatrix}$$



mRAdd(-2, A, 1, 3)

$$\begin{bmatrix} 1 & 2 & -4 & 3 \\ 0 & 3 & -2 & -1 \\ 0 & -3 & 13 & -8 \end{bmatrix}$$

In Example 7 in Section 1.1, you used Gaussian elimination with back-substitution to solve a system of linear equations. You will now learn the matrix version of Gaussian elimination. The two methods used in the next example are essentially the same. The basic difference is that with the matrix method there is no need to rewrite the variables over and over again.

EXAMPLE 3

Using Elementary Row Operations to Solve a System

Linear System

$$\begin{aligned} x - 2y + 3z &= 9 \\ -x + 3y &= -4 \\ 2x - 5y + 5z &= 17 \end{aligned}$$

Add the first equation to the second equation.

$$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ 2x - 5y + 5z &= 17 \end{aligned}$$

Add -2 times the first equation to the third equation.

$$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ -y - z &= -1 \end{aligned}$$

Add the second equation to the third equation.

$$\begin{aligned} x - 2y + 3z &= 9 \\ y + 3z &= 5 \\ 2z &= 4 \end{aligned}$$

Associated Augmented Matrix

$$\left[\begin{array}{cccc} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{array} \right]$$

Add the first row to the second row to produce a new second row.

$$\left[\begin{array}{cccc} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 2 & -5 & 5 & 17 \end{array} \right] \quad R_2 + R_1 \rightarrow R_2$$

Add -2 times the first row to the third row to produce a new third row.

$$\left[\begin{array}{cccc} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & -1 & -1 & -1 \end{array} \right] \quad R_3 + (-2)R_1 \rightarrow R_3$$

Add the second row to the third row to produce a new third row.

$$\left[\begin{array}{cccc} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{array} \right] \quad R_3 + R_2 \rightarrow R_3$$

Multiply the third equation by $\frac{1}{2}$.

$$\begin{aligned}x - 2y + 3z &= 9 \\y + 3z &= 5 \\z &= 2\end{aligned}$$

Multiply the third row by $\frac{1}{2}$ to produce a new third row.

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix} \quad \left(\frac{1}{2}\right)R_3 \rightarrow R_3$$

Now you can use back-substitution to find the solution, as in Example 6 in Section 1.1. The solution is $x = 1$, $y = -1$, and $z = 2$.

The last matrix in Example 3 is said to be in **row-echelon form**. The term *echelon* refers to the stair-step pattern formed by the nonzero elements of the matrix. To be in row-echelon form, a matrix must have the properties listed below.

Definition of Row-Echelon Form of a Matrix

A matrix in **row-echelon form** has the following properties.

1. All rows consisting entirely of zeros occur at the bottom of the matrix.
2. For each row that does not consist entirely of zeros, the first nonzero entry is 1 (called a **leading 1**).
3. For two successive (nonzero) rows, the leading 1 in the higher row is farther to the left than the leading 1 in the lower row.

REMARK: A matrix in row-echelon form is in **reduced row-echelon form** if every column that has a leading 1 has zeros in every position above and below its leading 1.

EXAMPLE 4 Row-Echelon Form

The matrices below are in row-echelon form.

$$(a) \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$$(b) \begin{bmatrix} 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 & -5 & 2 & -1 & 3 \\ 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$(d) \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The matrices shown in parts (b) and (d) are in *reduced* row-echelon form. The matrices listed below are not in row-echelon form.

$$(e) \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

$$(f) \begin{bmatrix} 1 & 2 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & -4 \end{bmatrix}$$

Technology Note

Use a graphing utility or a computer software program to find the reduced row-echelon form of the matrix in part (f) of Example 4. Keystrokes and programming syntax for these utilities/programs applicable to Example 4(f) are provided in the **Online Technology Guide**, available at college.hmco.com/pic/larsonELA6e.

It can be shown that every matrix is row-equivalent to a matrix in row-echelon form. For instance, in Example 4 you could change the matrix in part (e) to row-echelon form by multiplying the second row in the matrix by $\frac{1}{2}$.

The method of using Gaussian elimination with back-substitution to solve a system is as follows.

REMARK: For keystrokes and programming syntax regarding specific graphing utilities and computer software programs involving Example 4(f), please visit college.hmco.com/pic/larsonELA6e. Similar exercises and projects are also available on the website.

Gaussian Elimination with Back-Substitution

1. Write the augmented matrix of the system of linear equations.
2. Use elementary row operations to rewrite the augmented matrix in row-echelon form.
3. Write the system of linear equations corresponding to the matrix in row-echelon form, and use back-substitution to find the solution.

Gaussian elimination with back-substitution works well as an algorithmic method for solving systems of linear equations. For this algorithm, the order in which the elementary row operations are performed is important. Move from *left to right by columns*, changing all entries directly below the leading 1's to zeros.

EXAMPLE 5

Gaussian Elimination with Back-Substitution

Solve the system.

$$\begin{aligned} x_2 + x_3 - 2x_4 &= -3 \\ x_1 + 2x_2 - x_3 &= 2 \\ 2x_1 + 4x_2 + x_3 - 3x_4 &= -2 \\ x_1 - 4x_2 - 7x_3 - x_4 &= -19 \end{aligned}$$

SOLUTION

The augmented matrix for this system is

$$\left[\begin{array}{cccc|c} 0 & 1 & 1 & -2 & -3 \\ 1 & 2 & -1 & 0 & 2 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & -1 & -19 \end{array} \right]$$

Obtain a leading 1 in the upper left corner and zeros elsewhere in the first column.

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 2 & 4 & 1 & -3 & -2 \\ 1 & -4 & -7 & -1 & -19 \end{array} \right]$$

← The first two rows are interchanged. $R_1 \leftrightarrow R_2$

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 1 & -4 & -7 & -1 & -19 \end{array} \right]$$

← Adding -2 times the first row to the third row produces a new third row. $R_3 + (-2)R_1 \rightarrow R_3$

$$\begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 0 & -6 & -6 & -1 & -21 \end{bmatrix}$$

← Adding -1 times the first row to the fourth row produces a new fourth row. $R_4 + (-1)R_1 \rightarrow R_4$

Now that the first column is in the desired form, you should change the second column as shown below.

$$\begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 3 & -3 & -6 \\ 0 & 0 & 0 & -13 & -39 \end{bmatrix}$$

← Adding 6 times the second row to the fourth row produces a new fourth row. $R_4 + (6)R_2 \rightarrow R_4$

To write the third column in proper form, multiply the third row by $\frac{1}{3}$.

$$\begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & -13 & -39 \end{bmatrix}$$

← Multiplying the third row by $\frac{1}{3}$ produces a new third row. $(\frac{1}{3})R_3 \rightarrow R_3$

Similarly, to write the fourth column in proper form, you should multiply the fourth row by $-\frac{1}{13}$.

$$\begin{bmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 1 & -2 & -3 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$

← Multiplying the fourth row by $-\frac{1}{13}$ produces a new fourth row. $(-\frac{1}{13})R_4 \rightarrow R_4$

The matrix is now in row-echelon form, and the corresponding system of linear equations is as shown below.

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 2 \\ x_2 + x_3 - 2x_4 &= -3 \\ x_3 - x_4 &= -2 \\ x_4 &= 3 \end{aligned}$$

Using back-substitution, you can determine that the solution is

$$x_1 = -1, \quad x_2 = 2, \quad x_3 = 1, \quad x_4 = 3.$$

When solving a system of linear equations, remember that it is possible for the system to have no solution. If during the elimination process you obtain a row with all zeros except for the last entry, it is unnecessary to continue the elimination process. You can simply conclude that the system is inconsistent and has no solution.

EXAMPLE 6 A System with No Solution

Solve the system.

$$\begin{aligned}
 x_1 - x_2 + 2x_3 &= 4 \\
 x_1 \quad \quad + x_3 &= 6 \\
 2x_1 - 3x_2 + 5x_3 &= 4 \\
 3x_1 + 2x_2 - x_3 &= 1
 \end{aligned}$$

SOLUTION The augmented matrix for this system is

$$\begin{bmatrix} 1 & -1 & 2 & 4 \\ 1 & 0 & 1 & 6 \\ 2 & -3 & 5 & 4 \\ 3 & 2 & -1 & 1 \end{bmatrix}.$$

Apply Gaussian elimination to the augmented matrix.

$$\begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 2 & -3 & 5 & 4 \\ 3 & 2 & -1 & 1 \end{bmatrix} \quad R_2 + (-1)R_1 \rightarrow R_2$$

$$\begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & -1 & 1 & -4 \\ 3 & 2 & -1 & 1 \end{bmatrix} \quad R_3 + (-2)R_1 \rightarrow R_3$$

$$\begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & -1 & 1 & -4 \\ 0 & 5 & -7 & -11 \end{bmatrix} \quad R_4 + (-3)R_1 \rightarrow R_4$$

$$\begin{bmatrix} 1 & -1 & 2 & 4 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & -2 \\ 0 & 5 & -7 & -11 \end{bmatrix} \quad R_3 + R_2 \rightarrow R_3$$

Note that the third row of this matrix consists of all zeros except for the last entry. This means that the original system of linear equations is *inconsistent*. You can see why this is true by converting back to a system of linear equations.

$$\begin{array}{rcl}
 x_1 - x_2 + 2x_3 & = & 4 \\
 x_2 - x_3 & = & 2 \\
 0 & = & -2 \\
 5x_2 - 7x_3 & = & -11
 \end{array}$$

Because the third “equation” is a false statement, the system has no solution.

Discovery

Consider the system of linear equations.

$$\begin{array}{rcl}
 2x_1 + 3x_2 + 5x_3 & = & 0 \\
 -5x_1 + 6x_2 - 17x_3 & = & 0 \\
 7x_1 - 4x_2 + 3x_3 & = & 0
 \end{array}$$

Without doing any row operations, explain why this system is consistent.

The system below has more variables than equations. Why does it have an infinite number of solutions?

$$\begin{array}{rcl}
 2x_1 + 3x_2 + 5x_3 + 2x_4 & = & 0 \\
 -5x_1 + 6x_2 - 17x_3 - 3x_4 & = & 0 \\
 7x_1 - 4x_2 + 3x_3 + 13x_4 & = & 0
 \end{array}$$

Gauss-Jordan Elimination

With Gaussian elimination, you apply elementary row operations to a matrix to obtain a (row-equivalent) row-echelon form. A second method of elimination, called **Gauss-Jordan elimination** after Carl Gauss and Wilhelm Jordan (1842–1899), continues the reduction process until a *reduced* row-echelon form is obtained. This procedure is demonstrated in the next example.

EXAMPLE 7 Gauss-Jordan Elimination

Use Gauss-Jordan elimination to solve the system.

$$\begin{array}{rcl}
 x - 2y + 3z & = & 9 \\
 -x + 3y & = & -4 \\
 2x - 5y + 5z & = & 17
 \end{array}$$

SOLUTION In Example 3, Gaussian elimination was used to obtain the row-echelon form

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Now, rather than using back-substitution, apply elementary row operations until you obtain a matrix in reduced row-echelon form. To do this, you must produce zeros above each of the leading 1's, as follows.

$$\begin{bmatrix} 1 & 0 & 9 & 19 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix} \quad R_1 + (2)R_2 \rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & 9 & 19 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \quad R_2 + (-3)R_3 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \quad R_1 + (-9)R_3 \rightarrow R_1$$

Now, converting back to a system of linear equations, you have

$$\begin{aligned} x &= 1 \\ y &= -1 \\ z &= 2. \end{aligned}$$

The Gaussian and Gauss-Jordan elimination procedures employ an algorithmic approach easily adapted to computer use. These elimination procedures, however, make no effort to avoid fractional coefficients. For instance, if the system in Example 7 had been listed as

$$\begin{aligned} 2x - 5y + 5z &= 17 \\ x - 2y + 3z &= 9 \\ -x + 3y &= -4 \end{aligned}$$

both procedures would have required multiplying the first row by $\frac{1}{2}$, which would have introduced fractions in the first row. For hand computations, fractions can sometimes be avoided by judiciously choosing the order in which elementary row operations are applied.

REMARK: No matter which order you use, the reduced row-echelon form will be the same.

The next example demonstrates how Gauss-Jordan elimination can be used to solve a system with an infinite number of solutions.

EXAMPLE 8

A System with an Infinite Number of Solutions

Solve the system of linear equations.

$$\begin{aligned} 2x_1 + 4x_2 - 2x_3 &= 0 \\ 3x_1 + 5x_2 &= 1 \end{aligned}$$

SOLUTION

The augmented matrix of the system of linear equations is

$$\begin{bmatrix} 2 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{bmatrix}.$$

Using a graphing utility, a computer software program, or Gauss-Jordan elimination, you can verify that the reduced row-echelon form of the matrix is

$$\begin{bmatrix} 1 & 0 & 5 & 2 \\ 0 & 1 & -3 & -1 \end{bmatrix}.$$

The corresponding system of equations is

$$\begin{aligned} x_1 + 5x_3 &= 2 \\ x_2 - 3x_3 &= -1. \end{aligned}$$

Now, using the parameter t to represent the *nonleading* variable x_3 , you have

$$x_1 = 2 - 5t, \quad x_2 = -1 + 3t, \quad x_3 = t, \quad \text{where } t \text{ is any real number.}$$

REMARK: Note that in Example 8 an arbitrary parameter was assigned to the nonleading variable x_3 . You subsequently solved for the leading variables x_1 and x_2 as functions of t .

You have looked at two elimination methods for solving a system of linear equations. Which is better? To some degree the answer depends on personal preference. In real-life applications of linear algebra, systems of linear equations are usually solved by computer. Most computer programs use a form of Gaussian elimination, with special emphasis on ways to reduce rounding errors and minimize storage of data. Because the examples and exercises in this text are generally much simpler and focus on the underlying concepts, you will need to know both elimination methods.

Homogeneous Systems of Linear Equations

As the final topic of this section, you will look at systems of linear equations in which each of the constant terms is zero. We call such systems **homogeneous**. For example, a homogeneous system of m equations in n variables has the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= 0 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= 0. \end{aligned}$$

It is easy to see that a homogeneous system must have at least one solution. Specifically, if all variables in a homogeneous system have the value zero, then each of the equations must be satisfied. Such a solution is called **trivial** (or **obvious**). For instance, a homogeneous system of three equations in the three variables x_1 , x_2 , and x_3 must have $x_1 = 0$, $x_2 = 0$, and $x_3 = 0$ as a trivial solution.

EXAMPLE 9 Solving a Homogeneous System of Linear Equations

Solve the system of linear equations.

$$x_1 - x_2 + 3x_3 = 0$$

$$2x_1 + x_2 + 3x_3 = 0$$

SOLUTION Applying Gauss-Jordan elimination to the augmented matrix

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 2 & 1 & 3 & 0 \end{bmatrix}$$

yields the matrix shown below.

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 0 & 3 & -3 & 0 \end{bmatrix} \quad R_2 + (-2)R_1 \rightarrow R_2$$

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix} \quad \left(\frac{1}{3}\right)R_2 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix} \quad R_1 + R_2 \rightarrow R_1$$

The system of equations corresponding to this matrix is

$$x_1 + 2x_3 = 0$$

$$x_2 - x_3 = 0.$$

Using the parameter $t = x_3$, the solution set is

$$x_1 = -2t, \quad x_2 = t, \quad x_3 = t, \quad t \text{ is any real number.}$$

This system of equations has an infinite number of solutions, one of which is the trivial solution (given by $t = 0$).

Example 9 illustrates an important point about homogeneous systems of linear equations. You began with two equations in three variables and discovered that the system has an infinite number of solutions. In general, a homogeneous system with fewer equations than variables has an infinite number of solutions.

THEOREM 1.1
The Number of Solutions of
a Homogeneous System

Every homogeneous system of linear equations is consistent. Moreover, if the system has fewer equations than variables, then it must have an infinite number of solutions.

A proof of Theorem 1.1 can be done using the same procedures as those used in Example 9, but for a general matrix.

SECTION 1.2 Exercises

In Exercises 1–8, determine the size of the matrix.

1. $\begin{bmatrix} 1 & 2 & -4 \\ 3 & -4 & 6 \\ 0 & 1 & 2 \end{bmatrix}$

2. $\begin{bmatrix} 2 & -1 & 4 & 2 \\ 1 & 0 & 2 & -6 \end{bmatrix}$

3. $\begin{bmatrix} 2 & -1 & -1 & 1 \\ -6 & 2 & 0 & 1 \end{bmatrix}$

4. $[1 \quad 2 \quad 3 \quad 0 \quad 1]$

5. $[1 \quad 2 \quad 3 \quad 4 \quad -10]$

6. $[-1]$

7. $\begin{bmatrix} 8 & 6 & 4 & 1 & 3 \\ 2 & 1 & -7 & 4 & 1 \\ 1 & 1 & -1 & 2 & 1 \\ 1 & -1 & 2 & 0 & 0 \end{bmatrix}$

8. $\begin{bmatrix} 1 \\ 2 \\ -1 \\ -2 \end{bmatrix}$

In Exercises 9–14, determine whether the matrix is in row-echelon form. If it is, determine whether it is also in reduced row-echelon form.

9. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

10. $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 2 & 1 \end{bmatrix}$

11. $\begin{bmatrix} 2 & 0 & 1 & 3 \\ 0 & -1 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

12. $\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

13. $\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{bmatrix}$

14. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

In Exercises 15–22, find the solution set of the system of linear equations represented by the augmented matrix.

15. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

16. $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}$

17. $\begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

18. $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

19. $\begin{bmatrix} 2 & 1 & -1 & 3 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix}$

20. $\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & -2 & 1 & -2 \\ 1 & 0 & 1 & 0 \end{bmatrix}$

21. $\begin{bmatrix} 1 & 2 & 0 & 1 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 4 \end{bmatrix}$

22. $\begin{bmatrix} 1 & 2 & 0 & 1 & 3 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$

In Exercises 23–36, solve the system using either Gaussian elimination with back-substitution or Gauss-Jordan elimination.

23. $\begin{aligned} x + 2y &= 7 \\ 2x + y &= 8 \end{aligned}$

24. $\begin{aligned} 2x + 6y &= 16 \\ -2x - 6y &= -16 \end{aligned}$

25. $\begin{aligned} -x + 2y &= 1.5 \\ 2x - 4y &= 3 \end{aligned}$

26. $\begin{aligned} 2x - y &= -0.1 \\ 3x + 2y &= 1.6 \end{aligned}$

27. $\begin{aligned} -3x + 5y &= -22 \\ 3x + 4y &= 4 \\ 4x - 8y &= 32 \end{aligned}$

28. $\begin{aligned} x + 2y &= 0 \\ x + y &= 6 \\ 3x - 2y &= 8 \end{aligned}$

29. $\begin{aligned} x_1 - 3x_3 &= -2 \\ 3x_1 + x_2 - 2x_3 &= 5 \\ 2x_1 + 2x_2 + x_3 &= 4 \end{aligned}$

30. $\begin{aligned} 2x_1 - x_2 + 3x_3 &= 24 \\ 2x_2 - x_3 &= 14 \\ 7x_1 - 5x_2 &= 6 \end{aligned}$

31. $\begin{aligned} x_1 + x_2 - 5x_3 &= 3 \\ x_1 - 2x_3 &= 1 \\ 2x_1 - x_2 - x_3 &= 0 \end{aligned}$

32. $\begin{aligned} 2x_1 + 3x_3 &= 3 \\ 4x_1 - 3x_2 + 7x_3 &= 5 \\ 8x_1 - 9x_2 + 15x_3 &= 10 \end{aligned}$

33. $\begin{aligned} 4x + 12y - 7z - 20w &= 22 \\ 3x + 9y - 5z - 28w &= 30 \end{aligned}$

34. $\begin{aligned} x + 2y + z &= 8 \\ -3x - 6y - 3z &= -21 \end{aligned}$

35. $\begin{aligned} 3x + 3y + 12z &= 6 \\ x + y + 4z &= 2 \\ 2x + 5y + 20z &= 10 \\ -x + 2y + 8z &= 4 \end{aligned}$

36. $\begin{aligned} 2x + y - z + 2w &= -6 \\ 3x + 4y + w &= 1 \\ x + 5y + 2z + 6w &= -3 \\ 5x + 2y - z - w &= 3 \end{aligned}$



In Exercises 37–42, use a computer software program or graphing utility to solve the system of linear equations.

37. $\begin{aligned} x_1 - 2x_2 + 5x_3 - 3x_4 &= 23.6 \\ x_1 + 4x_2 - 7x_3 - 2x_4 &= 45.7 \\ 3x_1 - 5x_2 + 7x_3 + 4x_4 &= 29.9 \end{aligned}$

38. $\begin{aligned} 23.4x - 45.8y + 43.7z &= 87.2 \\ 86.4x + 12.3y - 56.9z &= 14.5 \\ 93.6x - 50.7y + 12.6z &= 44.4 \end{aligned}$

$$\begin{aligned}
 39. \quad & x_1 - x_2 + 2x_3 + 2x_4 + 6x_5 = 6 \\
 & 3x_1 - 2x_2 + 4x_3 + 4x_4 + 12x_5 = 14 \\
 & \quad x_2 - x_3 - x_4 - 3x_5 = -3 \\
 & 2x_1 - 2x_2 + 4x_3 + 5x_4 + 15x_5 = 10 \\
 & 2x_1 - 2x_2 + 4x_3 + 4x_4 + 13x_5 = 13
 \end{aligned}$$

$$\begin{aligned}
 40. \quad & x_1 + x_2 - 2x_3 + 3x_4 + 2x_5 = 9 \\
 & 3x_1 + 3x_2 - x_3 + x_4 + x_5 = 5 \\
 & 2x_1 + 2x_2 - x_3 + x_4 - 2x_5 = 1 \\
 & 4x_1 + 4x_2 + x_3 - 3x_5 = 4 \\
 & 8x_1 + 5x_2 - 2x_3 - x_4 + 2x_5 = 3
 \end{aligned}$$

$$\begin{aligned}
 41. \quad & 4x_1 - 3x_2 + x_3 - x_4 + 2x_5 - x_6 = 8 \\
 & x_1 - 2x_2 + x_3 - 3x_4 + x_5 - 4x_6 = 4 \\
 & 2x_1 + x_2 - 3x_3 + x_4 - 2x_5 + 5x_6 = 2 \\
 & -2x_1 + 3x_2 - x_3 + x_4 - x_5 + 2x_6 = -7 \\
 & x_1 - 3x_2 + x_3 - 2x_4 + x_5 - 2x_6 = 9 \\
 & 5x_1 - 4x_2 - x_3 - x_4 + 4x_5 + 5x_6 = 9
 \end{aligned}$$

$$\begin{aligned}
 42. \quad & x_1 + 2x_2 - 2x_3 + 2x_4 - x_5 + 3x_6 = 0 \\
 & 2x_1 - x_2 + 3x_3 + x_4 - 3x_5 + 2x_6 = 17 \\
 & x_1 + 3x_2 - 2x_3 + x_4 - 2x_5 - 3x_6 = -5 \\
 & 3x_1 - 2x_2 + x_3 - x_4 + 3x_5 - 2x_6 = -1 \\
 & -x_1 - 2x_2 + x_3 + 2x_4 - 2x_5 + 3x_6 = 10 \\
 & x_1 - 3x_2 + x_3 + 3x_4 - 2x_5 + x_6 = 11
 \end{aligned}$$

In Exercises 43–46, solve the homogeneous linear system corresponding to the coefficient matrix provided.

$$43. \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$44. \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$45. \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$46. \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$47. \text{ Consider the matrix } A = \begin{bmatrix} 1 & k & 2 \\ -3 & 4 & 1 \end{bmatrix}.$$

- If A is the *augmented* matrix of a system of linear equations, determine the number of equations and the number of variables.
- If A is the *augmented* matrix of a system of linear equations, find the value(s) of k such that the system is consistent.
- If A is the *coefficient* matrix of a *homogeneous* system of linear equations, determine the number of equations and the number of variables.

- If A is the *coefficient* matrix of a *homogeneous* system of linear equations, find the value(s) of k such that the system is consistent.

$$48. \text{ Consider the matrix } A = \begin{bmatrix} 2 & -1 & 3 \\ -4 & 2 & k \\ 4 & -2 & 6 \end{bmatrix}.$$

- If A is the *augmented* matrix of a system of linear equations, determine the number of equations and the number of variables.
- If A is the *augmented* matrix of a system of linear equations, find the value(s) of k such that the system is consistent.
- If A is the *coefficient* matrix of a *homogeneous* system of linear equations, determine the number of equations and the number of variables.
- If A is the *coefficient* matrix of a *homogeneous* system of linear equations, find the value(s) of k such that the system is consistent.

In Exercises 49 and 50, find values of a , b , and c (if possible) such that the system of linear equations has (a) a unique solution, (b) no solution, and (c) an infinite number of solutions.

$$\begin{array}{ll}
 49. \quad x + y = 2 & 50. \quad x + y = 0 \\
 \quad y + z = 2 & \quad y + z = 0 \\
 \quad x + z = 2 & \quad x + z = 0 \\
 ax + by + cz = 0 & ax + by + cz = 0
 \end{array}$$

51. The system below has one solution: $x = 1$, $y = -1$, and $z = 2$.

$$\begin{aligned}
 4x - 2y + 5z &= 16 && \text{Equation 1} \\
 x + y &= 0 && \text{Equation 2} \\
 -x - 3y + 2z &= 6 && \text{Equation 3}
 \end{aligned}$$

Solve the systems provided by (a) Equations 1 and 2, (b) Equations 1 and 3, and (c) Equations 2 and 3. (d) How many solutions does each of these systems have?

52. Assume the system below has a unique solution.

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 && \text{Equation 1} \\
 a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 && \text{Equation 2} \\
 a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 && \text{Equation 3}
 \end{aligned}$$

Does the system composed of Equations 1 and 2 have a unique solution, no solution, or an infinite number of solutions?

In Exercises 53 and 54, find the unique reduced row-echelon matrix that is row-equivalent to the matrix provided.

53. $\begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix}$

54. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

55. **Writing** Describe all possible 2×2 reduced row-echelon matrices. Support your answer with examples.

56. **Writing** Describe all possible 3×3 reduced row-echelon matrices. Support your answer with examples.

True or False? In Exercises 57 and 58, determine whether each statement is true or false. If a statement is true, give a reason or cite an appropriate statement from the text. If a statement is false, provide an example that shows the statement is not true in all cases or cite an appropriate statement from the text.

57. (a) A 6×3 matrix has six rows.
 (b) Every matrix is row-equivalent to a matrix in row-echelon form.
 (c) If the row-echelon form of the augmented matrix of a system of linear equations contains the row $[1 \ 0 \ 0 \ 0 \ 0]$, then the original system is inconsistent.
 (d) A homogeneous system of four linear equations in six variables has an infinite number of solutions.
58. (a) A 4×7 matrix has four columns.
 (b) Every matrix has a unique reduced row-echelon form.
 (c) A homogeneous system of four linear equations in four variables is always consistent.
 (d) Multiplying a row of a matrix by a constant is one of the elementary row operations.

In Exercises 59 and 60, determine conditions on a , b , c , and d such that the matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

will be row-equivalent to the given matrix.

59. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

60. $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

In Exercises 61 and 62, find all values of λ (the Greek letter lambda) such that the homogeneous system of linear equations will have nontrivial solutions.

61. $(\lambda - 2)x + \quad y = 0$
 $x + (\lambda - 2)y = 0$

62. $(\lambda - 1)x + 2y = 0$
 $x + \lambda y = 0$

63. **Writing** Is it possible for a system of linear equations with fewer equations than variables to have no solution? If so, give an example.

64. **Writing** Does a matrix have a unique row-echelon form? Illustrate your answer with examples. Is the reduced row-echelon form unique?

65. **Writing** Consider the 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

Perform the sequence of row operations.

- (a) Add (-1) times the second row to the first row.
 (b) Add 1 times the first row to the second row.
 (c) Add (-1) times the second row to the first row.
 (d) Multiply the first row by (-1) .

What happened to the original matrix? Describe, in general, how to interchange two rows of a matrix using only the second and third elementary row operations.

66. The augmented matrix represents a system of linear equations that has been reduced using Gauss-Jordan elimination. Write a system of equations with nonzero coefficients that is represented by the reduced matrix.

$$\begin{bmatrix} 1 & 0 & 3 & -2 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are many correct answers.

67. **Writing** Describe the row-echelon form of an augmented matrix that corresponds to a system of linear equations that is inconsistent.

68. **Writing** Describe the row-echelon form of an augmented matrix that corresponds to a system of linear equations that has infinitely many solutions.

69. **Writing** In your own words, describe the difference between a matrix in row-echelon form and a matrix in reduced row-echelon form.

1.3 Applications of Systems of Linear Equations

Systems of linear equations arise in a wide variety of applications and are one of the central themes in linear algebra. In this section you will look at two such applications, and you will see many more in subsequent chapters. The first application shows how to fit a polynomial function to a set of data points in the plane. The second application focuses on networks and Kirchhoff's Laws for electricity.

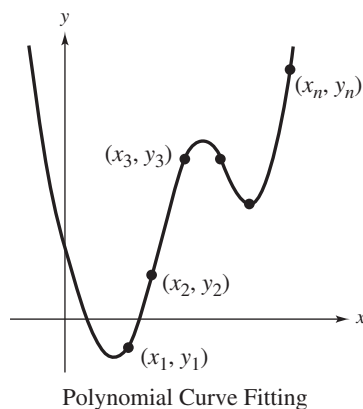


Figure 1.4

Polynomial Curve Fitting

Suppose a collection of data is represented by n points in the xy -plane,

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

and you are asked to find a polynomial function of degree $n - 1$

$$p(x) = a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1}$$

whose graph passes through the specified points. This procedure is called **polynomial curve fitting**. If all x -coordinates of the points are distinct, then there is precisely one polynomial function of degree $n - 1$ (or less) that fits the n points, as shown in Figure 1.4.

To solve for the n coefficients of $p(x)$, substitute each of the n points into the polynomial function and obtain n linear equations in n variables $a_0, a_1, a_2, \dots, a_{n-1}$.

$$a_0 + a_1x_1 + a_2x_1^2 + \dots + a_{n-1}x_1^{n-1} = y_1$$

$$a_0 + a_1x_2 + a_2x_2^2 + \dots + a_{n-1}x_2^{n-1} = y_2$$

$$\vdots$$

$$a_0 + a_1x_n + a_2x_n^2 + \dots + a_{n-1}x_n^{n-1} = y_n$$

This procedure is demonstrated with a second-degree polynomial in Example 1.

EXAMPLE 1

Polynomial Curve Fitting

Determine the polynomial $p(x) = a_0 + a_1x + a_2x^2$ whose graph passes through the points $(1, 4)$, $(2, 0)$, and $(3, 12)$.

SOLUTION

Substituting $x = 1, 2$, and 3 into $p(x)$ and equating the results to the respective y -values produces the system of linear equations in the variables a_0, a_1 , and a_2 shown below.

$$p(1) = a_0 + a_1(1) + a_2(1)^2 = a_0 + a_1 + a_2 = 4$$

$$p(2) = a_0 + a_1(2) + a_2(2)^2 = a_0 + 2a_1 + 4a_2 = 0$$

$$p(3) = a_0 + a_1(3) + a_2(3)^2 = a_0 + 3a_1 + 9a_2 = 12$$

The solution of this system is $a_0 = 24$, $a_1 = -28$, and $a_2 = 8$, so the polynomial function is

$$p(x) = 24 - 28x + 8x^2.$$



Simulation

Explore this concept further with an electronic simulation available on the website college.hmco.com/pic/larsonELA6e.

The graph of p is shown in Figure 1.5.

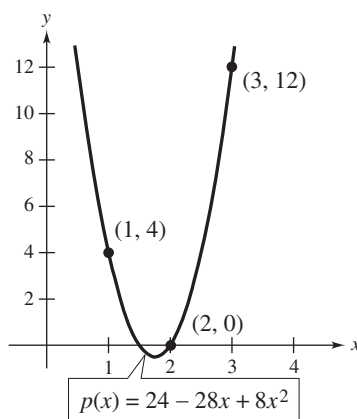


Figure 1.5

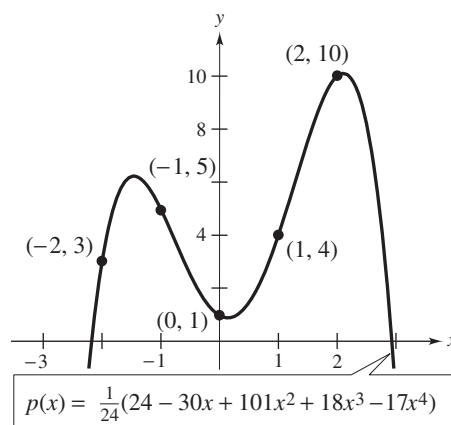


Figure 1.6

EXAMPLE 2 Polynomial Curve Fitting

Find a polynomial that fits the points $(-2, 3)$, $(-1, 5)$, $(0, 1)$, $(1, 4)$, and $(2, 10)$.

SOLUTION Because you are provided with five points, choose a fourth-degree polynomial function

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4.$$

Substituting the given points into $p(x)$ produces the system of linear equations listed below.

$$a_0 - 2a_1 + 4a_2 - 8a_3 + 16a_4 = 3$$

$$a_0 - a_1 + a_2 - a_3 + a_4 = 5$$

$$a_0 = 1$$

$$a_0 + a_1 + a_2 + a_3 + a_4 = 4$$

$$a_0 + 2a_1 + 4a_2 + 8a_3 + 16a_4 = 10$$

The solution of these equations is

$$a_0 = 1, \quad a_1 = -\frac{30}{24}, \quad a_2 = \frac{101}{24}, \quad a_3 = \frac{18}{24}, \quad a_4 = -\frac{17}{24}$$

which means the polynomial function is

$$\begin{aligned} p(x) &= 1 - \frac{30}{24}x + \frac{101}{24}x^2 + \frac{18}{24}x^3 - \frac{17}{24}x^4 \\ &= \frac{1}{24}(24 - 30x + 101x^2 + 18x^3 - 17x^4). \end{aligned}$$

The graph of p is shown in Figure 1.6.

The system of linear equations in Example 2 is relatively easy to solve because the x -values are small. With a set of points with large x -values, it is usually best to *translate* the values before attempting the curve-fitting procedure. This approach is demonstrated in the next example.

EXAMPLE 3 Translating Large x -Values Before Curve Fitting

Find a polynomial that fits the points

$$\begin{array}{ccccc} \overbrace{(x_1, y_1)} & \overbrace{(x_2, y_2)} & \overbrace{(x_3, y_3)} & \overbrace{(x_4, y_4)} & \overbrace{(x_5, y_5)} \\ (2006, 3), & (2007, 5), & (2008, 1), & (2009, 4), & (2010, 10). \end{array}$$

SOLUTION Because the given x -values are large, use the translation $z = x - 2008$ to obtain

$$\begin{array}{ccccc} \overbrace{(z_1, y_1)} & \overbrace{(z_2, y_2)} & \overbrace{(z_3, y_3)} & \overbrace{(z_4, y_4)} & \overbrace{(z_5, y_5)} \\ (-2, 3), & (-1, 5), & (0, 1), & (1, 4), & (2, 10). \end{array}$$

This is the same set of points as in Example 2. So, the polynomial that fits these points is

$$\begin{aligned} p(z) &= \frac{1}{24}(24 - 30z + 101z^2 + 18z^3 - 17z^4) \\ &= 1 - \frac{5}{4}z + \frac{101}{24}z^2 + \frac{3}{4}z^3 - \frac{17}{24}z^4. \end{aligned}$$

Letting $z = x - 2008$, you have

$$p(x) = 1 - \frac{5}{4}(x - 2008) + \frac{101}{24}(x - 2008)^2 + \frac{3}{4}(x - 2008)^3 - \frac{17}{24}(x - 2008)^4.$$

EXAMPLE 4 An Application of Curve Fitting

Find a polynomial that relates the periods of the first three planets to their mean distances from the sun, as shown in Table 1.1. Then test the accuracy of the fit by using the polynomial to calculate the period of Mars. (Distance is measured in astronomical units, and period is measured in years.) (Source: *CRC Handbook of Chemistry and Physics*)

TABLE 1.1

Planet	Mercury	Venus	Earth	Mars	Jupiter	Saturn
Mean Distance	0.387	0.723	1.0	1.523	5.203	9.541
Period	0.241	0.615	1.0	1.881	11.861	29.457

SOLUTION Begin by fitting a quadratic polynomial function

$$p(x) = a_0 + a_1x + a_2x^2$$

to the points (0.387, 0.241), (0.723, 0.615), and (1, 1). The system of linear equations obtained by substituting these points into $p(x)$ is

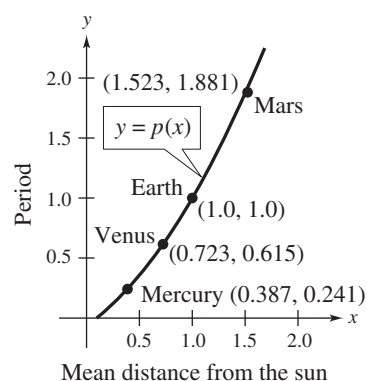


Figure 1.7

$$a_0 + 0.387a_1 + (0.387)^2a_2 = 0.241$$

$$a_0 + 0.723a_1 + (0.723)^2a_2 = 0.615$$

$$a_0 + a_1 + a_2 = 1.$$

The approximate solution of the system is

$$a_0 \approx -0.0634, \quad a_1 \approx 0.6119, \quad a_2 \approx 0.4515$$

which means that the polynomial function can be approximated by

$$p(x) = -0.0634 + 0.6119x + 0.4515x^2.$$

Using $p(x)$ to evaluate the period of Mars produces

$$p(1.523) \approx 1.916 \text{ years.}$$

This estimate is compared graphically with the actual period of Mars in Figure 1.7. Note that the actual period (from Table 1.1) is 1.881 years.

An important lesson may be learned from the application shown in Example 4: The polynomial that fits the given data points is not necessarily an accurate model for the relationship between x and y for x -values other than those corresponding to the given points. Generally, the farther the additional points are from the given points, the worse the fit. For instance, in Example 4 the mean distance of Jupiter is 5.203. The corresponding polynomial approximation for the period is 15.343 years—a poor estimate of Jupiter's actual period of 11.861 years.

The problem of curve fitting can be difficult. Types of functions other than polynomial functions often provide better fits. To see this, look again at the curve-fitting problem in Example 4. Taking the natural logarithms of the distances and periods of the first six planets produces the results shown in Table 1.2 and Figure 1.8.

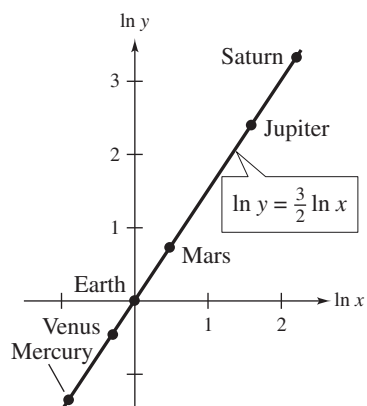


Figure 1.8

TABLE 1.2

Planet	Mercury	Venus	Earth	Mars	Jupiter	Saturn
Mean Distance (x)	0.387	0.723	1.0	1.523	5.203	9.541
Natural Log of Mean	-0.949	-0.324	0.0	0.421	1.649	2.256
Period (y)	0.241	0.615	1.0	1.881	11.861	29.457
Natural Log of Period	-1.423	-0.486	0.0	0.632	2.473	3.383

Now, fitting a polynomial to the logarithms of the distances and periods produces the *linear relationship* between $\ln x$ and $\ln y$ shown below.

$$\ln y = \frac{3}{2} \ln x$$

From this equation it follows that $y = x^{3/2}$, or $y^2 = x^3$.

In other words, the square of the period (in years) of each planet is equal to the cube of its mean distance (in astronomical units) from the sun. This relationship was first discovered by Johannes Kepler in 1619.

Network Analysis

Networks composed of branches and junctions are used as models in many diverse fields such as economics, traffic analysis, and electrical engineering.

In such models it is assumed that the total flow into a junction is equal to the total flow out of the junction. For example, because the junction shown in Figure 1.9 has 25 units flowing into it, there must be 25 units flowing out of it. This is represented by the linear equation

$$x_1 + x_2 = 25.$$

Because each junction in a network gives rise to a linear equation, you can analyze the flow through a network composed of several junctions by solving a system of linear equations. This procedure is illustrated in Example 5.

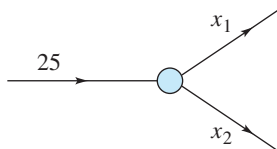


Figure 1.9

EXAMPLE 5 Analysis of a Network

Set up a system of linear equations to represent the network shown in Figure 1.10, and solve the system.

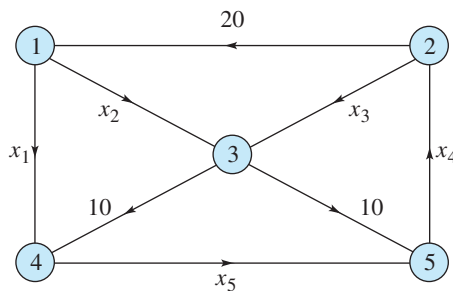


Figure 1.10

SOLUTION Each of the network's five junctions gives rise to a linear equation, as shown below.

$$\begin{array}{rclcl}
 x_1 + x_2 & & = & 20 & \text{Junction 1} \\
 & x_3 - x_4 & = & -20 & \text{Junction 2} \\
 & x_2 + x_3 & = & 20 & \text{Junction 3} \\
 x_1 & & - x_5 & = & -10 & \text{Junction 4} \\
 & -x_4 + x_5 & = & -10 & \text{Junction 5}
 \end{array}$$

The augmented matrix for this system is

$$\left[\begin{array}{cccccc} 1 & 1 & 0 & 0 & 0 & 20 \\ 0 & 0 & 1 & -1 & 0 & -20 \\ 0 & 1 & 1 & 0 & 0 & 20 \\ 1 & 0 & 0 & 0 & -1 & -10 \\ 0 & 0 & 0 & -1 & 1 & -10 \end{array} \right].$$

Gauss-Jordan elimination produces the matrix

$$\left[\begin{array}{cccccc} 1 & 0 & 0 & 0 & -1 & -10 \\ 0 & 1 & 0 & 0 & 1 & 30 \\ 0 & 0 & 1 & 0 & -1 & -10 \\ 0 & 0 & 0 & 1 & -1 & 10 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

From the matrix above, you can see that

$$x_1 - x_5 = -10, \quad x_2 + x_5 = 30, \quad x_3 - x_5 = -10, \quad \text{and} \quad x_4 - x_5 = 10.$$

Letting $t = x_5$, you have

$$x_1 = t - 10, \quad x_2 = -t + 30, \quad x_3 = t - 10, \quad x_4 = t + 10, \quad x_5 = t$$

where t is a real number, so this system has an infinite number of solutions.

In Example 5, suppose you could control the amount of flow along the branch labeled x_5 . Using the solution from Example 5, you could then control the flow represented by each of the other variables. For instance, letting $t = 10$ would reduce the flow of x_1 and x_3 to zero, as shown in Figure 1.11. Similarly, letting $t = 20$ would produce the network shown in Figure 1.12.

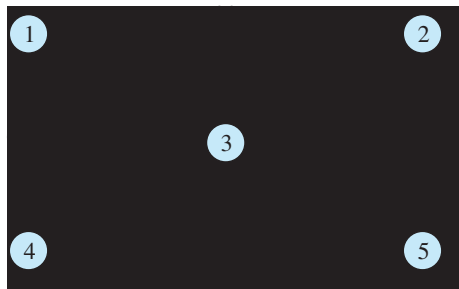


Figure 1.11

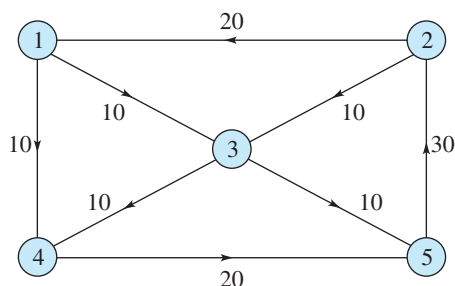


Figure 1.12

You can see how the type of network analysis demonstrated in Example 5 could be used in problems dealing with the flow of traffic through the streets of a city or the flow of water through an irrigation system.

An electrical network is another type of network where analysis is commonly applied. An analysis of such a system uses two properties of electrical networks known as **Kirchhoff's Laws**.

1. All the current flowing into a junction must flow out of it.
2. The sum of the products IR (I is current and R is resistance) around a closed path is equal to the total voltage in the path.

In an electrical network, current is measured in amps, resistance in ohms, and the product of current and resistance in volts. Batteries are represented by the symbol $\text{---}||\text{---}$. The larger vertical bar denotes where the current flows out of the terminal. Resistance is denoted by the symbol $\text{---}\wedge\wedge\wedge\text{---}$. The direction of the current is indicated by an arrow in the branch.

REMARK: A *closed* path is a sequence of branches such that the beginning point of the first branch coincides with the end point of the last branch.

EXAMPLE 6 Analysis of an Electrical Network

Determine the currents I_1 , I_2 , and I_3 for the electrical network shown in Figure 1.13.

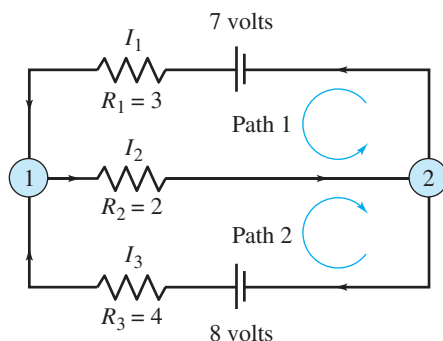


Figure 1.13

SOLUTION Applying Kirchhoff's first law to either junction produces

$$I_1 + I_3 = I_2 \quad \text{Junction 1 or Junction 2}$$

and applying Kirchhoff's second law to the two paths produces

$$R_1 I_1 + R_2 I_2 = 3I_1 + 2I_2 = 7 \quad \text{Path 1}$$

$$R_2 I_2 + R_3 I_3 = 2I_2 + 4I_3 = 8. \quad \text{Path 2}$$

So, you have the following system of three linear equations in the variables I_1 , I_2 , and I_3 .

$$I_1 - I_2 + I_3 = 0$$

$$3I_1 + 2I_2 = 7$$

$$2I_2 + 4I_3 = 8$$

Applying Gauss-Jordan elimination to the augmented matrix

$$\begin{bmatrix} 1 & -1 & 1 & 0 \\ 3 & 2 & 0 & 7 \\ 0 & 2 & 4 & 8 \end{bmatrix}$$

produces the reduced row-echelon form

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

which means $I_1 = 1$ amp, $I_2 = 2$ amps, and $I_3 = 1$ amp.

EXAMPLE 7 Analysis of an Electrical Network

Determine the currents I_1 , I_2 , I_3 , I_4 , I_5 , and I_6 for the electrical network shown in Figure 1.14.

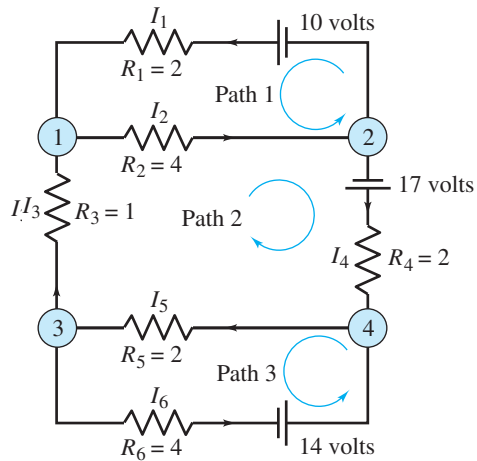


Figure 1.14

SOLUTION Applying Kirchhoff's first law to the four junctions produces

$$I_1 + I_3 = I_2 \quad \text{Junction 1}$$

$$I_1 + I_4 = I_2 \quad \text{Junction 2}$$

$$I_3 + I_6 = I_5 \quad \text{Junction 3}$$

$$I_4 + I_6 = I_5 \quad \text{Junction 4}$$

and applying Kirchhoff's second law to the three paths produces

$$2I_1 + 4I_2 = 10 \quad \text{Path 1}$$

$$4I_2 + I_3 + 2I_4 + 2I_5 = 17 \quad \text{Path 2}$$

$$2I_5 + 4I_6 = 14. \quad \text{Path 3}$$

You now have the following system of seven linear equations in the variables I_1 , I_2 , I_3 , I_4 , I_5 , and I_6 .

$$I_1 - I_2 + I_3 = 0$$

$$I_1 - I_2 + I_4 = 0$$

$$I_3 - I_5 + I_6 = 0$$

$$I_4 - I_5 + I_6 = 0$$

$$2I_1 + 4I_2 = 10$$

$$4I_2 + I_3 + 2I_4 + 2I_5 = 17$$

$$2I_5 + 4I_6 = 14$$

Using Gauss-Jordan elimination, a graphing utility, or a computer software program, you can solve this system to obtain

$$I_1 = 1, \quad I_2 = 2, \quad I_3 = 1, \quad I_4 = 1, \quad I_5 = 3, \quad \text{and} \quad I_6 = 2$$

meaning $I_1 = 1$ amp, $I_2 = 2$ amps, $I_3 = 1$ amp, $I_4 = 1$ amp, $I_5 = 3$ amps, and $I_6 = 2$ amps.

SECTION 1.3 Exercises

Polynomial Curve Fitting

In Exercises 1–6, (a) determine the polynomial function whose graph passes through the given points, and (b) sketch the graph of the polynomial function, showing the given points.

1. (2, 5), (3, 2), (4, 5)
2. (2, 4), (3, 4), (4, 4)
3. (2, 4), (3, 6), (5, 10)
4. (−1, 3), (0, 0), (1, 1), (4, 58)
5. (2006, 5), (2007, 7), (2008, 12) ($z = x - 2007$)
6. (2005, 150), (2006, 180), (2007, 240), (2008, 360) ($z = x - 2005$)

7. **Writing** Try to fit the graph of a polynomial function to the values shown in the table. What happens, and why?

x	1	2	3	3	4
y	1	1	2	3	4

8. The graph of a function f passes through the points $(0, 1)$, $(2, \frac{1}{3})$, and $(4, \frac{1}{5})$. Find a quadratic function whose graph passes through these points.
9. Find a polynomial function p of degree 2 or less that passes through the points $(0, 1)$, $(2, 3)$, and $(4, 5)$. Then sketch the graph of $y = 1/p(x)$ and compare this graph with the graph of the polynomial function found in Exercise 8.
10. **Calculus** The graph of a parabola passes through the points $(0, 1)$ and $(\frac{1}{2}, \frac{1}{2})$ and has a horizontal tangent at $(\frac{1}{2}, \frac{1}{2})$. Find an equation for the parabola and sketch its graph.
11. **Calculus** The graph of a cubic polynomial function has horizontal tangents at $(1, -2)$ and $(-1, 2)$. Find an equation for the cubic and sketch its graph.
12. Find an equation of the circle passing through the points $(1, 3)$, $(-2, 6)$, and $(4, 2)$.

13. The U.S. census lists the population of the United States as 227 million in 1980, 249 million in 1990, and 281 million in 2000. Fit a second-degree polynomial passing through these three points and use it to predict the population in 2010 and in 2020. (Source: U.S. Census Bureau)



14. The U.S. population figures for the years 1920, 1930, 1940, and 1950 are shown in the table. (Source: U.S. Census Bureau)

<i>Year</i>	1920	1930	1940	1950
<i>Population (in millions)</i>	106	123	132	151

- (a) Find a cubic polynomial that fits these data and use it to estimate the population in 1960.
- (b) The actual population in 1960 was 179 million. How does your estimate compare?



15. The net profits (in millions of dollars) for Microsoft from 2000 to 2007 are shown in the table. (Source: Microsoft Corporation)

<i>Year</i>	2000	2001	2002	2003
<i>Net Profit</i>	9421	10,003	10,384	10,526

<i>Year</i>	2004	2005	2006	2007
<i>Net Profit</i>	11,330	12,715	12,599	14,410

- (a) Set up a system of equations to fit the data for the years 2001, 2003, 2005, and 2007 to a cubic model.
- (b) Solve the system. Does the solution produce a reasonable model for predicting future net profits? Explain.

-  16. The sales (in billions of dollars) for Wal-Mart stores from 2000 to 2007 are shown in the table. (Source: Wal-Mart)

Year	2000	2001	2002	2003
Sales	191.3	217.8	244.5	256.3

Year	2004	2005	2006	2007
Sales	285.2	312.4	346.5	377.0

- (a) Set up a system of equations to fit the data for the years 2001, 2003, 2005, and 2007 to a cubic model.
 (b) Solve the system. Does the solution produce a reasonable model for predicting future sales? Explain.
17. Use $\sin 0 = 0$, $\sin(\pi/2) = 1$, and $\sin \pi = 0$ to estimate $\sin(\pi/3)$.
 18. Use $\log_2 1 = 0$, $\log_2 2 = 1$, and $\log_2 4 = 2$ to estimate $\log_2 3$.
19. **Guided Proof** Prove that if a polynomial function $p(x) = a_0 + a_1x + a_2x^2$ is zero for $x = -1$, $x = 0$, and $x = 1$, then $a_0 = a_1 = a_2 = 0$.
 Getting Started: Write a system of linear equations and solve the system for a_0 , a_1 , and a_2 .
 (i) Substitute $x = -1, 0$, and 1 into $p(x)$.
 (ii) Set the result equal to 0.
 (iii) Solve the resulting system of linear equations in the variables a_0 , a_1 , and a_2 .
20. The statement in Exercise 19 can be generalized: If a polynomial function $p(x) = a_0 + a_1x + \cdots + a_{n-1}x^{n-1}$ is zero for more than $n - 1$ x -values, then $a_0 = a_1 = \cdots = a_{n-1} = 0$. Use this result to prove that there is at most one polynomial function of degree $n - 1$ (or less) whose graph passes through n points in the plane with distinct x -coordinates.

Network Analysis

-  21. Water is flowing through a network of pipes (in thousands of cubic meters per hour), as shown in Figure 1.15.
- (a) Solve this system for the water flow represented by x_i , $i = 1, 2, \dots, 7$.
 (b) Find the water flow when $x_6 = x_7 = 0$.
 (c) Find the water flow when $x_5 = 1000$ and $x_6 = 0$.

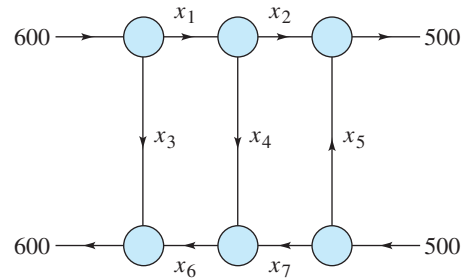


Figure 1.15

-  22. The flow of traffic (in vehicles per hour) through a network of streets is shown in Figure 1.16.

- (a) Solve this system for x_i , $i = 1, 2, \dots, 5$.
 (b) Find the traffic flow when $x_2 = 200$ and $x_3 = 50$.
 (c) Find the traffic flow when $x_2 = 150$ and $x_3 = 0$.

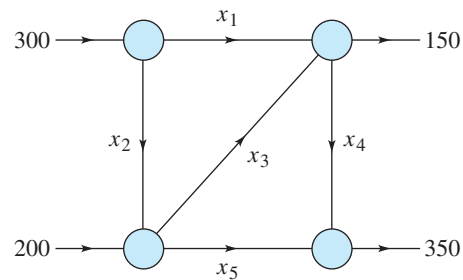


Figure 1.16

-  23. The flow of traffic (in vehicles per hour) through a network of streets is shown in Figure 1.17.

- (a) Solve this system for x_i , $i = 1, 2, 3, 4$.
 (b) Find the traffic flow when $x_4 = 0$.
 (c) Find the traffic flow when $x_4 = 100$.

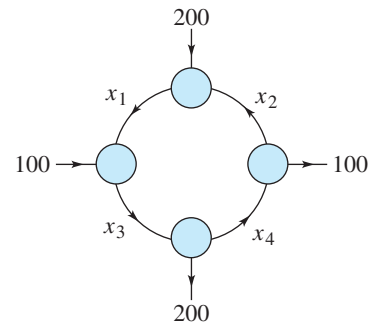


Figure 1.17

-  24. The flow of traffic (in vehicles per hour) through a network of streets is shown in Figure 1.18.
- Solve this system for x_i , $i = 1, 2, \dots, 5$.
 - Find the traffic flow when $x_3 = 0$ and $x_5 = 100$.
 - Find the traffic flow when $x_3 = x_5 = 100$.

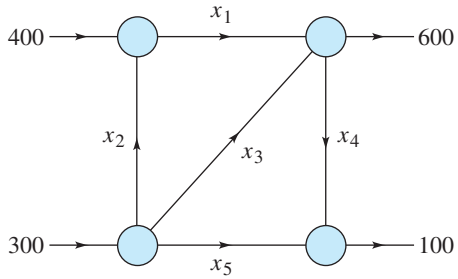


Figure 1.18

-  25. Determine the currents I_1 , I_2 , and I_3 for the electrical network shown in Figure 1.19.

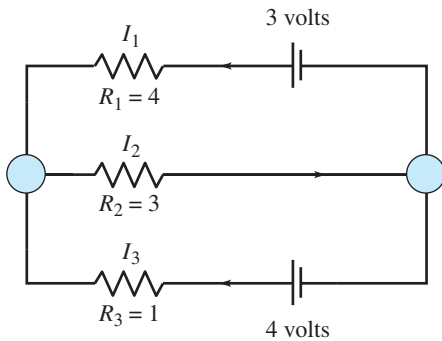


Figure 1.19

-  26. Determine the currents I_1 , I_2 , and I_3 for the electrical network shown in Figure 1.20.

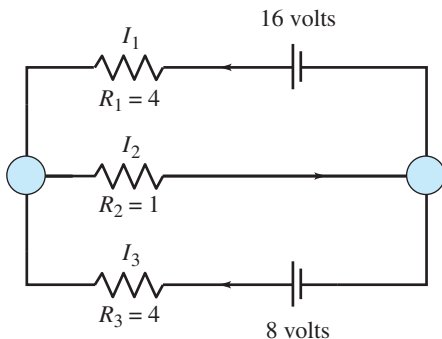


Figure 1.20

27. (a) Determine the currents I_1 , I_2 , and I_3 for the electrical network shown in Figure 1.21.
- (b) How is the result affected when A is changed to 2 volts and B is changed to 6 volts?

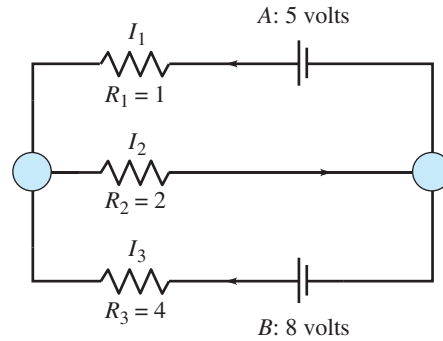


Figure 1.21

28. Determine the currents I_1 , I_2 , I_3 , I_4 , I_5 , and I_6 for the electrical network shown in Figure 1.22.

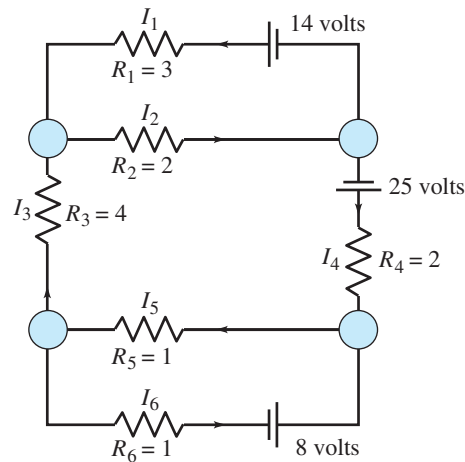


Figure 1.22

In Exercises 29–32, use a system of equations to write the partial fraction decomposition of the rational expression. Then solve the system using matrices.

- $\frac{4x^2}{(x+1)^2(x-1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$
- $\frac{8x^2}{(x-1)^2(x+1)} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$
- $\frac{20-x^2}{(x+2)(x-2)^2} = \frac{A}{x+2} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$

$$32. \frac{3x^2 - 7x - 12}{(x+4)(x-4)^2} = \frac{A}{x+4} + \frac{B}{x-4} + \frac{C}{(x-4)^2}$$

In Exercises 33 and 34, find the values of x , y , and λ that satisfy the system of equations. Such systems arise in certain problems of calculus, and λ is called the Lagrange multiplier.

$$\begin{array}{rcl} 33. & 2x & + \lambda & = 0 \\ & 2y + \lambda & = 0 \\ & x + y & - 4 & = 0 \end{array} \quad \begin{array}{rcl} 34. & 2y + 2\lambda & + 2 & = 0 \\ & 2x & + \lambda + 1 & = 0 \\ & 2x + y & - 100 & = 0 \end{array}$$

35. In Super Bowl XLI on February 4, 2007, the Indianapolis Colts beat the Chicago Bears by a score of 29 to 17. The total points scored came from 13 scoring plays, which were a combination of touchdowns, extra-point kicks, and field goals, worth 6, 1,

and 3 points, respectively. The numbers of field goals and extra-point kicks were equal. Write a system of equations to represent this event. Then determine the number of each type of scoring play. (Source: National Football League)

36. In the 2007 Fiesta Bowl Championship Series on January 8, 2007, the University of Florida Gators defeated the Ohio State University Buckeyes by a score of 41 to 14. The total points scored came from a combination of touchdowns, extra-point kicks, and field goals, worth 6, 1, and 3 points, respectively. The numbers of touchdowns and extra-point kicks were equal. The number of touchdowns was one more than three times the number of field goals. Write a system of equations to represent this event. Then determine the number of each type of scoring play. (Source: www.fiestabowl.org)

CHAPTER 1 Review Exercises

In Exercises 1–8, determine whether the equation is linear in the variables x and y .

1. $2x - y^2 = 4$
2. $2xy - 6y = 0$
3. $(\sin \pi)x + y = 2$
4. $e^{-2}x + 5y = 8$
5. $\frac{2}{x} + 4y = 3$
6. $\frac{4}{y} - x = 10$
7. $\frac{1}{2}x - \frac{1}{4}y = 0$
8. $\frac{3}{5}x + \frac{7}{10}y = 2$

In Exercises 9 and 10, find a parametric representation of the solution set of the linear equation.

9. $-4x + 2y - 6z = 1$
10. $3x_1 + 2x_2 - 4x_3 = 0$

In Exercises 11–22, solve the system of linear equations.

11. $x + y = 2$
 $3x - y = 0$
12. $x + y = -1$
 $3x + 2y = 0$
13. $3y = 2x$
 $y = x + 4$
14. $x = y + 3$
 $4x = y + 10$
15. $y + x = 0$
 $2x + y = 0$
16. $y = -4x$
 $y = x$
17. $x - y = 9$
 $-x + y = 1$
18. $40x_1 + 30x_2 = 24$
 $20x_1 + 15x_2 = -14$
19. $0.2x_1 + 0.3x_2 = 0.14$
 $0.4x_1 + 0.5x_2 = 0.20$
20. $0.2x - 0.1y = 0.07$
 $0.4x - 0.5y = -0.01$
21. $\frac{1}{2}x - \frac{1}{3}y = 0$
 $3x + 2(y + 5) = 10$
22. $\frac{1}{3}x + \frac{4}{7}y = 3$
 $2x + 3y = 15$

In Exercises 23 and 24, determine the size of the matrix.

$$23. \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 1 \end{bmatrix} \quad 24. \begin{bmatrix} 2 & 1 \\ -4 & -1 \\ 0 & 5 \end{bmatrix}$$

In Exercises 25–28, determine whether the matrix is in row-echelon form. If it is, determine whether it is also in reduced row-echelon form.

$$\begin{array}{ll} 25. \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} & 26. \begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ 27. \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & 28. \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

In Exercises 29 and 30, find the solution set of the system of linear equations represented by the augmented matrix.

$$29. \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad 30. \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

In Exercises 31–40, solve the system using either Gaussian elimination with back-substitution or Gauss-Jordan elimination.

31. $-x + y + 2z = 1$
 $2x + 3y + z = -2$
 $5x + 4y + 2z = 4$
32. $2x + 3y + z = 10$
 $2x - 3y - 3z = 22$
 $4x - 2y + 3z = -2$
33. $2x + 3y + 3z = 3$
 $6x + 6y + 12z = 13$
 $12x + 9y - z = 2$
34. $2x + 6z = -9$
 $3x - 2y + 11z = -16$
 $3x - y + 7z = -11$

$$\begin{array}{ll}
 35. \quad x - 2y + z = -6 & 36. \quad x + 2y + 6z = 1 \\
 \quad 2x - 3y = -7 & \quad 2x + 5y + 15z = 4 \\
 \quad -x + 3y - 3z = 11 & \quad 3x + y + 3z = -6 \\
 37. \quad 2x + y + 2z = 4 & 38. \quad 2x_1 + 5x_2 - 19x_3 = 34 \\
 \quad 2x + 2y = 5 & \quad 3x_1 + 8x_2 - 31x_3 = 54 \\
 \quad 2x - y + 6z = 2 &
 \end{array}$$

  39. $2x_1 + x_2 + x_3 + 2x_4 = -1$
 $5x_1 - 2x_2 + x_3 - 3x_4 = 0$
 $-x_1 + 3x_2 + 2x_3 + 2x_4 = 1$
 $3x_1 + 2x_2 + 3x_3 - 5x_4 = 12$

  40. $x_1 + 5x_2 + 3x_3 = 14$
 $4x_2 + 2x_3 + 5x_4 = 3$
 $3x_3 + 8x_4 + 6x_5 = 16$
 $2x_1 + 4x_2 - 2x_5 = 0$
 $2x_1 - x_3 = 0$

 In Exercises 41–46, use the matrix capabilities of a graphing utility to reduce the augmented matrix corresponding to the system of equations to solve the system.

$$\begin{array}{ll}
 41. \quad 3x + 3y + 12z = 6 & 42. \quad 2x + 10y + 2z = 6 \\
 \quad x + y + 4z = 2 & \quad x + 5y + 2z = 6 \\
 \quad 2x + 5y + 20z = 10 & \quad x + 5y + z = 3 \\
 \quad -x + 2y + 8z = 4 & \quad -3x - 15y + 3z = -9
 \end{array}$$

$$\begin{array}{l}
 43. \quad 2x + y - z + 2w = -6 \\
 \quad 3x + 4y + w = 1 \\
 \quad x + 5y + 2z + 6w = -3 \\
 \quad 5x + 2y - z - w = 3
 \end{array}$$

$$\begin{array}{l}
 44. \quad x + 2y + z + 4w = 11 \\
 \quad 3x + 6y + 5z + 12w = 30 \\
 \quad x + 3y - 3z + 2w = -5 \\
 \quad 6x - y - z + w = -9
 \end{array}$$

$$\begin{array}{ll}
 45. \quad x + y + z + w = 0 & 46. \quad x + 2y + z + 3w = 0 \\
 \quad 2x + 3y + z - 2w = 0 & \quad x - y + w = 0 \\
 \quad 3x + 5y + z = 0 & \quad 5y - z + 2w = 0
 \end{array}$$

In Exercises 47–50, solve the homogeneous system of linear equations.

$$\begin{array}{ll}
 47. \quad x_1 - 2x_2 - 8x_3 = 0 & 48. \quad 2x_1 + 4x_2 - 7x_3 = 0 \\
 \quad 3x_1 + 2x_2 = 0 & \quad x_1 - 3x_2 + 9x_3 = 0 \\
 \quad -x_1 + x_2 + 7x_3 = 0 & \quad 6x_1 + 9x_3 = 0
 \end{array}$$

$$\begin{array}{ll}
 49. \quad 2x_1 - 8x_2 + 4x_3 = 0 & 50. \quad x_1 + 3x_2 + 5x_3 = 0 \\
 \quad 3x_1 - 10x_2 + 7x_3 = 0 & \quad x_1 + 4x_2 + \frac{1}{2}x_3 = 0 \\
 \quad 10x_2 + 5x_3 = 0 &
 \end{array}$$

51. Determine the value of k such that the system of linear equations is inconsistent.

$$\begin{array}{l}
 kx + y = 0 \\
 x + ky = 1
 \end{array}$$

52. Determine the value of k such that the system of linear equations has exactly one solution.

$$\begin{array}{l}
 x - y + 2z = 0 \\
 -x + y - z = 0 \\
 x + ky + z = 0
 \end{array}$$

53. Find conditions on a and b such that the system of linear equations has (a) no solution, (b) exactly one solution, and (c) an infinite number of solutions.

$$\begin{array}{l}
 x + 2y = 3 \\
 ax + by = -9
 \end{array}$$

54. Find (if possible) conditions on a , b , and c such that the system of linear equations has (a) no solution, (b) exactly one solution, and (c) an infinite number of solutions.

$$\begin{array}{l}
 2x - y + z = a \\
 x + y + 2z = b \\
 3y + 3z = c
 \end{array}$$

55. **Writing** Describe a method for showing that two matrices are row-equivalent. Are the two matrices below row-equivalent?

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & 2 \\ 3 & 1 & 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 6 \\ 5 & 5 & 10 \end{bmatrix}$$

56. **Writing** Describe all possible 2×3 reduced row-echelon matrices. Support your answer with examples.

57. Let $n \geq 3$. Find the reduced row-echelon form of the $n \times n$ matrix.

$$\begin{bmatrix} 1 & 2 & 3 & \cdots & n \\ n+1 & n+2 & n+3 & \cdots & 2n \\ 2n+1 & 2n+2 & 2n+3 & \cdots & 3n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n^2 - n + 1 & n^2 - n + 2 & n^2 - n + 3 & \cdots & n^2 \end{bmatrix}$$

58. Find all values of λ for which the homogeneous system of linear equations has nontrivial solutions.

$$\begin{array}{l}
 (\lambda + 2)x_1 - 2x_2 + 3x_3 = 0 \\
 -2x_1 + (\lambda - 1)x_2 + 6x_3 = 0 \\
 x_1 + 2x_2 + \lambda x_3 = 0
 \end{array}$$

True or False? In Exercises 59 and 60, determine whether each statement is true or false. If a statement is true, give a reason or cite an appropriate statement from the text. If a statement is false, provide an example that shows the statement is not true in all cases or cite an appropriate statement from the text.

59. (a) The solution set of a linear equation can be parametrically represented in only one way.
 (b) A consistent system of linear equations can have an infinite number of solutions.
60. (a) A homogeneous system of linear equations must have at least one solution.
 (b) A system of linear equations with fewer equations than variables always has at least one solution.
61. The University of Tennessee Lady Volunteers defeated the Rutgers University Scarlet Knights 59 to 46. The Lady Volunteers' scoring resulted from a combination of three-point baskets, two-point baskets, and one-point free throws. There were three times as many two-point baskets as three-point baskets. The number of free throws was one less than the number of two-point baskets. (Source: National Collegiate Athletic Association)
 (a) Set up a system of linear equations to find the numbers of three-point baskets, two-point baskets, and one-point free throws scored by the Lady Volunteers.
 (b) Solve your system.
62. In Super Bowl I, on January 15, 1967, the Green Bay Packers defeated the Kansas City Chiefs by a score of 35 to 10. The total points scored came from a combination of touchdowns, extra-point kicks, and field goals, worth 6, 1, and 3 points, respectively. The numbers of touchdowns and extra-point kicks were equal. There were six times as many touchdowns as field goals. (Source: National Football League)
 (a) Set up a system of linear equations to find the numbers of touchdowns, extra-point kicks, and field goals that were scored.
 (b) Solve your system.

In Exercises 63 and 64, use a system of equations to write the partial fraction decomposition of the rational expression. Then solve the system using matrices.

63. $\frac{3x^2 - 3x - 2}{(x + 2)(x - 2)^2} = \frac{A}{x + 2} + \frac{B}{x - 2} + \frac{C}{(x - 2)^2}$
64. $\frac{3x^2 + 3x - 2}{(x + 1)^2(x - 1)} = \frac{A}{x + 1} + \frac{B}{x - 1} + \frac{C}{(x + 1)^2}$

Polynomial Curve Fitting

 In Exercises 65 and 66, (a) determine the polynomial whose graph passes through the given points, and (b) sketch the graph of the polynomial, showing the given points.

65. (2, 5), (3, 0), (4, 20)

66. (-1, -1), (0, 0), (1, 1), (2, 4)

 67. A company has sales (measured in millions) of \$50, \$60, and \$75 during three consecutive years. Find a quadratic function that fits these data, and use it to predict the sales during the fourth year.

68. The polynomial function $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ is zero when $x = 1, 2, 3$, and 4. What are the values of a_0, a_1, a_2 , and a_3 ?

 69. A wildlife management team studied the population of deer in one small tract of a wildlife preserve. The population and the number of years since the study began are shown in the table.

<i>Year</i>	0	4	80
<i>Population</i>	80	68	30

- (a) Set up a system of equations to fit the data to a quadratic polynomial function.
 (b) Solve your system.
 (c) Use a graphing utility to fit a quadratic model to the data.
 (d) Compare the quadratic polynomial function in part (b) with the model in part (c).
 (e) Cite the statement from the text that verifies your results.

 70. A research team studied the average monthly temperatures of a small lake over a period of about one year. The temperatures and the numbers of months since the study began are shown in the table.

<i>Month</i>	0	6	12
<i>Temperature</i>	40	73	52

- (a) Set up a system of equations to fit the data to a quadratic polynomial function.
 (b) Solve your system.
 (c) Use a graphing utility to fit a quadratic model to the data.
 (d) Compare the quadratic polynomial function in part (b) with the model in part (c).
 (e) Cite the statement from the text that verifies your results.

Network Analysis

71. Determine the currents I_1 , I_2 , and I_3 for the electrical network shown in Figure 1.23.

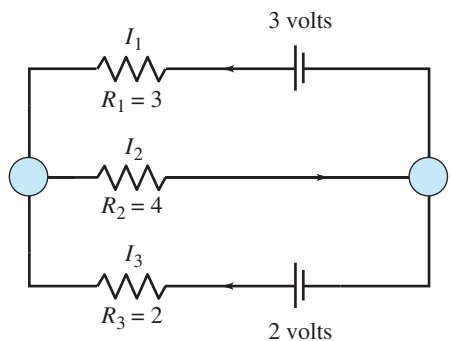


Figure 1.23

72. The flow through a network is shown in Figure 1.24.

- (a) Solve the system for x_i , $i = 1, 2, \dots, 6$.
 (b) Find the flow when $x_3 = 100$, $x_5 = 50$, and $x_6 = 50$.

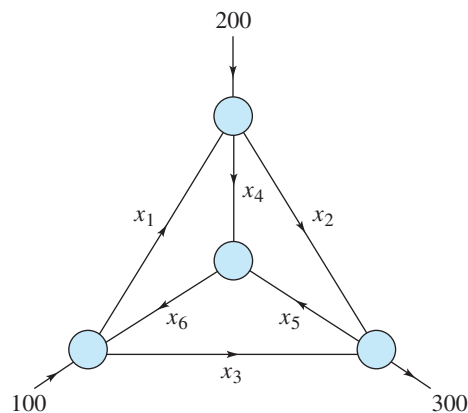


Figure 1.24

CHAPTER 1 Projects

1 Graphing Linear Equations

You saw in Section 1.1 that a system of two linear equations in two variables x and y can be represented geometrically as two lines in the plane. These lines can intersect at a point, coincide, or be parallel, as indicated in Figure 1.25.

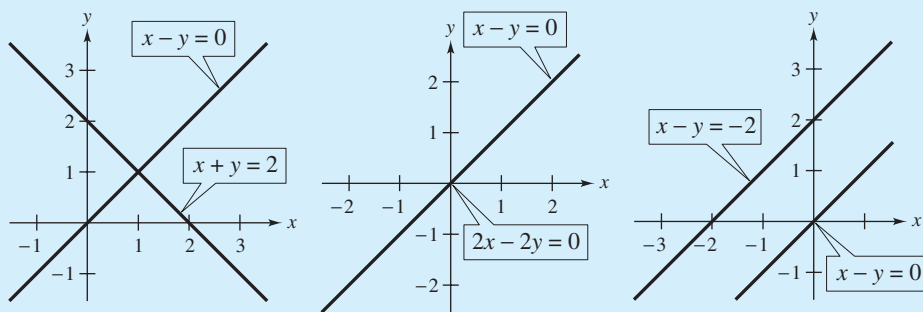


Figure 1.25

1. Consider the system below, where a and b are constants. Answer the questions that follow. For Questions (a)–(c), if an answer is yes, give an example. Otherwise, explain why the answer is no.

$$2x - y = 3$$

$$ax + by = 6$$

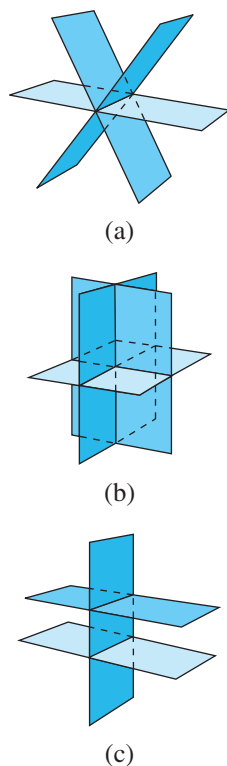


Figure 1.26

- Can you find values of a and b for which the resulting system has a unique solution?
 - Can you find values of a and b for which the resulting system has an infinite number of solutions?
 - Can you find values of a and b for which the resulting system has no solution?
 - Graph the resulting lines for each of the systems in parts (a), (b), and (c).
- Now consider a system of three linear equations in x , y , and z . Each equation represents a plane in the three-dimensional coordinate system.
 - Find an example of a system represented by three planes intersecting in a line, as shown in Figure 1.26(a).
 - Find an example of a system represented by three planes intersecting at a point, as shown in Figure 1.26(b).
 - Find an example of a system represented by three planes with no common intersection, as shown in Figure 1.26(c).
 - Are there other configurations of three planes not covered by the three examples in parts (a), (b), and (c)? Explain.

2 Underdetermined and Overdetermined Systems of Equations

The next system of linear equations is said to be **underdetermined** because there are more variables than equations.

$$\begin{aligned}x_1 + 2x_2 - 3x_3 &= 4 \\2x_1 - x_2 + 4x_3 &= -3\end{aligned}$$

Similarly, the following system is **overdetermined** because there are more equations than variables.

$$\begin{aligned}x_1 + 3x_2 &= 5 \\2x_1 - 2x_2 &= -3 \\-x_1 + 7x_2 &= 0\end{aligned}$$

You can explore whether the number of variables and the number of equations have any bearing on the consistency of a system of linear equations. For Exercises 1–4, if an answer is yes, give an example. Otherwise, explain why the answer is no.

- Can you find a consistent underdetermined linear system?
- Can you find a consistent overdetermined linear system?
- Can you find an inconsistent underdetermined linear system?
- Can you find an inconsistent overdetermined linear system?
- Explain why you would expect an overdetermined linear system to be inconsistent. Must this always be the case?
- Explain why you would expect an underdetermined linear system to have an infinite number of solutions. Must this always be the case?

2

Matrices

- 2.1 Operations with Matrices
- 2.2 Properties of Matrix Operations
- 2.3 The Inverse of a Matrix
- 2.4 Elementary Matrices
- 2.5 Applications of Matrix Operations

CHAPTER OBJECTIVES

- Write a system of linear equations represented by a matrix, as well as write the matrix form of a system of linear equations.
- Write and solve a system of linear equations in the form $Ax = b$.
- Use properties of matrix operations to solve matrix equations.
- Find the transpose of a matrix, the inverse of a matrix, and the inverse of a matrix product (if they exist).
- Factor a matrix into a product of elementary matrices, and determine when they are invertible.
- Find and use the LU -factorization of a matrix to solve a system of linear equations.
- Use a stochastic matrix to measure consumer preference.
- Use matrix multiplication to encode and decode messages.
- Use matrix algebra to analyze economic systems (Leontief input-output models).
- Use the method of least squares to find the least squares regression line for a set of data.

2.1 Operations with Matrices

In Section 1.2 you used matrices to solve systems of linear equations. Matrices, however, can be used to do much more than that. There is a rich mathematical theory of matrices, and its applications are numerous. This section and the next introduce some fundamentals of matrix theory.

It is standard mathematical convention to represent matrices in any one of the following three ways.

1. A matrix can be denoted by an uppercase letter such as

$$A, B, C, \dots$$

2. A matrix can be denoted by a representative element enclosed in brackets, such as

$$[a_{ij}], [b_{ij}], [c_{ij}], \dots$$

3. A matrix can be denoted by a rectangular array of numbers

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}.$$

As mentioned in Chapter 1, the matrices in this text are primarily *real matrices*. That is, their entries contain real numbers.

Two matrices are said to be **equal** if their corresponding entries are equal.

Definition of Equality of Matrices

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are **equal** if they have the same size ($m \times n$) and

$$a_{ij} = b_{ij}$$

for $1 \leq i \leq m$ and $1 \leq j \leq n$.

EXAMPLE 1

Equality of Matrices

Consider the four matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 3 \end{bmatrix},$$

$$C = [1 \quad 3], \quad \text{and} \quad D = \begin{bmatrix} 1 & 2 \\ x & 4 \end{bmatrix}.$$

Matrices A and B are **not** equal because they are of different sizes. Similarly, B and C are not equal. Matrices A and D are equal if and only if $x = 3$.

REMARK: The phrase “if and only if” means the statement is true in both directions. For example, “ p if and only if q ” means that p implies q and q implies p .

A matrix that has only one column, such as matrix B in Example 1, is called a **column matrix** or **column vector**. Similarly, a matrix that has only one row, such as matrix C in Example 1, is called a **row matrix** or **row vector**. Boldface lowercase letters are often used to designate column matrices and row matrices. For instance, matrix A in Example 1 can be partitioned into the two column matrices $\mathbf{a}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$, as follows.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & \vdots & 2 \\ 3 & \vdots & 4 \end{bmatrix} = [\mathbf{a}_1 \quad \mathbf{a}_2]$$

Matrix Addition

You can **add** two matrices (of the same size) by adding their corresponding entries.

Definition of Matrix Addition

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are matrices of size $m \times n$, then their **sum** is the $m \times n$ matrix given by

$$A + B = [a_{ij} + b_{ij}].$$

The sum of two matrices of different sizes is undefined.

EXAMPLE 2

Addition of Matrices

$$(a) \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -1+1 & 2+3 \\ 0-1 & 1+2 \end{bmatrix} = \begin{bmatrix} 0 & 5 \\ -1 & 3 \end{bmatrix}$$

$$(b) \begin{bmatrix} 0 & 1 & -2 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 2 & 3 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(d) The sum of

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 0 & -1 \\ 3 & -2 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 \\ -1 & 3 \\ 2 & 4 \end{bmatrix}$$

is undefined.

Scalar Multiplication

When working with matrices, real numbers are referred to as **scalars**. You can multiply a matrix A by a scalar c by multiplying each entry in A by c .

Definition of Scalar Multiplication

If $A = [a_{ij}]$ is an $m \times n$ matrix and c is a scalar, then the **scalar multiple** of A by c is the $m \times n$ matrix given by

$$cA = [ca_{ij}].$$

You can use $-A$ to represent the scalar product $(-1)A$. If A and B are of the same size, $A - B$ represents the sum of A and $(-1)B$. That is,

$$A - B = A + (-1)B. \quad \text{Subtraction of matrices}$$

EXAMPLE 3 Scalar Multiplication and Matrix Subtraction

For the matrices

$$A = \begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix}$$

find (a) $3A$, (b) $-B$, and (c) $3A - B$.

SOLUTION (a) $3A = 3 \begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 3(1) & 3(2) & 3(4) \\ 3(-3) & 3(0) & 3(-1) \\ 3(2) & 3(1) & 3(2) \end{bmatrix} = \begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix}$

(b) $-B = (-1) \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ -1 & 4 & -3 \\ 1 & -3 & -2 \end{bmatrix}$

(c) $3A - B = \begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 6 & 12 \\ -10 & 4 & -6 \\ 7 & 0 & 4 \end{bmatrix}$

Matrix Multiplication

REMARK: It is often convenient to rewrite a matrix B as cA by factoring c out of every entry in matrix B . For instance, the scalar $\frac{1}{2}$ has been factored out of the matrix below.

$$\begin{bmatrix} \frac{1}{2} & -\frac{3}{2} \\ \frac{5}{2} & \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -3 \\ 5 & 1 \end{bmatrix}$$

$B = cA$

The third basic matrix operation is **matrix multiplication**. To see the usefulness of this operation, consider the following application in which matrices are helpful for organizing information.

A football stadium has three concession areas, located in the south, north, and west stands. The top-selling items are peanuts, hot dogs, and soda. Sales for a certain day are recorded in the first matrix below, and the prices (in dollars) of the three items are given in the second matrix.

Number of Items Sold			
	Peanuts	Hot Dogs	Soda
Selling Price			
South stand	120	250	305
North stand	207	140	419
West stand	29	120	190

$\begin{bmatrix} 2.00 \\ 3.00 \\ 2.75 \end{bmatrix}$ Peanuts
Hot Dogs
Soda

To calculate the total sales of the three top-selling items at the south stand, you can multiply each entry in the first row of the matrix on the left by the corresponding entry in the price column matrix on the right and add the results. The south stand sales are

$$(120)(2.00) + (250)(3.00) + (305)(2.75) = \$1828.75. \quad \text{South stand sales}$$

Similarly, you can calculate the sales for the other two stands as follows.

$$(207)(2.00) + (140)(3.00) + (419)(2.75) = \$1986.25 \quad \text{North stand sales}$$

$$(29)(2.00) + (120)(3.00) + (190)(2.75) = \$940.50 \quad \text{West stand sales}$$

The preceding computations are examples of matrix multiplication. You can write the product of the 3×3 matrix indicating the number of items sold and the 3×1 matrix indicating the selling prices as follows.

$$\begin{bmatrix} 120 & 250 & 305 \\ 207 & 140 & 419 \\ 29 & 120 & 190 \end{bmatrix} \begin{bmatrix} 2.00 \\ 3.00 \\ 2.75 \end{bmatrix} = \begin{bmatrix} 1828.75 \\ 1986.25 \\ 940.50 \end{bmatrix}$$

The product of these matrices is the 3×1 matrix giving the total sales for each of the three stands.

The general definition of the product of two matrices shown below is based on the ideas just developed. Although at first glance this definition may seem unusual, you will see that it has many practical applications.

Definition of Matrix Multiplication

If $A = [a_{ij}]$ is an $m \times n$ matrix and $B = [b_{ij}]$ is an $n \times p$ matrix, then the **product** AB is an $m \times p$ matrix

$$AB = [c_{ij}]$$

where

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj}.$$

This definition means that the entry in the i th row and the j th column of the product AB is obtained by multiplying the entries in the i th row of A by the corresponding entries in the j th column of B and then adding the results. The next example illustrates this process.

EXAMPLE 4 Finding the Product of Two Matrices

Find the product AB , where

$$A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}.$$

SOLUTION First note that the product AB is defined because A has size 3×2 and B has size 2×2 . Moreover, the product AB has size 3×2 and will take the form

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}.$$

To find c_{11} (the entry in the first row and first column of the product), multiply corresponding entries in the first row of A and the first column of B . That is,

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}.$$

$c_{11} = (-1)(-3) + (3)(-4) = -9$

Similarly, to find c_{12} , multiply corresponding entries in the first row of A and the second column of B to obtain

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}.$$

$c_{12} = (-1)(2) + (3)(1) = 1$

Continuing this pattern produces the results shown below.

$$\begin{aligned} c_{21} &= (4)(-3) + (-2)(-4) = -4 \\ c_{22} &= (4)(2) + (-2)(1) = 6 \\ c_{31} &= (5)(-3) + (0)(-4) = -15 \\ c_{32} &= (5)(2) + (0)(1) = 10 \end{aligned}$$

The product is

$$AB = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{bmatrix}.$$

Be sure you understand that for the product of two matrices to be defined, the number of columns of the first matrix must equal the number of rows of the second matrix. That is,

$$\begin{array}{ccc} A & B & = AB \\ m \times n & n \times p & m \times p \\ \uparrow \quad \uparrow \quad \uparrow & & \\ \text{equal} & & \\ \text{size of } AB & & \end{array}$$

So, the product BA is not defined for matrices such as A and B in Example 4.

HISTORICAL NOTE

Arthur Cayley
(1821–1895)

showed signs of mathematical genius at an early age, but ironically wasn't able to find a position as a mathematician upon graduating from college. Ultimately, however, Cayley made major contributions to linear algebra. To read about his work, visit college.hmco.com/pic/larsonELA6e.

The general pattern for matrix multiplication is as follows. To obtain the element in the i th row and the j th column of the product AB , use the i th row of A and the j th column of B .

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & a_{i3} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2p} \\ b_{31} & b_{32} & \cdots & b_{3j} & \cdots & b_{3p} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nj} & \cdots & b_{np} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1j} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2j} & \cdots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ c_{i1} & c_{i2} & \cdots & c_{ij} & \cdots & c_{ip} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mj} & \cdots & c_{mp} \end{bmatrix}$$

$a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj} = c_{ij}$

Discovery

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}.$$

Calculate $A + B$ and $B + A$.

In general, is the operation of matrix addition commutative? Now calculate AB and BA . Is matrix multiplication commutative?

EXAMPLE 5 Matrix Multiplication

$$\text{(a)} \quad \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -2 \end{bmatrix} \begin{bmatrix} -2 & 4 & 2 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} -5 & 7 & -1 \\ -3 & 6 & 6 \end{bmatrix}$$

$2 \times 3 \qquad 3 \times 3 \qquad 2 \times 3$

$$\text{(b)} \quad \begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix}$$

$2 \times 2 \qquad 2 \times 2 \qquad 2 \times 2$

$$\text{(c)} \quad \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$2 \times 2 \qquad 2 \times 2 \qquad 2 \times 2$

$$\text{(d)} \quad [1 \quad -2 \quad -3] \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = [1]$$

$1 \times 3 \qquad 3 \times 1 \qquad 1 \times 1$

$$(e) \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 2 & -4 & -6 \\ -1 & 2 & 3 \\ 1 & -2 & -3 \end{bmatrix}$$

3×1 1×3 3×3

REMARK: Note the difference between the two products in parts (d) and (e) of Example 5. In general, matrix multiplication is not commutative. It is usually not true that the product AB is equal to the product BA . (See Section 2.2 for further discussion of the noncommutativity of matrix multiplication.)

Technology Note

Most graphing utilities and computer software programs can perform matrix addition, scalar multiplication, and matrix multiplication. If you are using a graphing utility, your screens for Example 5(c) may look like:

A B $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ $\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$	A*B $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
---	---

Keystrokes and programming syntax for these utilities/programs applicable to Example 5(c) are provided in the **Online Technology Guide**, available at college.hmco.com/pic/larsonELA6e.

Systems of Linear Equations

One practical application of matrix multiplication is representing a system of linear equations. Note how the system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

can be written as the matrix equation $A\mathbf{x} = \mathbf{b}$, where A is the coefficient matrix of the system, and $\mathbf{x}$ and $\mathbf{b}$ are column matrices. You can write the system as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

A $\mathbf{x}$ $=$ $\mathbf{b}$

EXAMPLE 6 Solving a System of Linear Equations

Solve the matrix equation $A\mathbf{x} = \mathbf{0}$, where

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 3 & -2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \text{and} \quad \mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

SOLUTION As a system of linear equations, $A\mathbf{x} = \mathbf{0}$ looks like

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_1 + 3x_2 - 2x_3 &= 0. \end{aligned}$$

Using Gauss-Jordan elimination on the augmented matrix of this system, you obtain

$$\begin{bmatrix} 1 & 0 & -\frac{1}{7} & 0 \\ 0 & 1 & -\frac{4}{7} & 0 \end{bmatrix}.$$

So, the system has an infinite number of solutions. Here a convenient choice of a parameter is $x_3 = 7t$, and you can write the solution set as

$$x_1 = t, \quad x_2 = 4t, \quad x_3 = 7t, \quad t \text{ is any real number.}$$

In matrix terminology, you have found that the matrix equation

$$\begin{bmatrix} 1 & -2 & 1 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

has an infinite number of solutions represented by

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ 4t \\ 7t \end{bmatrix} = t \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}, \quad t \text{ is any scalar.}$$

That is, any scalar multiple of the column matrix on the right is a solution.

Partitioned Matrices

The system $A\mathbf{x} = \mathbf{b}$ can be represented in a more convenient way by partitioning the matrices A and $\mathbf{x}$ in the following manner. If

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

are the coefficient matrix, the column matrix of unknowns, and the right-hand side, respectively, of the $m \times n$ linear system $A\mathbf{x} = \mathbf{b}$, then you can write

$$\begin{aligned}
 A\mathbf{x} &= \mathbf{b} \\
 \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} &= \mathbf{b} \\
 \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix} &= \mathbf{b} \\
 x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} &= \mathbf{b}.
 \end{aligned}$$

In other words,

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b},$$

where $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are the columns of the matrix A . The expression

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

is called a **linear combination** of the column matrices $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ with **coefficients** $x_1, x_2, \dots, x_n$.

In general, the matrix product $A\mathbf{x}$ is a linear combination of the column vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ that form the coefficient matrix A . Furthermore, the system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ can be expressed as such a linear combination, where the coefficients of the linear combination are a solution of the system.

EXAMPLE 7

Solving a System of Linear Equations

The linear system

$$\begin{aligned}
 x_1 + 2x_2 + 3x_3 &= 0 \\
 4x_1 + 5x_2 + 6x_3 &= 3 \\
 7x_1 + 8x_2 + 9x_3 &= 6
 \end{aligned}$$

Technology Note

Many real-life applications of linear systems involve enormous numbers of equations and variables. For example, a flight crew scheduling problem for American Airlines required the manipulation of matrices with 837 rows and more than 12,750,000 columns. This application of *linear programming* required that the problem be partitioned into smaller pieces and then solved on a CRAY Y-MP supercomputer.

(Source: Very-Large Scale Linear Programming, A Case Study in Combining Interior Point and Simplex Methods, Bixby, Robert E., et al., *Operations Research*, 40, no. 5, 1992.)

can be rewritten as a matrix equation $A\mathbf{x} = \mathbf{b}$, as follows.

$$x_1 \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix}$$

Using Gaussian elimination, you can show that this system has an infinite number of solutions, one of which is $x_1 = 1$, $x_2 = 1$, $x_3 = -1$.

$$1 \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} + (-1) \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix}$$

That is, $\mathbf{b}$ can be expressed as a linear combination of the columns of A . This representation of one column vector in terms of others is a fundamental theme of linear algebra.

Just as you partitioned A into columns and $\mathbf{x}$ into rows, it is often useful to consider an $m \times n$ matrix partitioned into smaller matrices. For example, the matrix on the left below can be partitioned as shown below at the right.

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ -1 & -2 & 2 & 1 \end{bmatrix} \quad \left[\begin{array}{cc|cc} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ -1 & -2 & 2 & 1 \end{array} \right]$$

The matrix could also be partitioned into column matrices

$$\left[\begin{array}{c|c|c|c} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ -1 & -2 & 2 & 1 \end{array} \right] = [\mathbf{c}_1 \quad \mathbf{c}_2 \quad \mathbf{c}_3 \quad \mathbf{c}_4]$$

or row matrices

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ -1 & -2 & 2 & 1 \end{array} \right] = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \mathbf{r}_3 \end{bmatrix}.$$

SECTION 2.1 Exercises

In Exercises 1–6, find (a) $A + B$, (b) $A - B$, (c) $2A$, (d) $2A - B$, and (e) $B + \frac{1}{2}A$.

1. $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -1 \\ -1 & 8 \end{bmatrix}$

2. $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -3 & -2 \\ 4 & 2 \end{bmatrix}$

3. $A = \begin{bmatrix} 6 & -1 \\ 2 & 4 \\ -3 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ -1 & 5 \\ 1 & 10 \end{bmatrix}$

4. $A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & -1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -3 & 4 \\ -3 & 1 & -2 \end{bmatrix}$

5. $A = \begin{bmatrix} 3 & 2 & -1 \\ 2 & 4 & 5 \\ 0 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 2 & 1 \\ 5 & 4 & 2 \\ 2 & 1 & 0 \end{bmatrix}$

6. $A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & -1 \\ 2 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 6 & 2 \\ 4 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix}$

7. Find (a) c_{21} and (b) c_{13} , where $C = 2A - 3B$,

$$A = \begin{bmatrix} 5 & 4 & 4 \\ -3 & 1 & 2 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 1 & 2 & -7 \\ 0 & -5 & 1 \end{bmatrix}.$$

8. Find (a) c_{23} and (b) c_{32} , where $C = 5A + 2B$,

$$A = \begin{bmatrix} 4 & 11 & -9 \\ 0 & 3 & 2 \\ -3 & 1 & 1 \end{bmatrix}, \text{ and } B = \begin{bmatrix} 1 & 0 & 5 \\ -4 & 6 & 11 \\ -6 & 4 & 9 \end{bmatrix}.$$

9. Solve for x , y , and z in the matrix equation

$$4 \begin{bmatrix} x & y \\ z & -1 \end{bmatrix} = 2 \begin{bmatrix} y & z \\ -x & 1 \end{bmatrix} + 2 \begin{bmatrix} 4 & x \\ 5 & -x \end{bmatrix}.$$

10. Solve for x , y , z , and w in the matrix equation

$$\begin{bmatrix} w & x \\ y & x \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 2 & -1 \end{bmatrix} + 2 \begin{bmatrix} y & w \\ z & x \end{bmatrix}.$$

In Exercises 11–18, find (a) AB and (b) BA (if they are defined).

11. $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -1 \\ -1 & 8 \end{bmatrix}$

12. $A = \begin{bmatrix} 1 & -1 & 7 \\ 2 & -1 & 8 \\ 3 & 1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 1 & -3 & 2 \end{bmatrix}$

13. $A = \begin{bmatrix} 2 & 1 \\ -3 & 4 \\ 1 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 & 0 \\ 4 & 0 & 2 \\ 8 & -1 & 7 \end{bmatrix}$

14. $A = \begin{bmatrix} 3 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$

15. $A = \begin{bmatrix} -1 & 3 \\ 4 & -5 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 0 & 7 \end{bmatrix}$

16. $A = \begin{bmatrix} 0 & -1 & 0 \\ 4 & 0 & 2 \\ 8 & -1 & 7 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$

17. $A = \begin{bmatrix} 6 \\ -2 \\ 1 \\ 6 \end{bmatrix}$, $B = \begin{bmatrix} 10 & 12 \end{bmatrix}$

18. $A = \begin{bmatrix} 1 & 0 & 3 & -2 & 4 \\ 6 & 13 & 8 & -17 & 20 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 6 \\ 4 & 2 \end{bmatrix}$

In Exercises 19 and 20, find (a) $2A + B$, (b) $3B - A$, (c) AB , and (d) BA (if they are defined).

19. $A = \begin{bmatrix} 2 & -2 & 4 & 1 & 0 & 3 \\ -1 & 4 & 2 & -2 & -1 & 3 \\ 3 & -3 & 1 & 2 & 3 & -4 \\ 2 & -1 & 3 & 0 & 1 & 2 \\ 5 & 1 & -2 & -4 & 1 & 3 \\ 2 & 2 & 3 & -4 & -1 & -2 \end{bmatrix}$

$$B = \begin{bmatrix} 1 & 2 & -3 & 4 & 1 & 2 \\ 2 & -3 & 1 & 3 & -1 & 2 \\ 0 & -2 & -3 & 0 & 1 & -1 \\ 1 & 2 & 3 & 2 & 1 & -1 \\ 2 & -1 & -3 & 0 & 4 & 2 \\ 1 & -2 & 4 & -2 & -4 & -1 \end{bmatrix}$$

20. $A = \begin{bmatrix} 2 & 1 & 3 & 2 & -1 \\ 3 & -1 & 0 & 1 & 2 \\ 2 & 1 & -3 & 3 & -2 \\ -4 & 0 & 2 & -3 & 1 \\ 1 & 0 & -1 & 2 & 4 \\ 2 & -3 & 2 & 1 & -4 \end{bmatrix}$

$$B = \begin{bmatrix} 5 & 2 & 1 & 3 & 2 & 1 \\ -4 & -2 & 2 & -1 & 3 & -1 \\ 4 & 0 & 1 & 3 & -2 & 1 \\ -1 & 2 & -3 & -1 & 2 & 3 \\ -2 & 1 & 4 & 3 & -2 & 2 \\ 1 & -2 & 3 & 4 & -2 & -1 \end{bmatrix}$$

In Exercises 21–28, let A , B , C , D , and E be matrices with the provided orders.

$A: 3 \times 4$ $B: 3 \times 4$ $C: 4 \times 2$ $D: 4 \times 2$ $E: 4 \times 3$

If defined, determine the size of the matrix. If not defined, provide an explanation.

21. $A + B$ 22. $C + E$ 23. $\frac{1}{2}D$ 24. $-4A$
25. AC 26. BE 27. $E - 2A$ 28. $2D + C$

In Exercises 29–36, write the system of linear equations in the form $A\mathbf{x} = \mathbf{b}$ and solve this matrix equation for $\mathbf{x}$.

29. $-x_1 + x_2 = 4$ 30. $2x_1 + 3x_2 = 5$
 $-2x_1 + x_2 = 0$ $x_1 + 4x_2 = 10$
31. $-2x_1 - 3x_2 = -4$ 32. $-4x_1 + 9x_2 = -13$
 $6x_1 + x_2 = -36$ $x_1 - 3x_2 = 12$

$$\begin{array}{ll}
33. \begin{cases} x_1 - 2x_2 + 3x_3 = 9 \\ -x_1 + 3x_2 - x_3 = -6 \\ 2x_1 - 5x_2 + 5x_3 = 17 \end{cases} & 34. \begin{cases} x_1 + x_2 - 3x_3 = -1 \\ -x_1 + 2x_2 = 1 \\ x_1 - x_2 + x_3 = 2 \end{cases} \\
35. \begin{cases} x_1 - 5x_2 + 2x_3 = -20 \\ -3x_1 + x_2 - x_3 = 8 \\ -2x_2 + 5x_3 = -16 \end{cases} & \\
36. \begin{cases} x_1 - x_2 + 4x_3 = 17 \\ x_1 + 3x_2 = -11 \\ -6x_2 + 5x_3 = 40 \end{cases} &
\end{array}$$

In Exercises 37 and 38, solve the matrix equation for A .

$$37. \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad 38. \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

In Exercises 39 and 40, solve the matrix equation for a , b , c , and d .

$$39. \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 6 & 3 \\ 19 & 2 \end{bmatrix}$$

$$40. \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 17 \\ 4 & -1 \end{bmatrix}$$

41. Find conditions on w , x , y , and z such that $AB = BA$ for the matrices below.

$$A = \begin{bmatrix} w & x \\ y & z \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

42. Verify $AB = BA$ for the following matrices.

$$A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix}$$

A square matrix

$$A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & a_{nn} \end{bmatrix}$$

is called a **diagonal matrix** if all entries that are not on the main diagonal are zero. In Exercises 43 and 44, find the product AA for the diagonal matrix.

$$43. A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$44. A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

In Exercises 45 and 46, find the products AB and BA for the diagonal matrices.

$$45. A = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} -5 & 0 \\ 0 & 4 \end{bmatrix}$$

$$46. A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} -7 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

47. **Guided Proof** Prove that if A and B are diagonal matrices (of the same size), then $AB = BA$.

Getting Started: To prove that the matrices AB and BA are equal, you need to show that their corresponding entries are equal.

- Begin your proof by letting $A = [a_{ij}]$ and $B = [b_{ij}]$ be two diagonal $n \times n$ matrices.
- The ij th entry of the product AB is $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$.
- Evaluate the entries c_{ij} for the two cases $i \neq j$ and $i = j$.
- Repeat this analysis for the product BA .

48. **Writing** Let A and B be 3×3 matrices, where A is diagonal.

- Describe the product AB . Illustrate your answer with examples.
- Describe the product BA . Illustrate your answer with examples.
- How do the results in parts (a) and (b) change if the diagonal entries of A are all equal?

In Exercises 49–52, find the trace of the matrix. The **trace** of an $n \times n$ matrix A is the sum of the main diagonal entries. That is, $\text{Tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$.

$$49. \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & 4 \\ 3 & 1 & 3 \end{bmatrix}$$

$$50. \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$51. \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 4 & 2 & 1 & 0 \\ 0 & 0 & 5 & 1 \end{bmatrix}$$

$$52. \begin{bmatrix} 1 & 4 & 3 & 2 \\ 4 & 0 & 6 & 1 \\ 3 & 6 & 2 & 1 \\ 2 & 1 & 1 & -3 \end{bmatrix}$$

53. Prove that each statement is true if A and B are square matrices of order n and c is a scalar.

- $\text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$
- $\text{Tr}(cA) = c\text{Tr}(A)$

54. Prove that if A and B are square matrices of order n , then $\text{Tr}(AB) = \text{Tr}(BA)$.

55. Show that the matrix equation has no solution.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

56. Show that no 2×2 matrices A and B exist that satisfy the matrix equation

$$AB - BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

57. Let $i = \sqrt{-1}$ and let $A = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$.

- (a) Find A^2 , A^3 , and A^4 . Identify any similarities among i^2 , i^3 , and i^4 .
 (b) Find and identify B^2 .

58. **Guided Proof** Prove that if the product AB is a square matrix, then the product BA is defined.

Getting Started: To prove that the product BA is defined, you need to show that the number of columns of B equals the number of rows of A .

- (i) Begin your proof by noting that the number of columns of A equals the number of rows of B .
 (ii) You can then assume that A has size $m \times n$ and B has size $n \times p$.
 (iii) Use the hypothesis that the product AB is a square matrix.

59. Prove that if both products AB and BA are defined, then AB and BA are square matrices.

60. Let A and B be two matrices such that the product AB is defined. Show that if A has two identical rows, then the corresponding two rows of AB are also identical.

61. Let A and B be $n \times n$ matrices. Show that if the i th row of A has all zero entries, then the i th row of AB will have all zero entries. Give an example using 2×2 matrices to show that the converse is not true.

62. The columns of matrix T show the coordinates of the vertices of a triangle. Matrix A is a transformation matrix.

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \end{bmatrix}$$

- (a) Find AT and AAT . Then sketch the original triangle and the two transformed triangles. What transformation does A represent?
 (b) A triangle is determined by AAT . Describe the transformation process that produces the triangle determined by AT and then the triangle determined by T .

63. A corporation has three factories, each of which manufactures acoustic guitars and electric guitars. The number of guitars of type i produced at factory j in one day is represented by a_{ij} in the matrix

$$A = \begin{bmatrix} 70 & 50 & 25 \\ 35 & 100 & 70 \end{bmatrix}.$$

Find the production levels if production is increased by 20%.

64. A corporation has four factories, each of which manufactures sport utility vehicles and pickup trucks. The number of vehicles of type i produced at factory j in one day is represented by a_{ij} in the matrix

$$A = \begin{bmatrix} 100 & 90 & 70 & 30 \\ 40 & 20 & 60 & 60 \end{bmatrix}.$$

Find the production levels if production is increased by 10%.

65. A fruit grower raises two crops, apples and peaches. Each of these crops is shipped to three different outlets. The number of units of crop i that are shipped to outlet j is represented by a_{ij} in the matrix

$$A = \begin{bmatrix} 125 & 100 & 75 \\ 100 & 175 & 125 \end{bmatrix}.$$

The profit per unit is represented by the matrix

$$B = [\$3.50 \quad \$6.00].$$

Find the product BA and state what each entry of the product represents.



66. A company manufactures tables and chairs at two locations. Matrix C gives the total cost of manufacturing each product at each location.

$$C = \begin{array}{cc} & \begin{array}{cc} \text{Location 1} & \text{Location 2} \end{array} \\ \begin{array}{c} \text{Tables} \\ \text{Chairs} \end{array} & \begin{bmatrix} 627 & 681 \\ 135 & 150 \end{bmatrix} \end{array}$$

- (a) If labor accounts for about $\frac{2}{3}$ of the total cost, determine the matrix L that gives the labor cost for each product at each location. What matrix operation did you use?
 (b) Find the matrix M that gives material costs for each product at each location. (Assume there are only labor and material costs.)

True or False? In Exercises 67 and 68, determine whether each statement is true or false. If a statement is true, give a reason or cite an appropriate statement from the text. If a statement is false, provide an example that shows the statement is not true in all cases or cite an appropriate statement from the text.

67. (a) For the product of two matrices to be defined, the number of columns of the first matrix must equal the number of rows of the second matrix.
- (b) The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ can be expressed as a linear combination, where the coefficients of the linear combination are a solution of the system.
68. (a) If A is an $m \times n$ matrix and B is an $n \times r$ matrix, then the product AB is an $m \times r$ matrix.
- (b) The matrix equation $A\mathbf{x} = \mathbf{b}$, where A is the coefficient matrix and $\mathbf{x}$ and $\mathbf{b}$ are column matrices, can be used to represent a system of linear equations.

69. **Writing** The matrix

$$P = \begin{matrix} & \begin{matrix} \text{To } R & \text{To } D & \text{To } I \end{matrix} \\ \begin{matrix} \text{From } R \\ \text{From } D \\ \text{From } I \end{matrix} & \begin{bmatrix} 0.75 & 0.15 & 0.10 \\ 0.20 & 0.60 & 0.20 \\ 0.30 & 0.40 & 0.30 \end{bmatrix} \end{matrix}$$

represents the proportions of a voting population that change from party i to party j in a given election. That is, p_{ij} ($i \neq j$) represents the proportion of the voting population that changes from party i to party j , and p_{ii} represents the proportion that remains loyal to party i from one election to the next. Find the product of P with itself. What does this product represent?

70. The matrices show the numbers of people (in thousands) who lived in various regions of the United States in 2005 and the numbers of people (in thousands) projected to live in those regions in 2015. The regional populations are separated into three age categories. (Source: U.S. Census Bureau)

	2005		
	0–17	18–64	65+
Northeast	12,607	34,418	6286
Midwest	16,131	41,395	7177
South	26,728	63,911	11,689
Mountain	5306	12,679	2020
Pacific	12,524	30,741	4519

	2015		
	0–17	18–64	65+
Northeast	12,441	35,289	8835
Midwest	16,363	42,250	9955
South	29,373	73,496	17,572
Mountain	5263	14,231	3337
Pacific	12,826	33,292	7086

- (a) The total population in 2005 was 288,131,000 and the projected total population in 2015 is 321,609,000. Rewrite the matrices to give the information as percents of the total population.
- (b) Write a matrix that gives the projected changes in the percents of the population in the various regions and age groups from 2005 to 2015.
- (c) Based on the result of part (b), which age group(s) is (are) projected to show relative growth from 2005 to 2015?

In Exercises 71 and 72, perform the indicated block multiplication of matrices A and B . If matrices A and B have each been partitioned into four submatrices

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix},$$

then you can **block multiply** A and B , provided the sizes of the submatrices are such that the matrix multiplications and additions are defined.

$$\begin{aligned} AB &= \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} \\ &= \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix} \end{aligned}$$

$$71. A = \left[\begin{array}{cc|cc} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 \end{array} \right], \quad B = \left[\begin{array}{cc|c} 1 & 2 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 3 \end{array} \right]$$

$$72. A = \left[\begin{array}{cc|cc} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{array} \right], \quad B = \left[\begin{array}{cc|cc} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{array} \right]$$

In Exercises 73–76, express the column matrix $\mathbf{b}$ as a linear combination of the columns of A .

$$73. A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & -3 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ 7 \end{bmatrix}$$

$$74. A = \begin{bmatrix} 1 & 2 & 4 \\ -1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

$$75. A = \begin{bmatrix} 1 & 1 & -5 \\ 1 & 0 & -1 \\ 2 & -1 & -1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$$

$$76. A = \begin{bmatrix} -3 & 5 \\ 3 & 4 \\ 4 & -8 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -22 \\ 4 \\ 32 \end{bmatrix}$$

2.2 Properties of Matrix Operations

In Section 2.1 you concentrated on the mechanics of the three basic matrix operations: matrix addition, scalar multiplication, and matrix multiplication. This section begins to develop the **algebra of matrices**. You will see that this algebra shares many (but not all) of the properties of the algebra of real numbers. Several properties of matrix addition and scalar multiplication are listed below.

THEOREM 2.1 Properties of Matrix Addition and Scalar Multiplication

If A , B , and C are $m \times n$ matrices and c and d are scalars, then the following properties are true.

- | | |
|--------------------------------|--|
| 1. $A + B = B + A$ | Commutative property of addition |
| 2. $A + (B + C) = (A + B) + C$ | Associative property of addition |
| 3. $(cd)A = c(dA)$ | Associative property of multiplication |
| 4. $1A = A$ | Multiplicative identity |
| 5. $c(A + B) = cA + cB$ | Distributive property |
| 6. $(c + d)A = cA + dA$ | Distributive property |

PROOF

The proofs of these six properties follow directly from the definitions of matrix addition, scalar multiplication, and the corresponding properties of real numbers. For example, to prove the commutative property of *matrix addition*, let $A = [a_{ij}]$ and $B = [b_{ij}]$. Then, using the commutative property of *addition of real numbers*, write

$$A + B = [a_{ij} + b_{ij}] = [b_{ij} + a_{ij}] = B + A.$$

Similarly, to prove Property 5, use the distributive property (for real numbers) of multiplication over addition to write

$$c(A + B) = [c(a_{ij} + b_{ij})] = [ca_{ij} + cb_{ij}] = cA + cB.$$

The proofs of the remaining four properties are left as exercises. (See Exercises 47–50.)

In the preceding section, matrix addition was defined as the sum of *two* matrices, making it a binary operation. The associative property of matrix addition now allows you to write expressions such as $A + B + C$ as $(A + B) + C$ or as $A + (B + C)$. This same reasoning applies to sums of four or more matrices.

EXAMPLE 1 Addition of More than Two Matrices

By adding corresponding entries, you can obtain the sum of four matrices shown below.

$$\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

One important property of the addition of real numbers is that the number 0 serves as the additive identity. That is, $c + 0 = c$ for any real number c . For matrices, a similar property holds. Specifically, if A is an $m \times n$ matrix and O_{mn} is the $m \times n$ matrix consisting entirely of zeros, then $A + O_{mn} = A$. The matrix O_{mn} is called a **zero matrix**, and it serves as the **additive identity** for the set of all $m \times n$ matrices. For example, the following matrix serves as the additive identity for the set of all 2×3 matrices.

$$O_{23} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

When the size of the matrix is understood, you may denote a zero matrix simply by 0.

The following properties of zero matrices are easy to prove, and their proofs are left as an exercise. (See Exercise 51.)

THEOREM 2.2 Properties of Zero Matrices

If A is an $m \times n$ matrix and c is a scalar, then the following properties are true.

1. $A + O_{mn} = A$
2. $A + (-A) = O_{mn}$
3. If $cA = O_{mn}$, then $c = 0$ or $A = O_{mn}$.

REMARK: Property 2 can be described by saying that matrix $-A$ is the **additive inverse** of A .

The algebra of real numbers and the algebra of matrices have many similarities. For example, compare the solutions below.

Real Numbers (Solve for x .)	$m \times n$ Matrices (Solve for X .)
$x + a = b$	$X + A = B$
$x + a + (-a) = b + (-a)$	$X + A + (-A) = B + (-A)$
$x + 0 = b - a$	$X + O = B - A$
$x = b - a$	$X = B - A$

The process of solving a matrix equation is demonstrated in Example 2.

EXAMPLE 2 Solving a Matrix Equation

Solve for X in the equation $3X + A = B$, where

$$A = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -3 & 4 \\ 2 & 1 \end{bmatrix}.$$

SOLUTION Begin by solving the equation for X to obtain

$$3X = B - A \quad \rightarrow \quad X = \frac{1}{3}(B - A).$$

Now, using the given matrices A and B , you have

$$X = \frac{1}{3} \left(\begin{bmatrix} -3 & 4 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} \right) = \frac{1}{3} \begin{bmatrix} -4 & 6 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} -\frac{4}{3} & 2 \\ \frac{2}{3} & -\frac{2}{3} \end{bmatrix}.$$

Properties of Matrix Multiplication

In the next theorem, the algebra of matrices is extended to include some useful properties of matrix multiplication. The proof of Property 2 is presented below. The proofs of the remaining properties are left as an exercise. (See Exercise 52.)

THEOREM 2.3
Properties of
Matrix Multiplication

If A , B , and C are matrices (with sizes such that the given matrix products are defined) and c is a scalar, then the following properties are true.

1. $A(BC) = (AB)C$
2. $A(B + C) = AB + AC$
3. $(A + B)C = AC + BC$
4. $c(AB) = (cA)B = A(cB)$

PROOF

To prove Property 2, show that the matrices $A(B + C)$ and $AB + AC$ are equal by showing that their corresponding entries are equal. Assume A has size $m \times n$, B has size $n \times p$, and C has size $n \times p$. Using the definition of matrix multiplication, the entry in the i th row and j th column of $A(B + C)$ is $a_{i1}(b_{1j} + c_{1j}) + \cdots + a_{in}(b_{nj} + c_{nj})$. Moreover, the entry in the i th row and j th column of $AB + AC$ is

$$(a_{i1}b_{1j} + \cdots + a_{in}b_{nj}) + (a_{i1}c_{1j} + \cdots + a_{in}c_{nj}).$$

By distributing and regrouping, you can see that these two ij th entries are equal. So,

$$A(B + C) = AB + AC.$$

The associative property of matrix multiplication permits you to write such matrix products as ABC without ambiguity, as demonstrated in Example 3.

EXAMPLE 3 Matrix Multiplication Is Associative

Find the matrix product ABC by grouping the factors first as $(AB)C$ and then as $A(BC)$. Show that the same result is obtained from both processes.

$$A = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}$$

SOLUTION Grouping the factors as $(AB)C$, you have

$$\begin{aligned} (AB)C &= \left(\begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix} \right) \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix} \\ &= \begin{bmatrix} -5 & 4 & 0 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 17 & 4 \\ 13 & 14 \end{bmatrix}. \end{aligned}$$

Grouping the factors as $A(BC)$, you obtain the same result.

$$\begin{aligned} A(BC) &= \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ -7 & 2 \end{bmatrix} = \begin{bmatrix} 17 & 4 \\ 13 & 14 \end{bmatrix} \end{aligned}$$

Note that no commutative property for matrix multiplication is listed in Theorem 2.3. Although the product AB is defined, it can easily happen that A and B are not of the proper sizes to define the product BA . For instance, if A is of size 2×3 and B is of size 3×3 , then the product AB is defined but the product BA is not. The next example shows that even if both products AB and BA are defined, they may not be equal.

EXAMPLE 4 Noncommutativity of Matrix Multiplication

Show that AB and BA are not equal for the matrices

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}.$$

$$\textbf{SOLUTION} \quad AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 4 & -4 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 7 \\ 4 & -2 \end{bmatrix}$$

$$AB \neq BA$$

Do not conclude from Example 4 that the matrix products AB and BA are *never* the same. Sometimes they are the same. For example, try multiplying the following matrices, first in the order AB and then in the order BA .

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -2 & 4 \\ 2 & -2 \end{bmatrix}$$

You will see that the two products are equal. The point is this: Although AB and BA are sometimes equal, AB and BA are usually not equal.

Another important quality of matrix algebra is that it does not have a general cancellation property for matrix multiplication. That is, if $AC = BC$, it is not necessarily true that $A = B$. This is demonstrated in Example 5. (In the next section you will see that, for some special types of matrices, cancellation is valid.)

EXAMPLE 5**An Example in Which Cancellation Is Not Valid**

Show that $AC = BC$.

$$A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$$

$$\text{SOLUTION} \quad AC = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix}$$

$$BC = \begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix}$$

$AC = BC$, even though $A \neq B$.

You will now look at a special type of *square* matrix that has 1's on the main diagonal and 0's elsewhere.

$$I_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

For instance, if $n = 1, 2$, or 3 , we have

$$I_1 = [1], \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

1 × 1

2 × 2

3 × 3

When the order of the matrix is understood to be n , you may denote I_n simply as I .

As stated in Theorem 2.4 on the next page, the matrix I_n serves as the **identity** for matrix multiplication; it is called the **identity matrix of order n** . The proof of this theorem is left as an exercise. (See Exercise 53.)

THEOREM 2.4
Properties of the
Identity Matrix

If A is a matrix of size $m \times n$, then the following properties are true.

1. $AI_n = A$
2. $I_m A = A$

As a special case of this theorem, note that if A is a *square* matrix of order n , then

$$AI_n = I_n A = A.$$

EXAMPLE 6 **Multiplication by an Identity Matrix**

$$\begin{aligned} \text{(a)} \quad & \begin{bmatrix} 3 & -2 \\ 4 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 4 & 0 \\ -1 & 1 \end{bmatrix} \\ \text{(b)} \quad & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} \end{aligned}$$

For repeated multiplication of *square* matrices, you can use the same exponential notation used with real numbers. That is, $A^1 = A$, $A^2 = AA$, and for a positive integer k , A^k is

$$A^k = \underbrace{AA \cdots A}_{k \text{ factors}}.$$

It is convenient also to define $A^0 = I_n$ (where A is a square matrix of order n). These definitions allow you to establish the properties

$$1. A^j A^k = A^{j+k} \quad \text{and} \quad 2. (A^j)^k = A^{jk}$$

where j and k are nonnegative integers.

EXAMPLE 7 **Repeated Multiplication of a Square Matrix**

Find A^3 for the matrix $A = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix}$.

SOLUTION
$$A^3 = \left(\begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} \right) \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 6 & -3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ 3 & -6 \end{bmatrix}$$

In Section 1.1 you saw that a system of linear equations must have exactly one solution, an infinite number of solutions, or no solution. Using the matrix algebra developed so far, you can now prove that this is true.

THEOREM 2.5
**Number of Solutions of a
 System of Linear Equations**

For a system of linear equations in n variables, precisely one of the following is true.

1. The system has exactly one solution.
2. The system has an infinite number of solutions.
3. The system has no solution.

PROOF

Represent the system by the matrix equation $A\mathbf{x} = \mathbf{b}$. If the system has exactly one solution or no solution, then there is nothing to prove. So, you can assume that the system has at least two distinct solutions $\mathbf{x}_1$ and $\mathbf{x}_2$. The proof will be complete if you can show that this assumption implies that the system has an infinite number of solutions. Because $\mathbf{x}_1$ and $\mathbf{x}_2$ are solutions, you have $A\mathbf{x}_1 = A\mathbf{x}_2 = \mathbf{b}$ and $A(\mathbf{x}_1 - \mathbf{x}_2) = \mathbf{0}$. This implies that the (nonzero) column matrix $\mathbf{x}_h = \mathbf{x}_1 - \mathbf{x}_2$ is a solution of the homogeneous system of linear equations $A\mathbf{x} = \mathbf{0}$. It can now be said that for any scalar c ,

$$A(\mathbf{x}_1 + c\mathbf{x}_h) = A\mathbf{x}_1 + A(c\mathbf{x}_h) = \mathbf{b} + c(A\mathbf{x}_h) = \mathbf{b} + c\mathbf{0} = \mathbf{b}.$$

So $\mathbf{x}_1 + c\mathbf{x}_h$ is a solution of $A\mathbf{x} = \mathbf{b}$ for any scalar c . Because there are an infinite number of possible values of c and each value produces a different solution, you can conclude that the system has an infinite number of solutions.

The Transpose of a Matrix

The **transpose** of a matrix is formed by writing its columns as rows. For instance, if A is the $m \times n$ matrix shown by

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix},$$

Size: $m \times n$

then the transpose, denoted by A^T , is the $n \times m$ matrix below

$$A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} & \cdots & a_{m1} \\ a_{12} & a_{22} & a_{32} & \cdots & a_{m2} \\ a_{13} & a_{23} & a_{33} & \cdots & a_{m3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & a_{mn} \end{bmatrix}.$$

Size: $n \times m$

EXAMPLE 8 The Transpose of a Matrix

Find the transpose of each matrix.

$$(a) A = \begin{bmatrix} 2 \\ 8 \end{bmatrix} \quad (b) B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad (c) C = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (d) D = \begin{bmatrix} 0 & 1 \\ 2 & 4 \\ 1 & -1 \end{bmatrix}$$

SOLUTION

$$(a) A^T = [2 \quad 8] \quad (b) B^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \quad (c) C^T = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(d) D^T = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 4 & -1 \end{bmatrix}$$

Discovery

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$.

Calculate $(AB)^T$, $A^T B^T$, and $B^T A^T$. Make a conjecture about the transpose of a product of two square matrices. Select two other square matrices to check your conjecture.

REMARK: Note that the square matrix in part (c) of Example 8 is equal to its transpose. Such a matrix is called **symmetric**. A matrix A is symmetric if $A = A^T$. From this definition it is clear that a symmetric matrix must be square. Also, if $A = [a_{ij}]$ is a symmetric matrix, then $a_{ij} = a_{ji}$ for all $i \neq j$.

THEOREM 2.6
Properties of
Transposes

If A and B are matrices (with sizes such that the given matrix operations are defined) and c is a scalar, then the following properties are true.

1. $(A^T)^T = A$ Transpose of a transpose
2. $(A + B)^T = A^T + B^T$ Transpose of a sum
3. $(cA)^T = c(A^T)$ Transpose of a scalar multiple
4. $(AB)^T = B^T A^T$ Transpose of a product

PROOF

Because the transpose operation interchanges rows and columns, Property 1 seems to make sense. To prove Property 1, let A be an $m \times n$ matrix. Observe that A^T has size $n \times m$ and $(A^T)^T$ has size $m \times n$, the same as A . To show that $(A^T)^T = A$ you must show that the ij th entries are the same. Let a_{ij} be the ij th entry of A . Then a_{ij} is the ji th entry of A^T , and the ij th entry of $(A^T)^T$. This proves Property 1. The proofs of the remaining properties are left as an exercise. (See Exercise 54.)

REMARK: Remember that you *reverse the order* of multiplication when forming the transpose of a product. That is, the transpose of AB is $(AB)^T = B^T A^T$ and is *not* usually equal to $A^T B^T$.

Properties 2 and 4 can be generalized to cover sums or products of any finite number of matrices. For instance, the transpose of the sum of three matrices is

$$(A + B + C)^T = A^T + B^T + C^T,$$

and the transpose of the product of three matrices is

$$(ABC)^T = C^T B^T A^T.$$

EXAMPLE 9 Finding the Transpose of a Product

Show that $(AB)^T$ and $B^T A^T$ are equal.

$$A = \begin{bmatrix} 2 & 1 & -2 \\ -1 & 0 & 3 \\ 0 & -2 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 1 \\ 2 & -1 \\ 3 & 0 \end{bmatrix}$$

$$\text{SOLUTION} \quad AB = \begin{bmatrix} 2 & 1 & -2 \\ -1 & 0 & 3 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & -1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 6 & -1 \\ -1 & 2 \end{bmatrix}$$

$$(AB)^T = \begin{bmatrix} 2 & 6 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$B^T A^T = \begin{bmatrix} 3 & 2 & 3 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -2 \\ -2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 6 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$(AB)^T = B^T A^T$$

EXAMPLE 10 The Product of a Matrix and Its Transpose

For the matrix

$$A = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -2 & -1 \end{bmatrix}$$

find the product AA^T and show that it is symmetric.

SOLUTION Because

$$AA^T = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 3 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 10 & -6 & -5 \\ -6 & 4 & 2 \\ -5 & 2 & 5 \end{bmatrix}$$

it follows that $AA^T = (AA^T)^T$, so AA^T is symmetric.

REMARK: The property demonstrated in Example 10 is true in general. That is, for any matrix A , the matrix given by $B = AA^T$ is symmetric. You are asked to prove this result in Exercise 55.

SECTION 2.2 Exercises

In Exercises 1–6, perform the indicated operations when $a = 3$, $b = -4$, and

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}, \quad O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

1. $aA + bB$ 2. $A + B$ 3. $ab(B)$
4. $(a + b)B$ 5. $(a - b)(A - B)$ 6. $(ab)O$

7. Solve for X when

$$A = \begin{bmatrix} -4 & 0 \\ 1 & -5 \\ -3 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ -2 & 1 \\ 4 & 4 \end{bmatrix}.$$

- (a) $3X + 2A = B$ (b) $2A - 5B = 3X$
(c) $X - 3A + 2B = O$ (d) $6X - 4A - 3B = O$

8. Solve for X when

$$A = \begin{bmatrix} -2 & -1 \\ 1 & 0 \\ 3 & -4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 3 \\ 2 & 0 \\ -4 & -1 \end{bmatrix}.$$

- (a) $X = 3A - 2B$ (b) $2X = 2A - B$
(c) $2X + 3A = B$ (d) $2A + 4B = -2X$

In Exercises 9–14, perform the indicated operations, provided that $c = -2$ and

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix},$$

$$O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

9. $B(CA)$ 10. $C(BC)$ 11. $(B + C)A$
12. $B(C + O)$ 13. $(cB)(C + C)$ 14. $B(cA)$

In Exercises 15 and 16, demonstrate that if $AC = BC$, then A is not necessarily equal to B for the following matrices.

15. $A = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, C = \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}$

16. $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 4 \\ 3 & -2 & 1 \end{bmatrix}, B = \begin{bmatrix} 4 & -6 & 3 \\ 5 & 4 & 4 \\ -1 & 0 & 1 \end{bmatrix},$
 $C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 4 & -2 & 3 \end{bmatrix}$

In Exercises 17 and 18, demonstrate that if $AB = O$, then it is not necessarily true that $A = O$ or $B = O$ for the following matrices.

17. $A = \begin{bmatrix} 3 & 3 \\ 4 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

18. $A = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -\frac{1}{2} & 1 \end{bmatrix}$

In Exercises 19–22, perform the indicated operations when

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \quad \text{and} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

19. A^2 20. A^4
21. $A(I + A)$ 22. $A + IA$

In Exercises 23–28, find (a) A^T , (b) A^TA , and (c) AA^T .

23. $A = \begin{bmatrix} 4 & 2 & 1 \\ 0 & 2 & -1 \end{bmatrix}$ 24. $A = \begin{bmatrix} 1 & -1 \\ 3 & 4 \\ 0 & -2 \end{bmatrix}$

25. $A = \begin{bmatrix} 2 & 1 & -3 \\ 1 & 4 & 1 \\ 0 & 2 & 1 \end{bmatrix}$ 26. $A = \begin{bmatrix} -7 & 11 & 12 \\ 4 & -3 & 1 \\ 6 & -1 & 3 \end{bmatrix}$

27. $A = \begin{bmatrix} 0 & -4 & 3 & 2 \\ 8 & 4 & 0 & 1 \\ -2 & 3 & 5 & 1 \\ 0 & 0 & -3 & 2 \end{bmatrix}$

28. $A = \begin{bmatrix} 4 & -3 & 2 & 0 \\ 2 & 0 & 11 & -1 \\ -1 & -2 & 0 & 3 \\ 14 & -2 & 12 & -9 \\ 6 & 8 & -5 & 4 \end{bmatrix}$

Writing In Exercises 29 and 30, explain why the formula is *not* valid for matrices. Illustrate your argument with examples.

29. $(A + B)(A - B) = A^2 - B^2$

30. $(A + B)(A + B) = A^2 + 2AB + B^2$

In Exercises 31–34, verify that $(AB)^T = B^T A^T$.

31. $A = \begin{bmatrix} -1 & 1 & -2 \\ 2 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 0 \\ 1 & 2 \\ 1 & -1 \end{bmatrix}$

32. $A = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & -1 \\ 2 & 1 \end{bmatrix}$

33. $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \\ -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & -1 \end{bmatrix}$

34. $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & -2 \\ 0 & 1 & 3 \end{bmatrix}$

True or False? In Exercises 35 and 36, determine whether each statement is true or false. If a statement is true, give a reason or cite an appropriate statement from the text. If a statement is false, provide an example that shows the statement is not true in all cases or cite an appropriate statement from the text.

35. (a) Matrix addition is commutative.
 (b) Matrix multiplication is associative.
 (c) The transpose of the product of two matrices equals the product of their transposes; that is, $(AB)^T = A^T B^T$.
 (d) For any matrix C , the matrix CC^T is symmetric.
36. (a) Matrix multiplication is commutative.
 (b) Every matrix A has an additive inverse.
 (c) If the matrices A , B , and C satisfy $AB = AC$, then $B = C$.
 (d) The transpose of the sum of two matrices equals the sum of their transposes.

37. Consider the matrices shown below.

$$X = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad Y = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad W = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- (a) Find scalars a and b such that $Z = aX + bY$.
 (b) Show that there do not exist scalars a and b such that $W = aX + bY$.

- (c) Show that if $aX + bY + cW = O$, then $a = b = c = 0$.
 (d) Find scalars a , b , and c , not all equal to zero, such that $aX + bY + cZ = O$.

38. Consider the matrices shown below.

$$X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad Y = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix}, \quad W = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- (a) Find scalars a and b such that $Z = aX + bY$.
 (b) Show that there do not exist scalars a and b such that $W = aX + bY$.
 (c) Show that if $aX + bY + cW = O$, then $a = b = c = 0$.
 (d) Find scalars a , b , and c , not all equal to zero, such that $aX + bY + cZ = O$.

In Exercises 39 and 40, compute the power of A for the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

39. A^{19}

40. A^{20}

An n th root of a matrix B is a matrix A such that $A^n = B$. In Exercises 41 and 42, find the n th root of the matrix B .

41. $B = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix}$, $n = 2$ 42. $B = \begin{bmatrix} 8 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 27 \end{bmatrix}$, $n = 3$

In Exercises 43–46, use the given definition to find $f(A)$: If f is the polynomial function,

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

then for an $n \times n$ matrix A , $f(A)$ is defined to be

$$f(A) = a_0I_n + a_1A + a_2A^2 + \cdots + a_nA^n.$$

43. $f(x) = x^2 - 5x + 2$, $A = \begin{bmatrix} 2 & 0 \\ 4 & 5 \end{bmatrix}$

44. $f(x) = x^2 - 7x + 6$, $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

45. $f(x) = x^2 - 3x + 2$, $A = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$

46. $f(x) = x^3 - 2x^2 + 5x - 10$, $A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 2 \\ -1 & 1 & 3 \end{bmatrix}$

- 47. Guided Proof** Prove the associative property of matrix addition: $A + (B + C) = (A + B) + C$.

Getting Started: To prove that $A + (B + C)$ and $(A + B) + C$ are equal, show that their corresponding entries are the same.

- Begin your proof by letting A , B , and C be $m \times n$ matrices.
 - Observe that the ij th entry of $B + C$ is $b_{ij} + c_{ij}$.
 - Furthermore, the ij th entry of $A + (B + C)$ is $a_{ij} + (b_{ij} + c_{ij})$.
 - Determine the ij th entry of $(A + B) + C$.
- 48.** Prove the associative property of scalar multiplication: $(cd)A = c(dA)$.

- 49.** Prove that the scalar 1 is the identity for scalar multiplication: $1A = A$.

- 50.** Prove the following distributive property: $(c + d)A = cA + dA$.

- 51.** Prove Theorem 2.2.

- 52.** Complete the proof of Theorem 2.3.

- Prove the associative property of multiplication: $A(BC) = (AB)C$.
- Prove the distributive property: $(A + B)C = AC + BC$.
- Prove the property: $c(AB) = (cA)B = A(cB)$.

- 53.** Prove Theorem 2.4.

- 54.** Prove Properties 2, 3, and 4 of Theorem 2.6.

- 55. Guided Proof** Prove that if A is an $m \times n$ matrix, then AA^T and A^TA are symmetric matrices.

Getting Started: To prove that AA^T is symmetric, you need to show that it is equal to its transpose, $(AA^T)^T = AA^T$.

- Begin your proof with the left-hand matrix expression $(AA^T)^T$.
 - Use the properties of the transpose operation to show that it can be simplified to equal the right-hand expression, AA^T .
 - Repeat this analysis for the product A^TA .
- 56.** Give an example of two 2×2 matrices A and B such that $(AB)^T \neq A^TB^T$.

In Exercises 57–60, determine whether the matrix is symmetric, skew-symmetric, or neither. A square matrix is called **skew-symmetric** if $A^T = -A$.

57. $A = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$ **58.** $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

59. $A = \begin{bmatrix} 0 & 2 & 1 \\ 2 & 0 & 3 \\ 1 & 3 & 0 \end{bmatrix}$ **60.** $A = \begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & -3 \\ 1 & 3 & 0 \end{bmatrix}$

- 61.** Prove that the main diagonal of a skew-symmetric matrix consists entirely of zeros.

- 62.** Prove that if A and B are $n \times n$ skew-symmetric matrices, then $A + B$ is skew-symmetric.

- 63.** Let A be a square matrix of order n .

- Show that $\frac{1}{2}(A + A^T)$ is symmetric.
- Show that $\frac{1}{2}(A - A^T)$ is skew-symmetric.
- Prove that A can be written as the sum of a symmetric matrix B and a skew-symmetric matrix C , $A = B + C$.
- Write the matrix

$$A = \begin{bmatrix} 2 & 5 & 3 \\ -3 & 6 & 0 \\ 4 & 1 & 1 \end{bmatrix}$$

as the sum of a skew-symmetric matrix and a symmetric matrix.

- 64.** Prove that if A is an $n \times n$ matrix, then $A - A^T$ is skew-symmetric.

- 65.** Let A and B be two $n \times n$ symmetric matrices.

- Give an example to show that the product AB is not necessarily symmetric.
- Prove that AB is symmetric if and only if $AB = BA$.

-  **66.** Consider matrices of the form

$$A = \begin{bmatrix} 0 & a_{12} & a_{13} & a_{14} & \cdots & a_{1n} \\ 0 & 0 & a_{23} & a_{24} & \cdots & a_{2n} \\ 0 & 0 & 0 & a_{34} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a_{(n-1)n} \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{bmatrix}.$$

- Write a 2×2 matrix and a 3×3 matrix in the form of A .
- Use a graphing utility or computer software program to raise each of the matrices to higher powers. Describe the result.
- Use the result of part (b) to make a conjecture about powers of A if A is a 4×4 matrix. Use a graphing utility to test your conjecture.
- Use the results of parts (b) and (c) to make a conjecture about powers of A if A is an $n \times n$ matrix.

2.3 The Inverse of a Matrix

Section 2.2 discussed some of the similarities between the algebra of real numbers and the algebra of matrices. This section further develops the algebra of matrices to include the solutions of matrix equations involving matrix multiplication. To begin, consider the real number equation $ax = b$. To solve this equation for x , multiply both sides of the equation by a^{-1} (provided $a \neq 0$).

$$\begin{aligned} ax &= b \\ (a^{-1}a)x &= a^{-1}b \\ (1)x &= a^{-1}b \\ x &= a^{-1}b \end{aligned}$$

The number a^{-1} is called the *multiplicative inverse* of a because $a^{-1}a$ yields 1 (the identity element for multiplication). The definition of a multiplicative inverse of a matrix is similar.

Definition of the Inverse of a Matrix

An $n \times n$ matrix A is **invertible** (or **nonsingular**) if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n$$

where I_n is the identity matrix of order n . The matrix B is called the (multiplicative) **inverse** of A . A matrix that does not have an inverse is called **noninvertible** (or **singular**).

Nonsquare matrices do not have inverses. To see this, note that if A is of size $m \times n$ and B is of size $n \times m$ (where $m \neq n$), then the products AB and BA are of different sizes and cannot be equal to each other. Indeed, not all square matrices possess inverses. (See Example 4.) The next theorem, however, tells you that if a matrix *does* possess an inverse, then that inverse is unique.

THEOREM 2.7 Uniqueness of an Inverse Matrix

If A is an invertible matrix, then its inverse is unique. The inverse of A is denoted by A^{-1} .

PROOF

Because A is invertible, you know it has at least one inverse B such that

$$AB = I = BA.$$

Suppose A has another inverse C such that

$$AC = I = CA.$$

Then you can show that B and C are equal, as follows.

$$\begin{aligned} AB &= I \\ C(AB) &= CI \\ (CA)B &= C \\ IB &= C \\ B &= C \end{aligned}$$

Consequently $B = C$, and it follows that the inverse of a matrix is unique.

Because the inverse A^{-1} of an invertible matrix A is unique, you can call it *the* inverse of A and write

$$AA^{-1} = A^{-1}A = I.$$

EXAMPLE 1 The Inverse of a Matrix

Show that B is the inverse of A , where

$$A = \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}.$$

SOLUTION Using the definition of an inverse matrix, you can show that B is the inverse of A by showing that $AB = I = BA$, as follows.

$$\begin{aligned} AB &= \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 + 2 & 2 - 2 \\ -1 + 1 & 2 - 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ BA &= \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 + 2 & 2 - 2 \\ -1 + 1 & 2 - 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

REMARK: Recall that it is not always true that $AB = BA$, even if both products are defined. If A and B are both square matrices and $AB = I_n$, however, then it can be shown that $BA = I_n$. Although the proof of this fact is omitted, it implies that in Example 1 you needed only to check that $AB = I_2$.

The next example shows how to use a system of equations to find the inverse of a matrix.

EXAMPLE 2 Finding the Inverse of a Matrix

Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}.$$

SOLUTION To find the inverse of A , try to solve the matrix equation $AX = I$ for X .

$$\begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x_{11} + 4x_{21} & x_{12} + 4x_{22} \\ -x_{11} - 3x_{21} & -x_{12} - 3x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Now, by equating corresponding entries, you obtain the two systems of linear equations shown below.

$$\begin{array}{rcl} x_{11} + 4x_{21} & = & 1 \\ -x_{11} - 3x_{21} & = & 0 \end{array} \qquad \begin{array}{rcl} x_{12} + 4x_{22} & = & 0 \\ -x_{12} - 3x_{22} & = & 1 \end{array}$$

Solving the first system, you find that the first column of X is $x_{11} = -3$ and $x_{21} = 1$. Similarly, solving the second system, you find that the second column of X is $x_{12} = -4$ and $x_{22} = 1$. The inverse of A is

$$X = A^{-1} = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}.$$

Try using matrix multiplication to check this result.

Generalizing the method used to solve Example 2 provides a convenient method for finding an inverse. Notice first that the two systems of linear equations

$$\begin{array}{rcl} x_{11} + 4x_{21} & = & 1 \\ -x_{11} - 3x_{21} & = & 0 \end{array} \qquad \begin{array}{rcl} x_{12} + 4x_{22} & = & 0 \\ -x_{12} - 3x_{22} & = & 1 \end{array}$$

have the *same coefficient matrix*. Rather than solve the two systems represented by

$$\begin{bmatrix} 1 & 4 & \vdots & 1 \\ -1 & -3 & \vdots & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 4 & \vdots & 0 \\ -1 & -3 & \vdots & 1 \end{bmatrix}$$

separately, you can solve them simultaneously. You can do this by **adjoining** the identity matrix to the coefficient matrix to obtain

$$\begin{bmatrix} 1 & 4 & \vdots & 1 & 0 \\ -1 & -3 & \vdots & 0 & 1 \end{bmatrix}.$$

By applying Gauss-Jordan elimination to this matrix, you can solve *both* systems with a single elimination process, as follows.

$$\begin{bmatrix} 1 & 4 & \vdots & 1 & 0 \\ 0 & 1 & \vdots & 1 & 1 \end{bmatrix} \qquad R_2 + R_1 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 0 & \vdots & -3 & -4 \\ 0 & 1 & \vdots & 1 & 1 \end{bmatrix} \qquad R_1 + (-4)R_2 \rightarrow R_1$$

Applying Gauss-Jordan elimination to the “doubly augmented” matrix $[A : I]$, you obtain the matrix $[I : A^{-1}]$.

$$\begin{array}{ccc}
 \begin{bmatrix} 1 & 4 & \vdots & 1 & 0 \\ -1 & -3 & \vdots & 0 & 1 \end{bmatrix} & \longrightarrow & \begin{bmatrix} 1 & 0 & \vdots & -3 & -4 \\ 0 & 1 & \vdots & 1 & 1 \end{bmatrix} \\
 \text{\textcolor{teal}{A}} & & \text{\textcolor{teal}{I}} \qquad \text{\textcolor{teal}{I}} \qquad \text{\textcolor{teal}{A}^{-1}}
 \end{array}$$

This procedure (or algorithm) works for an arbitrary $n \times n$ matrix. If A cannot be row reduced to I_n , then A is noninvertible (or singular). This procedure will be formally justified in the next section, after the concept of an elementary matrix is introduced. For now the algorithm is summarized as follows.

Finding the Inverse of a Matrix by Gauss-Jordan Elimination

Let A be a square matrix of order n .

1. Write the $n \times 2n$ matrix that consists of the given matrix A on the left and the $n \times n$ identity matrix I on the right to obtain $[A : I]$. Note that you separate the matrices A and I by a dotted line. This process is called **adjoining** matrix I to matrix A .
2. If possible, row reduce A to I using elementary row operations on the *entire* matrix $[A : I]$. The result will be the matrix $[I : A^{-1}]$. If this is not possible, then A is noninvertible (or singular).
3. Check your work by multiplying AA^{-1} and $A^{-1}A$ to see that $AA^{-1} = I = A^{-1}A$.

EXAMPLE 3 Finding the Inverse of a Matrix

Find the inverse of the matrix.

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{bmatrix}$$

SOLUTION Begin by adjoining the identity matrix to A to form the matrix

$$[A : I] = \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -6 & 2 & 3 & 0 & 0 & 1 \end{array} \right].$$

Now, using elementary row operations, rewrite this matrix in the form $[I : A^{-1}]$, as follows.

$$\begin{array}{l}
 \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ -6 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \qquad R_2 + (-1)R_1 \rightarrow R_2 \\
 \\
 \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & -4 & 3 & 6 & 0 & 1 \end{array} \right] \qquad R_3 + (6)R_1 \rightarrow R_3 \\
 \\
 \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 4 & 1 \end{array} \right] \qquad R_3 + (4)R_2 \rightarrow R_3
 \end{array}$$

$$\begin{aligned}
 &\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & \vdots & 1 & 0 & 0 \\ 0 & 1 & -1 & \vdots & -1 & 1 & 0 \\ 0 & 0 & 1 & \vdots & -2 & -4 & -1 \end{array} \right] && (-1)R_3 \rightarrow R_3 \\
 &\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & \vdots & 1 & 0 & 0 \\ 0 & 1 & 0 & \vdots & -3 & -3 & -1 \\ 0 & 0 & 1 & \vdots & -2 & -4 & -1 \end{array} \right] && R_2 + R_3 \rightarrow R_2 \\
 &\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \vdots & -2 & -3 & -1 \\ 0 & 1 & 0 & \vdots & -3 & -3 & -1 \\ 0 & 0 & 1 & \vdots & -2 & -4 & -1 \end{array} \right] && R_1 + R_2 \rightarrow R_1
 \end{aligned}$$

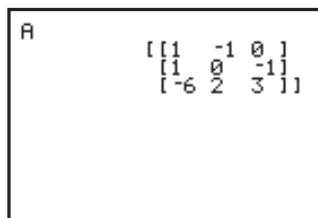
The matrix A is invertible, and its inverse is

$$A^{-1} = \begin{bmatrix} -2 & -3 & -1 \\ -3 & -3 & -1 \\ -2 & -4 & -1 \end{bmatrix}.$$

Try confirming this by showing that $AA^{-1} = I = A^{-1}A$.

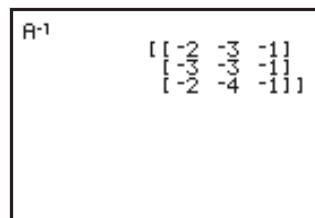
Technology Note

Most graphing utilities and computer software programs can calculate the inverse of a square matrix. If you are using a graphing utility, your screens for Example 3 may look like the images below. Keystrokes and programming syntax for these utilities/programs applicable to Example 3 are provided in the **Online Technology Guide**, available at college.hmco.com/pic/larsonELA6e.



A

$$\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{bmatrix}$$



A⁻¹

$$\begin{bmatrix} -2 & -3 & -1 \\ -3 & -3 & -1 \\ -2 & -4 & -1 \end{bmatrix}$$

The process shown in Example 3 applies to any $n \times n$ matrix and will find the inverse of matrix A , if possible. If matrix A has no inverse, the process will also tell you that. The next example applies the process to a singular matrix (one that has no inverse).

EXAMPLE 4 A Singular Matrix

Show that the matrix has no inverse.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ -2 & 3 & -2 \end{bmatrix}$$

SOLUTION Adjoin the identity matrix to A to form

$$[A \vdots I] = \begin{bmatrix} 1 & 2 & 0 & \vdots & 1 & 0 & 0 \\ 3 & -1 & 2 & \vdots & 0 & 1 & 0 \\ -2 & 3 & -2 & \vdots & 0 & 0 & 1 \end{bmatrix}$$

and apply Gauss-Jordan elimination as follows.

$$\begin{bmatrix} 1 & 2 & 0 & \vdots & 1 & 0 & 0 \\ 0 & -7 & 2 & \vdots & -3 & 1 & 0 \\ -2 & 3 & -2 & \vdots & 0 & 0 & 1 \end{bmatrix} \quad R_2 + (-3)R_1 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 2 & 0 & \vdots & 1 & 0 & 0 \\ 0 & -7 & 2 & \vdots & -3 & 1 & 0 \\ 0 & 7 & -2 & \vdots & 2 & 0 & 1 \end{bmatrix} \quad R_3 + (2)R_1 \rightarrow R_3$$

Now, notice that adding the second row to the third row produces a row of zeros on the left side of the matrix.

$$\begin{bmatrix} 1 & 2 & 0 & \vdots & 1 & 0 & 0 \\ 0 & -7 & 2 & \vdots & -3 & 1 & 0 \\ 0 & 0 & 0 & \vdots & -1 & 1 & 1 \end{bmatrix} \quad R_3 + R_2 \rightarrow R_3$$

Because the “ A portion” of the matrix has a row of zeros, you can conclude that it is not possible to rewrite the matrix $[A \vdots I]$ in the form $[I \vdots A^{-1}]$. This means that A has no inverse, or is noninvertible (or singular).

Using Gauss-Jordan elimination to find the inverse of a matrix works well (even as a computer technique) for matrices of size 3×3 or greater. For 2×2 matrices, however, you can use a formula to find the inverse instead of using Gauss-Jordan elimination. This simple formula is explained as follows.

If A is a 2×2 matrix represented by

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

then A is invertible if and only if $ad - bc \neq 0$. Moreover, if $ad - bc \neq 0$, then the inverse is represented by

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Try verifying this inverse by finding the product AA^{-1} .

REMARK: The denominator $ad - bc$ is called the **determinant** of A . You will study determinants in detail in Chapter 3.

EXAMPLE 5 Finding the Inverse of a 2 x 2 Matrix

If possible, find the inverse of each matrix.

$$(a) A = \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix} \quad (b) B = \begin{bmatrix} 3 & -1 \\ -6 & 2 \end{bmatrix}$$

SOLUTION

- (a) For the matrix A , apply the formula for the inverse of a 2×2 matrix to obtain $ad - bc = (3)(2) - (-1)(-2) = 4$. Because this quantity is not zero, the inverse is formed by interchanging the entries on the main diagonal and changing the signs of the other two entries, as follows.

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{3}{4} \end{bmatrix}$$

- (b) For the matrix B , you have $ad - bc = (3)(2) - (-1)(-6) = 0$, which means that B is noninvertible.

Properties of Inverses

Some important properties of inverse matrices are listed below.

THEOREM 2.8
Properties of
Inverse Matrices

If A is an invertible matrix, k is a positive integer, and c is a scalar not equal to zero, then A^{-1} , A^k , cA , and A^T are invertible and the following are true.

1. $(A^{-1})^{-1} = A$
2. $(A^k)^{-1} = A^{-1}A^{-1} \cdots A^{-1} = (A^{-1})^k$
3. $(cA)^{-1} = \frac{1}{c}A^{-1}$, $c \neq 0$
4. $(A^T)^{-1} = (A^{-1})^T$

PROOF

The key to the proofs of Properties 1, 3, and 4 is the fact that the inverse of a matrix is unique (Theorem 2.7). That is, if $BC = CB = I$, then C is the inverse of B .

Property 1 states that the inverse of A^{-1} is A itself. To prove this, observe that $A^{-1}A = AA^{-1} = I$, which means that A is the inverse of A^{-1} . Thus, $A = (A^{-1})^{-1}$.

Similarly, Property 3 states that $\frac{1}{c}A^{-1}$ is the inverse of (cA) , $c \neq 0$. To prove this, use the properties of scalar multiplication given in Theorems 2.1 and 2.3, as follows.

$$(cA)\left(\frac{1}{c}A^{-1}\right) = \left(c\frac{1}{c}\right)AA^{-1} = (1)I = I$$

and

$$\left(\frac{1}{c}A^{-1}\right)(cA) = \left(\frac{1}{c}c\right)A^{-1}A = (1)I = I$$

So $\frac{1}{c}A^{-1}$ is the inverse of (cA) , which implies that

$$\frac{1}{c}A^{-1} = (cA)^{-1}.$$

Properties 2 and 4 are left for you to prove. (See Exercises 47 and 48.)

For nonsingular matrices, the exponential notation used for repeated multiplication of *square* matrices can be extended to include exponents that are negative integers. This may be done by defining A^{-k} to be

$$A^{-k} = \underbrace{A^{-1}A^{-1} \cdots A^{-1}}_{k \text{ factors}} = (A^{-1})^k.$$

Using this convention you can show that the properties $A^jA^k = A^{j+k}$ and $(A^j)^k = A^{jk}$ hold true for any integers j and k .

EXAMPLE 6

The Inverse of the Square of a Matrix

Compute A^{-2} in two different ways and show that the results are equal.

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$$

SOLUTION One way to find A^{-2} is to find $(A^2)^{-1}$ by squaring the matrix A to obtain

$$A^2 = \begin{bmatrix} 3 & 5 \\ 10 & 18 \end{bmatrix}$$

and using the formula for the inverse of a 2×2 matrix to obtain

$$\begin{aligned} (A^2)^{-1} &= \frac{1}{4} \begin{bmatrix} 18 & -5 \\ -10 & 3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{9}{2} & -\frac{5}{4} \\ -\frac{5}{2} & \frac{3}{4} \end{bmatrix}. \end{aligned}$$

Another way to find A^{-2} is to find $(A^{-1})^2$ by finding A^{-1}

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}$$

and then squaring this matrix to obtain

$$(A^{-1})^2 = \begin{bmatrix} \frac{9}{2} & -\frac{5}{4} \\ -\frac{5}{2} & \frac{3}{4} \end{bmatrix}.$$

Note that each method produces the same result.

Discovery

Let $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 1 & -1 \end{bmatrix}$.

Calculate $(AB)^{-1}$, $A^{-1}B^{-1}$, and $B^{-1}A^{-1}$. Make a conjecture about the inverse of a product of two nonsingular matrices. Select two other nonsingular matrices and see whether your conjecture holds.

The next theorem gives a formula for computing the inverse of a product of two matrices.

THEOREM 2.9 The Inverse of a Product

If A and B are invertible matrices of size n , then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

PROOF

To show that $B^{-1}A^{-1}$ is the inverse of AB , you need only show that it conforms to the definition of an inverse matrix. That is,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = A(I)A^{-1} = (AI)A^{-1} = AA^{-1} = I.$$

In a similar way you can show that $(B^{-1}A^{-1})(AB) = I$ and conclude that AB is invertible and has the indicated inverse.

Theorem 2.9 states that the inverse of a product of two invertible matrices is the product of their inverses taken in the *reverse* order. This can be generalized to include the product of several invertible matrices:

$$(A_1A_2A_3 \cdots A_n)^{-1} = A_n^{-1} \cdots A_3^{-1}A_2^{-1}A_1^{-1}.$$

(See Example 4 in Appendix A.)

EXAMPLE 7

Finding the Inverse of a Matrix Product

Find $(AB)^{-1}$ for the matrices

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 3 \end{bmatrix}$$

using the fact that A^{-1} and B^{-1} are represented by

$$A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B^{-1} = \begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 0 \\ \frac{2}{3} & 0 & -\frac{1}{3} \end{bmatrix}.$$

SOLUTION Using Theorem 2.9 produces

$$(AB)^{-1} = \overset{B^{-1}}{\quad} \overset{A^{-1}}{\quad} = \begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 0 \\ \frac{2}{3} & 0 & -\frac{1}{3} \end{bmatrix} \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & -5 & -2 \\ -8 & 4 & 3 \\ 5 & -2 & -\frac{7}{3} \end{bmatrix}.$$

REMARK: Note that you *reverse the order* of multiplication to find the inverse of AB . That is, $(AB)^{-1} = B^{-1}A^{-1}$, and the inverse of AB is usually *not* equal to $A^{-1}B^{-1}$.

One important property in the algebra of real numbers is the cancellation property. That is, if $ac = bc$ ($c \neq 0$), then $a = b$. *Invertible* matrices have similar cancellation properties.

THEOREM 2.10 Cancellation Properties

If C is an invertible matrix, then the following properties hold.

1. If $AC = BC$, then $A = B$. Right cancellation property
2. If $CA = CB$, then $A = B$. Left cancellation property

PROOF To prove Property 1, use the fact that C is invertible and write

$$\begin{aligned} AC &= BC \\ (AC)C^{-1} &= (BC)C^{-1} \\ A(CC^{-1}) &= B(CC^{-1}) \\ AI &= BI \\ A &= B. \end{aligned}$$

The second property can be proved in a similar way; this is left to you. (See Exercise 50.)

Be sure to remember that Theorem 2.10 can be applied only if C is an *invertible* matrix. If C is not invertible, then cancellation is not usually valid. For instance, Example 5 in Section 2.2 gives an example of a matrix equation $AC = BC$ in which $A \neq B$, because C is not invertible in the example.

Systems of Equations

In Theorem 2.5 you were able to prove that a system of linear equations can have exactly one solution, an infinite number of solutions, or no solution. For *square* systems (those having the same number of equations as variables), you can use the theorem below to determine whether the system has a unique solution.

THEOREM 2.11 Systems of Equations with Unique Solutions

If A is an invertible matrix, then the system of linear equations $A\mathbf{x} = \mathbf{b}$ has a unique solution given by

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

PROOF

Because A is nonsingular, the steps shown below are valid.

$$\begin{aligned} A\mathbf{x} &= \mathbf{b} \\ A^{-1}A\mathbf{x} &= A^{-1}\mathbf{b} \\ I\mathbf{x} &= A^{-1}\mathbf{b} \\ \mathbf{x} &= A^{-1}\mathbf{b} \end{aligned}$$

This solution is unique because if $\mathbf{x}_1$ and $\mathbf{x}_2$ were two solutions, you could apply the cancellation property to the equation $A\mathbf{x}_1 = \mathbf{b} = A\mathbf{x}_2$ to conclude that $\mathbf{x}_1 = \mathbf{x}_2$.

Theorem 2.11 is theoretically important, but it is not very practical for solving a system of linear equations. It would require more work to find A^{-1} and then multiply by $\mathbf{b}$ than simply to solve the system using Gaussian elimination with back-substitution. A situation in which you might consider using Theorem 2.11 as a computational technique would be one in which you have *several* systems of linear equations, all of which have the same coefficient matrix A . In such a case, you could find the inverse matrix once and then solve each system by computing the product $A^{-1}\mathbf{b}$. This is demonstrated in Example 8.

EXAMPLE 8

Solving a System of Equations Using an Inverse Matrix

Use an inverse matrix to solve each system.

(a) $2x + 3y + z = -1$	(b) $2x + 3y + z = 4$	(c) $2x + 3y + z = 0$
$3x + 3y + z = 1$	$3x + 3y + z = 8$	$3x + 3y + z = 0$
$2x + 4y + z = -2$	$2x + 4y + z = 5$	$2x + 4y + z = 0$

SOLUTION

First note that the coefficient matrix for each system is

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 3 & 1 \\ 2 & 4 & 1 \end{bmatrix}.$$

Using Gauss-Jordan elimination, you can find A^{-1} to be

$$A^{-1} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix}.$$

To solve each system, use matrix multiplication, as follows.

$$(a) \mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$$

The solution is $x = 2$, $y = -1$, and $z = -2$.

$$(b) \mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix} \begin{bmatrix} 4 \\ 8 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ -7 \end{bmatrix}$$

The solution is $x = 4$, $y = 1$, and $z = -7$.

$$(c) \mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The solution is trivial: $x = 0$, $y = 0$, and $z = 0$.

SECTION 2.3 Exercises

In Exercises 1–4, show that B is the inverse of A .

$$1. A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

$$2. A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}, B = \begin{bmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix}$$

$$3. A = \begin{bmatrix} -2 & 2 & 3 \\ 1 & -1 & 0 \\ 0 & 1 & 4 \end{bmatrix}, B = \frac{1}{3} \begin{bmatrix} -4 & -5 & 3 \\ -4 & -8 & 3 \\ 1 & 2 & 0 \end{bmatrix}$$

$$4. A = \begin{bmatrix} 2 & -17 & 11 \\ -1 & 11 & -7 \\ 0 & 3 & -2 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix}$$

In Exercises 5–24, find the inverse of the matrix (if it exists).

$$5. \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$$

$$6. \begin{bmatrix} 1 & -2 \\ 2 & -3 \end{bmatrix}$$

$$7. \begin{bmatrix} -7 & 33 \\ 4 & -19 \end{bmatrix}$$

$$8. \begin{bmatrix} -1 & 1 \\ 3 & -3 \end{bmatrix}$$

$$9. \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 4 \\ 3 & 6 & 5 \end{bmatrix}$$

$$10. \begin{bmatrix} 1 & 2 & 2 \\ 3 & 7 & 9 \\ -1 & -4 & -7 \end{bmatrix}$$

$$11. \begin{bmatrix} 1 & 2 & -1 \\ 3 & 7 & -10 \\ 7 & 16 & -21 \end{bmatrix}$$

$$12. \begin{bmatrix} 10 & 5 & -7 \\ -5 & 1 & 4 \\ 3 & 2 & -2 \end{bmatrix}$$

$$13. \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 0 \\ -2 & 0 & 3 \end{bmatrix}$$

$$15. \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ -0.3 & 0.2 & 0.2 \\ 0.5 & 0.5 & 0.5 \end{bmatrix}$$

$$17. \begin{bmatrix} 1 & 0 & 0 \\ 3 & 4 & 0 \\ 2 & 5 & 5 \end{bmatrix}$$

$$19. \begin{bmatrix} -8 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 \end{bmatrix}$$



$$21. \begin{bmatrix} 1 & -2 & -1 & -2 \\ 3 & -5 & -2 & -3 \\ 2 & -5 & -2 & -5 \\ -1 & 4 & 4 & 11 \end{bmatrix}$$



$$23. \begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 2 & 0 & 4 \\ 1 & 0 & 3 & 0 \\ 0 & 2 & 0 & 4 \end{bmatrix}$$

$$14. \begin{bmatrix} 3 & 2 & 5 \\ 2 & 2 & 4 \\ -4 & 4 & 0 \end{bmatrix}$$

$$16. \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$18. \begin{bmatrix} 1 & 0 & 0 \\ 3 & 0 & 0 \\ 2 & 5 & 5 \end{bmatrix}$$

$$20. \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$



$$22. \begin{bmatrix} 4 & 8 & -7 & 14 \\ 2 & 5 & -4 & 6 \\ 0 & 2 & 1 & -7 \\ 3 & 6 & -5 & 10 \end{bmatrix}$$



$$24. \begin{bmatrix} 1 & 3 & -2 & 0 \\ 0 & 2 & 4 & 6 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

In Exercises 25–28, use an inverse matrix to solve each system of linear equations.

$$25. (a) x + 2y = -1$$

$$x - 2y = 3$$

$$(b) x + 2y = 10$$

$$x - 2y = -6$$

$$(c) x + 2y = -3$$

$$x - 2y = 0$$

$$27. (a) x_1 + 2x_2 + x_3 = 2$$

$$x_1 + 2x_2 - x_3 = 4$$

$$x_1 - 2x_2 + x_3 = -2$$

$$(b) x_1 + 2x_2 + x_3 = 1$$

$$x_1 + 2x_2 - x_3 = 3$$

$$x_1 - 2x_2 + x_3 = -3$$

$$26. (a) 2x - y = -3$$

$$2x + y = 7$$

$$(b) 2x - y = -1$$

$$2x + y = -3$$

$$(c) 2x - y = 6$$

$$2x + y = 10$$

$$28. (a) x_1 + x_2 - 2x_3 = 0$$

$$x_1 - 2x_2 + x_3 = 0$$

$$x_1 - x_2 - x_3 = -1$$

$$(b) x_1 + x_2 - 2x_3 = -1$$

$$x_1 - 2x_2 + x_3 = 2$$

$$x_1 - x_2 - x_3 = 0$$

 In Exercises 29–32, use a graphing utility or computer software program with matrix capabilities to solve the system of linear equations using an inverse matrix.

$$29. x_1 + 2x_2 - x_3 + 3x_4 - x_5 = -3$$

$$x_1 - 3x_2 + x_3 + 2x_4 - x_5 = -3$$

$$2x_1 + x_2 + x_3 - 3x_4 + x_5 = 6$$

$$x_1 - x_2 + 2x_3 + x_4 - x_5 = 2$$

$$2x_1 + x_2 - x_3 + 2x_4 + x_5 = -3$$

$$30. x_1 + x_2 - x_3 + 3x_4 - x_5 = 3$$

$$2x_1 + x_2 + x_3 + x_4 + x_5 = 4$$

$$x_1 + x_2 - x_3 + 2x_4 - x_5 = 3$$

$$2x_1 + x_2 + 4x_3 + x_4 - x_5 = -1$$

$$3x_1 + x_2 + x_3 - 2x_4 + x_5 = 5$$

$$31. 2x_1 - 3x_2 + x_3 - 2x_4 + x_5 - 4x_6 = 20$$

$$3x_1 + x_2 - 4x_3 + x_4 - x_5 + 2x_6 = -16$$

$$4x_1 + x_2 - 3x_3 + 4x_4 - x_5 + 2x_6 = -12$$

$$-5x_1 - x_2 + 4x_3 + 2x_4 - 5x_5 + 3x_6 = -2$$

$$x_1 + x_2 - 3x_3 + 4x_4 - 3x_5 + x_6 = -15$$

$$3x_1 - x_2 + 2x_3 - 3x_4 + 2x_5 - 6x_6 = 25$$

$$32. 4x_1 - 2x_2 + 4x_3 + 2x_4 - 5x_5 - x_6 = 1$$

$$3x_1 + 6x_2 - 5x_3 - 6x_4 + 3x_5 + 3x_6 = -11$$

$$2x_1 - 3x_2 + x_3 + 3x_4 - x_5 - 2x_6 = 0$$

$$-x_1 + 4x_2 - 4x_3 - 6x_4 + 2x_5 + 4x_6 = -9$$

$$3x_1 - x_2 + 5x_3 + 2x_4 - 3x_5 - 5x_6 = 1$$

$$-2x_1 + 3x_2 - 4x_3 - 6x_4 + x_5 + 2x_6 = -12$$

In Exercises 33–36, use the inverse matrices to find (a) $(AB)^{-1}$, (b) $(A^T)^{-1}$, (c) A^{-2} , and (d) $(2A)^{-1}$.

$$33. A^{-1} = \begin{bmatrix} 2 & 5 \\ -7 & 6 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 7 & -3 \\ 2 & 0 \end{bmatrix}$$

$$34. A^{-1} = \begin{bmatrix} -\frac{2}{7} & \frac{1}{7} \\ \frac{3}{7} & \frac{2}{7} \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} \frac{5}{11} & \frac{2}{11} \\ \frac{3}{11} & -\frac{1}{11} \end{bmatrix}$$

$$35. A^{-1} = \begin{bmatrix} 1 & -\frac{1}{2} & \frac{3}{4} \\ \frac{3}{2} & \frac{1}{2} & -2 \\ \frac{1}{4} & 1 & \frac{1}{2} \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 2 & 4 & \frac{5}{2} \\ -\frac{3}{4} & 2 & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & 2 \end{bmatrix}$$

$$36. A^{-1} = \begin{bmatrix} 1 & -4 & 2 \\ 0 & 1 & 3 \\ 4 & 2 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 6 & 5 & -3 \\ -2 & 4 & -1 \\ 1 & 3 & 4 \end{bmatrix}$$

In Exercises 37 and 38, find x such that the matrix is equal to its own inverse.

$$37. A = \begin{bmatrix} 3 & x \\ -2 & -3 \end{bmatrix} \quad 38. A = \begin{bmatrix} 2 & x \\ -1 & -2 \end{bmatrix}$$

In Exercises 39 and 40, find x such that the matrix is singular.

$$39. A = \begin{bmatrix} 4 & x \\ -2 & -3 \end{bmatrix} \quad 40. A = \begin{bmatrix} x & 2 \\ -3 & 4 \end{bmatrix}$$

In Exercises 41 and 42, find A provided that

$$41. (2A)^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad 42. (4A)^{-1} = \begin{bmatrix} 2 & 4 \\ -3 & 2 \end{bmatrix}$$

In Exercises 43 and 44, show that the matrix is invertible and find its inverse.

$$43. A = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix} \quad 44. A = \begin{bmatrix} \sec \theta & \tan \theta \\ \tan \theta & \sec \theta \end{bmatrix}$$

True or False? In Exercises 45 and 46, determine whether each statement is true or false. If a statement is true, give a reason or cite an appropriate statement from the text. If a statement is false, provide an example that shows the statement is not true in all cases or cite an appropriate statement from the text.

45. (a) The inverse of a nonsingular matrix is unique.
 (b) If the matrices A , B , and C satisfy $BA = CA$ and A is invertible, then $B = C$.
 (c) The inverse of the product of two matrices is the product of their inverses; that is, $(AB)^{-1} = A^{-1}B^{-1}$.
 (d) If A can be row reduced to the identity matrix, then A is nonsingular.
46. (a) The product of four invertible 7×7 matrices is invertible.
 (b) The transpose of the inverse of a nonsingular matrix is equal to the inverse of the transpose.
 (c) The matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if $ab - dc \neq 0$.
 (d) If A is a square matrix, then the system of linear equations $A\mathbf{x} = \mathbf{b}$ has a unique solution.

47. Prove Property 2 of Theorem 2.8: If A is an invertible matrix and k is a positive integer, then

$$(A^k)^{-1} = \underbrace{A^{-1}A^{-1} \cdots A^{-1}}_{k \text{ factors}} = (A^{-1})^k$$

48. Prove Property 4 of Theorem 2.8: If A is an invertible matrix, then $(A^T)^{-1} = (A^{-1})^T$.

49. **Guided Proof** Prove that the inverse of a symmetric nonsingular matrix is symmetric.

Getting Started: To prove that the inverse of A is symmetric, you need to show that $(A^{-1})^T = A^{-1}$.

- (i) Let A be a symmetric, nonsingular matrix.
 (ii) This means that $A^T = A$ and A^{-1} exists.
 (iii) Use the properties of the transpose to show that $(A^{-1})^T$ is equal to A^{-1} .

50. Prove Property 2 of Theorem 2.10: If C is an invertible matrix such that $CA = CB$, then $A = B$.

51. Prove that if $A^2 = A$, then $I - 2A = (I - 2A)^{-1}$.

52. Prove that if A , B , and C are square matrices and $ABC = I$, then B is invertible and $B^{-1} = CA$.

53. Prove that if A is invertible and $AB = O$, then $B = O$.

54. **Guided Proof** Prove that if $A^2 = A$, then either $A = I$ or A is singular.

Getting Started: You must show that either A is singular or A equals the identity matrix.

- (i) Begin your proof by observing that A is either singular or nonsingular.
 (ii) If A is singular, then you are done.
 (iii) If A is nonsingular, then use the inverse matrix A^{-1} and the hypothesis $A^2 = A$ to show that $A = I$.

55. **Writing** Is the sum of two invertible matrices invertible? Explain why or why not. Illustrate your conclusion with appropriate examples.

56. **Writing** Under what conditions will the diagonal matrix

$$A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

be invertible? If A is invertible, find its inverse.

57. Use the result of Exercise 56 to find A^{-1} for each matrix.

$$(a) A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$(b) A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix}$$

58. Let $A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$.

- (a) Show that $A^2 - 2A + 5I = O$, where I is the identity matrix of order 2.
 (b) Show that $A^{-1} = \frac{1}{5}(2I - A)$.
 (c) Show that, in general, for any square matrix satisfying $A^2 - 2A + 5I = O$, the inverse of A is given by $A^{-1} = \frac{1}{5}(2I - A)$.

59. Let $\mathbf{u}$ be an $n \times 1$ column matrix satisfying $\mathbf{u}^T \mathbf{u} = 1$. The $n \times n$ matrix $H = I_n - 2\mathbf{u}\mathbf{u}^T$ is called a **Householder matrix**.

(a) Prove that H is symmetric and nonsingular.

(b) Let $\mathbf{u} = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix}$. Show that $\mathbf{u}^T \mathbf{u} = 1$ and calculate the

Householder matrix H .

60. Prove that if the matrix $I - AB$ is nonsingular, then so is $I - BA$.

61. Let A , D , and P be $n \times n$ matrices satisfying $AP = PD$. If P is nonsingular, solve this equation for A . Must it be true that $A = D$?

62. Let A , D , and P be $n \times n$ matrices satisfying $P^{-1}AP = D$. Solve this equation for A . Must it be true that $A = D$?

63. Find an example of a singular 2×2 matrix satisfying $A^2 = A$.

2.4 Elementary Matrices

In Section 1.2, the three elementary row operations for matrices listed below were introduced.

1. Interchange two rows.
2. Multiply a row by a nonzero constant.
3. Add a multiple of a row to another row.

In this section, you will see how matrix multiplication can be used to perform these operations.

Definition of an Elementary Matrix

An $n \times n$ matrix is called an **elementary matrix** if it can be obtained from the identity matrix I_n by a single elementary row operation.

REMARK: The identity matrix I_n is elementary by this definition because it can be obtained from itself by multiplying any one of its rows by 1.

EXAMPLE 1

Elementary Matrices and Nonelementary Matrices

Which of the following matrices are elementary? For those that are, describe the corresponding elementary row operation.

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

(e) $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

(f) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

SOLUTION

(a) This matrix is elementary. It can be obtained by multiplying the second row of I_3 by 3.

(b) This matrix is *not* elementary because it is not square.

(c) This matrix is *not* elementary because it was obtained by multiplying the third row of I_3 by 0 (row multiplication must be by a *nonzero* constant).

- (d) This matrix *is* elementary. It can be obtained by interchanging the second and third rows of I_3 .
- (e) This matrix *is* elementary. It can be obtained by multiplying the first row of I_2 by 2 and adding the result to the second row.
- (f) This matrix is *not* elementary because two elementary row operations are required to obtain it from I_3 .

Elementary matrices are useful because they enable you to use matrix multiplication to perform elementary row operations, as demonstrated in Example 2.

EXAMPLE 2 Elementary Matrices and Elementary Row Operations

- (a) In the matrix product below, E is the elementary matrix in which the first two rows of I_3 have been interchanged.

$$\begin{array}{c} E \\ \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \begin{array}{c} A \\ \begin{bmatrix} 0 & 2 & 1 \\ 1 & -3 & 6 \\ 3 & 2 & -1 \end{bmatrix} \end{array} = \begin{bmatrix} 1 & -3 & 6 \\ 0 & 2 & 1 \\ 3 & 2 & -1 \end{bmatrix}$$

Note that the first two rows of A have been interchanged by multiplying *on the left* by E .

- (b) In the next matrix product, E is the elementary matrix in which the second row of I_3 has been multiplied by $\frac{1}{2}$.

$$\begin{array}{c} E \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \begin{array}{c} A \\ \begin{bmatrix} 1 & 0 & -4 & 1 \\ 0 & 2 & 6 & -4 \\ 0 & 1 & 3 & 1 \end{bmatrix} \end{array} = \begin{bmatrix} 1 & 0 & -4 & 1 \\ 0 & 1 & 3 & -2 \\ 0 & 1 & 3 & 1 \end{bmatrix}$$

Here the size of A is 3×4 . A could, however, be any $3 \times n$ matrix and multiplication on the left by E would still result in multiplying the second row of A by $\frac{1}{2}$.

- (c) In the product shown below, E is the elementary matrix in which 2 times the first row of I_3 has been added to the second row.

$$\begin{array}{c} E \\ \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \begin{array}{c} A \\ \begin{bmatrix} 1 & 0 & -1 \\ -2 & -2 & 3 \\ 0 & 4 & 5 \end{bmatrix} \end{array} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -2 & 1 \\ 0 & 4 & 5 \end{bmatrix}$$

Note that in the product EA , 2 times the first row of A has been added to the second row.

In each of the three products in Example 2, you were able to perform elementary row operations by multiplying *on the left* by an elementary matrix. This property of elementary matrices is generalized in the next theorem, which is stated without proof.

THEOREM 2.12
Representing Elementary
Row Operations

Let E be the elementary matrix obtained by performing an elementary row operation on I_m . If that same elementary row operation is performed on an $m \times n$ matrix A , then the resulting matrix is given by the product EA .

REMARK: Be sure to remember that in Theorem 2.12, A is multiplied *on the left* by the elementary matrix E . Right multiplication by elementary matrices, which involves column operations, will not be considered in this text.

Most applications of elementary row operations require a sequence of operations. For instance, Gaussian elimination usually requires several elementary row operations to row reduce a matrix A . For elementary matrices, this sequence translates into multiplication (on the left) by several elementary matrices. The order of multiplication is important; the elementary matrix immediately to the left of A corresponds to the row operation performed first. This process is demonstrated in Example 3.

EXAMPLE 3 **Using Elementary Matrices**

Find a sequence of elementary matrices that can be used to write the matrix A in row-echelon form.

$$A = \begin{bmatrix} 0 & 1 & 3 & 5 \\ 1 & -3 & 0 & 2 \\ 2 & -6 & 2 & 0 \end{bmatrix}$$

SOLUTION

Matrix	Elementary Row Operation	Elementary Matrix
$\begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 2 & -6 & 2 & 0 \end{bmatrix}$	$R_1 \leftrightarrow R_2$	$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$\begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & -4 \end{bmatrix}$	$R_3 + (-2)R_1 \rightarrow R_3$	$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$
$\begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & -2 \end{bmatrix}$	$(\frac{1}{2})R_3 \rightarrow R_3$	$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$

The three elementary matrices E_1 , E_2 , and E_3 can be used to perform the same elimination.

$$\begin{aligned}
B = E_3 E_2 E_1 A &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 3 & 5 \\ 1 & -3 & 0 & 2 \\ 2 & -6 & 2 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 2 & -6 & 2 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & -4 \end{bmatrix} = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & -2 \end{bmatrix}
\end{aligned}$$

REMARK: The procedure demonstrated in Example 3 is primarily of theoretical interest. In other words, this procedure is not suggested as a practical method for performing Gaussian elimination.

The two matrices in Example 3

$$A = \begin{bmatrix} 0 & 1 & 3 & 5 \\ 1 & -3 & 0 & 2 \\ 2 & -6 & 2 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

are row-equivalent because you can obtain B by performing a sequence of row operations on A . That is, $B = E_3 E_2 E_1 A$.

The definition of row-equivalent matrices can be restated using elementary matrices, as follows.

Definition of Row Equivalence

Let A and B be $m \times n$ matrices. Matrix B is **row-equivalent** to A if there exists a finite number of elementary matrices $E_1, E_2, \dots, E_k$ such that

$$B = E_k E_{k-1} \cdots E_2 E_1 A.$$

You know from Section 2.3 that not all square matrices are invertible. Every elementary matrix, however, is invertible. Moreover, the inverse of an elementary matrix is itself an elementary matrix.

THEOREM 2.13 Elementary Matrices Are Invertible

If E is an elementary matrix, then E^{-1} exists and is an elementary matrix.

To find the inverse of an elementary matrix E , simply reverse the elementary row operation used to obtain E . For instance, you can find the inverse of each of the three elementary matrices shown in Example 3 as follows.

Elementary Matrix		Inverse Matrix	
$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$R_1 \leftrightarrow R_2$	$E_1^{-1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$R_1 \leftrightarrow R_2$
$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$	$R_3 + (-2)R_1 \rightarrow R_3$	$E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$	$R_3 + (2)R_1 \rightarrow R_3$
$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$	$(\frac{1}{2})R_3 \rightarrow R_3$	$E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$	$(2)R_3 \rightarrow R_3$

The following theorem states that every invertible matrix can be written as the product of elementary matrices.

THEOREM 2.14
A Property of
Invertible Matrices

A square matrix A is invertible if and only if it can be written as the product of elementary matrices.

PROOF

The phrase “if and only if” means that there are actually two parts to the theorem. On the one hand, you have to show that *if* A is invertible, *then* it can be written as the product of elementary matrices. Then you have to show that *if* A can be written as the product of elementary matrices, *then* A is invertible.

To prove the theorem in one direction, assume A is the product of elementary matrices. Then, because every elementary matrix is invertible and the product of invertible matrices is invertible, it follows that A is invertible.

To prove the theorem in the other direction, assume A is invertible. From Theorem 2.11 you know that the system of linear equations represented by $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. But this implies that the augmented matrix $[A : \mathbf{0}]$ can be rewritten in the form $[I : \mathbf{0}]$ (using elementary row operations corresponding to $E_1, E_2, \dots$, and E_k). We now have $E_k \cdots E_3 E_2 E_1 A = I$ and it follows that $A = E_1^{-1} E_2^{-1} E_3^{-1} \cdots E_k^{-1}$. A can be written as the product of elementary matrices, and the proof is complete.

The first part of this proof is illustrated in Example 4.

EXAMPLE 4

Writing a Matrix as the Product of Elementary Matrices

Find a sequence of elementary matrices whose product is

$$A = \begin{bmatrix} -1 & -2 \\ 3 & 8 \end{bmatrix}.$$

SOLUTION Begin by finding a sequence of elementary row operations that can be used to rewrite A in reduced row-echelon form.

Matrix	Elementary Row Operation	Elementary Matrix
$\begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$	$(-1)R_1 \rightarrow R_1$	$E_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
$\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$	$R_2 + (-3)R_1 \rightarrow R_2$	$E_2 = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$
$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$	$(\frac{1}{2})R_2 \rightarrow R_2$	$E_3 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$
$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$R_1 + (-2)R_2 \rightarrow R_1$	$E_4 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$

Now, from the matrix product $E_4 E_3 E_2 E_1 A = I$, solve for A to obtain $A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1}$. This implies that A is a product of elementary matrices.

$$A = \overset{E_1^{-1}}{\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}} \overset{E_2^{-1}}{\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}} \overset{E_3^{-1}}{\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}} \overset{E_4^{-1}}{\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}} = \begin{bmatrix} -1 & -2 \\ 3 & 8 \end{bmatrix}$$

In Section 2.3 you learned a process for finding the inverse of a nonsingular matrix A . There, you used Gauss-Jordan elimination to reduce the augmented matrix $[A : I]$ to $[I : A^{-1}]$. You can now use Theorem 2.14 to justify this procedure. Specifically, the proof of Theorem 2.14 allows you to write the product

$$I = E_k \cdots E_3 E_2 E_1 A.$$

Multiplying both sides of this equation (on the right) by A^{-1} , we can write

$$A^{-1} = E_k \cdots E_3 E_2 E_1 I.$$

In other words, a sequence of elementary matrices that reduces A to the identity also can be used to reduce the identity I to A^{-1} . Applying the corresponding sequence of elementary row operations to the matrices A and I simultaneously, you have

$$E_k \cdots E_3 E_2 E_1 [A : I] = [I : A^{-1}].$$

Of course, if A is singular, then no such sequence can be found.

The next theorem ties together some important relationships between $n \times n$ matrices and systems of linear equations. The essential parts of this theorem have already been proved (see Theorems 2.11 and 2.14); it is left to you to fill in the other parts of the proof.

THEOREM 2.15
Equivalent Conditions

If A is an $n \times n$ matrix, then the following statements are equivalent.

1. A is invertible.
2. $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ column matrix $\mathbf{b}$.
3. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
4. A is row-equivalent to I_n .
5. A can be written as the product of elementary matrices.

The LU-Factorization

Solving systems of linear equations is the most important application of linear algebra. At the heart of the most efficient and modern algorithms for solving linear systems, $A\mathbf{x} = \mathbf{b}$ is the so-called *LU*-factorization, in which the square matrix A is expressed as a product, $A = LU$. In this product, the square matrix L is **lower triangular**, which means all the entries above the main diagonal are zero. The square matrix U is **upper triangular**, which means all the entries below the main diagonal are zero.

$$\begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

3×3 lower triangular matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

3×3 upper triangular matrix

By writing $A\mathbf{x} = LU\mathbf{x}$ and letting $U\mathbf{x} = \mathbf{y}$, you can solve for $\mathbf{x}$ in two stages. First solve $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$; then solve $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$. Each system is easy to solve because the coefficient matrices are triangular. In particular, neither system requires any row operations.

Definition of
LU-Factorization

If the $n \times n$ matrix A can be written as the product of a lower triangular matrix L and an upper triangular matrix U , then $A = LU$ is an **LU-factorization** of A .

EXAMPLE 5

LU-Factorizations

$$(a) \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix} = LU$$

is an *LU*-factorization of the matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$ as the product of the lower

triangular matrix $L = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and the upper triangular matrix $U = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}$.

$$(b) A = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix} = LU$$

is an LU -factorization of the matrix A .

If a square matrix A can be row reduced to an upper triangular matrix U using only the row operation of adding a multiple of one row to another row below it, then it is easy to find an LU -factorization of the matrix A . All you need to do is keep track of the individual row operations, as indicated in the example below.

EXAMPLE 6 Finding the LU -Factorizations of a Matrix

Find the LU -factorization of the matrix $A = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{bmatrix}$.

SOLUTION Begin by row reducing A to upper triangular form while keeping track of the elementary matrices used for each row operation.

Matrix	Elementary Row Operation	Elementary Matrix
$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & -4 & 2 \end{bmatrix}$	$R_3 + (-2)R_1 \rightarrow R_3$	$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$
$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix}$	$R_3 + (4)R_2 \rightarrow R_3$	$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}$

The matrix U on the left is upper triangular, and it follows that $E_2 E_1 A = U$, or $A = E_1^{-1} E_2^{-1} U$. Because the product of the lower triangular matrices

$$E_1^{-1} E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix}$$

is again a lower triangular matrix L , the factorization $A = LU$ is complete. Notice that this is the same LU -factorization that is shown in Example 5(b) at the top of this page.

In general, if A can be row reduced to an upper triangular matrix U using only the row operation of adding a multiple of one row to another row, then A has an LU -factorization.

$$\begin{aligned} E_k \cdot \cdots \cdot E_2 E_1 A &= U \\ A &= E_1^{-1} E_2^{-1} \cdot \cdots \cdot E_k^{-1} U \\ A &= LU \end{aligned}$$

Here L is the product of the inverses of the elementary matrices used in the row reduction.

Note that the multipliers in Example 6 are -2 and 4 , which are the negatives of the corresponding entries in L . This is true in general. If U can be obtained from A using only the row operation of adding a multiple of one row to another row below, then the matrix L is lower triangular with 1's along the diagonal. Furthermore, the negative of each multiplier is in the same position as that of the corresponding zero in U .

Once you have obtained an LU -factorization of a matrix A , you can then solve the system of n linear equations in n variables $A\mathbf{x} = \mathbf{b}$ very efficiently in two steps.

1. Write $\mathbf{y} = U\mathbf{x}$ and solve $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$.
2. Solve $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$.

The column matrix $\mathbf{x}$ is the solution of the original system because

$$A\mathbf{x} = LU\mathbf{x} = L\mathbf{y} = \mathbf{b}.$$

The second step in this algorithm is just back-substitution, because the matrix U is upper triangular. The first step is similar, except that it starts at the top of the matrix, because L is lower triangular. For this reason, the first step is often called **forward substitution**.

EXAMPLE 7 Solving a Linear System Using LU -Factorization

Solve the linear system.

$$\begin{aligned} x_1 - 3x_2 &= -5 \\ x_2 + 3x_3 &= -1 \\ 2x_1 - 10x_2 + 2x_3 &= -20 \end{aligned}$$

SOLUTION You obtained the LU -factorization of the coefficient matrix A in Example 6.

$$A = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix}$$

First, let $\mathbf{y} = U\mathbf{x}$ and solve the system $L\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \\ -20 \end{bmatrix}$$

This system is easy to solve using forward substitution. Starting with the first equation, you have $y_1 = -5$. The second equation gives $y_2 = -1$. Finally, from the third equation,

$$\begin{aligned} 2y_1 - 4y_2 + y_3 &= -20 \\ y_3 &= -20 - 2y_1 + 4y_2 \\ y_3 &= -20 - 2(-5) + 4(-1) \\ y_3 &= -14. \end{aligned}$$

The solution of $L\mathbf{y} = \mathbf{b}$ is

$$\mathbf{y} = \begin{bmatrix} -5 \\ -1 \\ -14 \end{bmatrix}.$$

Now solve the system $U\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ using back-substitution.

$$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \\ -14 \end{bmatrix}$$

From the bottom equation, $x_3 = -1$. Then, the second equation gives $x_2 + 3(-1) = -1$, or $x_2 = 2$. Finally, the first equation is $x_1 - 3(2) = -5$, or $x_1 = 1$. So, the solution of the original system of equations is

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}.$$

SECTION 2.4 Exercises

In Exercises 1–8, determine whether the matrix is elementary. If it is, state the elementary row operation used to produce it.

1. $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

2. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

4. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

5. $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

6. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

7. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -5 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

8. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -3 & 1 \end{bmatrix}$

In Exercises 9–12, let A , B , and C be

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \\ -1 & 2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 1 & 2 \\ 1 & 2 & -3 \end{bmatrix},$$

$$C = \begin{bmatrix} 0 & 4 & -3 \\ 0 & 1 & 2 \\ -1 & 2 & 0 \end{bmatrix}.$$

9. Find an elementary matrix E such that $EA = B$.

10. Find an elementary matrix E such that $EA = C$.

11. Find an elementary matrix E such that $EB = A$.

12. Find an elementary matrix E such that $EC = A$.

In Exercises 13–20, find the inverse of the elementary matrix.

13. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

14. $\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$

15. $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

16. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$

17. $\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}, \quad k \neq 0$

18. $\begin{bmatrix} k & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k \neq 0$

19. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

20. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & k & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

In Exercises 21–24, find the inverse of the matrix using elementary matrices.

$$21. \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$$

$$22. \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

$$23. \begin{bmatrix} 1 & 0 & -1 \\ 0 & 6 & -1 \\ 0 & 0 & 4 \end{bmatrix}$$

$$24. \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

In Exercises 25–32, factor the matrix A into a product of elementary matrices.

$$25. A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

$$26. A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$27. A = \begin{bmatrix} 4 & -1 \\ 3 & -1 \end{bmatrix}$$

$$28. A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$29. A = \begin{bmatrix} 1 & -2 & 0 \\ -1 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$30. A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 1 & 3 & 4 \end{bmatrix}$$

$$31. A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$32. A = \begin{bmatrix} 4 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & 2 \\ 1 & 0 & 0 & -2 \end{bmatrix}$$

True or False? In Exercises 33 and 34, determine whether each statement is true or false. If a statement is true, give a reason or cite an appropriate statement from the text. If a statement is false, provide an example that shows the statement is not true in all cases or cite an appropriate statement from the text.

33. (a) The identity matrix is an elementary matrix.
 (b) If E is an elementary matrix, then $2E$ is an elementary matrix.
 (c) The matrix A is row-equivalent to the matrix B if there exists a finite number of elementary matrices $E_1, E_2, \dots, E_k$ such that $A = E_k E_{k-1} \cdots E_2 E_1 B$.
 (d) The inverse of an elementary matrix is an elementary matrix.
34. (a) The zero matrix is an elementary matrix.
 (b) A square matrix is nonsingular if it can be written as the product of elementary matrices.
 (c) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution if and only if $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ column matrix $\mathbf{b}$.

35. **Writing** E is the elementary matrix obtained by interchanging two rows in I_n . A is an $n \times n$ matrix.
 (a) How will EA compare with A ?
 (b) Find E^2 .

36. **Writing** E is the elementary matrix obtained by multiplying a row in I_n by a nonzero constant c . A is an $n \times n$ matrix.
 (a) How will EA compare with A ?
 (b) Find E^2 .

37. Use elementary matrices to find the inverse of

$$A = \begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & c \end{bmatrix}, \quad c \neq 0.$$

38. Use elementary matrices to find the inverse of

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & c \end{bmatrix}, \quad c \neq 0.$$

39. **Writing** Is the product of two elementary matrices always elementary? Explain why or why not and provide appropriate examples to illustrate your conclusion.
 40. **Writing** Is the sum of two elementary matrices always elementary? Explain why or why not and provide appropriate examples to illustrate your conclusion.

In Exercises 41–44, find the LU -factorization of the matrix.

$$41. \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$$

$$42. \begin{bmatrix} -2 & 1 \\ -6 & 4 \end{bmatrix}$$

$$43. \begin{bmatrix} 3 & 0 & 1 \\ 6 & 1 & 1 \\ -3 & 1 & 0 \end{bmatrix}$$

$$44. \begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 1 \\ 10 & 12 & 3 \end{bmatrix}$$

In Exercises 45 and 46, solve the linear system $A\mathbf{x} = \mathbf{b}$ by

- (a) finding the LU -factorization of the coefficient matrix A ,
 (b) solving the lower triangular system $L\mathbf{y} = \mathbf{b}$, and
 (c) solving the upper triangular system $U\mathbf{x} = \mathbf{y}$.
45.
$$\begin{aligned} 2x + y &= 1 \\ y - z &= 2 \\ -2x + y + z &= -2 \end{aligned}$$
46.
$$\begin{aligned} 2x_1 &= 4 \\ -2x_1 + x_2 - x_3 &= -4 \\ 6x_1 + 2x_2 + x_3 &= 15 \\ -x_4 &= -1 \end{aligned}$$
47. **Writing** Suppose you needed to solve many systems of linear equations $A\mathbf{x} = \mathbf{b}_i$, each having the same coefficient matrix A . Explain how you could use the LU -factorization technique to make the task easier, rather than solving each system individually using Gaussian elimination.

48. (a) Show that the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

does not have an LU -factorization.

- (b) Find the
- LU
- factorization of the matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

that has 1's along the main diagonal of L . Are there any restrictions on the matrix A ?

In Exercises 49–54, determine whether the matrix is idempotent. A square matrix A is **idempotent** if $A^2 = A$.

49.
$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

50.
$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

51.
$$\begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$$

52.
$$\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$$

53.
$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

54.
$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

55. Determine
- a
- and
- b
- such that
- A
- is idempotent.

$$A = \begin{bmatrix} 1 & 0 \\ a & b \end{bmatrix}$$

56. Determine conditions on
- a
- ,
- b
- , and
- c
- such that
- A
- is idempotent.

$$A = \begin{bmatrix} a & 0 \\ b & c \end{bmatrix}$$

57. Prove that if
- A
- is an
- $n \times n$
- matrix that is idempotent and invertible, then
- $A = I_n$
- .

- 58.
- Guided Proof**
- Prove that
- A
- is idempotent if and only if
- A^T
- is idempotent.

Getting Started: The phrase “if and only if” means that you have to prove two statements:

1. If A is idempotent, then A^T is idempotent.
2. If A^T is idempotent, then A is idempotent.
 - (i) Begin your proof of the first statement by assuming that A is idempotent.
 - (ii) This means that $A^2 = A$.
 - (iii) Use the properties of the transpose to show that A^T is idempotent.
 - (iv) Begin your proof of the second statement by assuming that A^T is idempotent.

59. Prove that if
- A
- and
- B
- are idempotent and
- $AB = BA$
- , then
- AB
- is idempotent.

60. Prove that if
- A
- is row-equivalent to
- B
- , then
- B
- is row-equivalent to
- A
- .

- 61.
- Guided Proof**
- Prove that if
- A
- is row-equivalent to
- B
- and
- B
- is row-equivalent to
- C
- , then
- A
- is row-equivalent to
- C
- .

Getting Started: To prove that A is row-equivalent to C , you have to find elementary matrices $E_1, \dots, E_k$ such that $A = E_k \cdot \dots \cdot E_1 C$.

- (i) Begin your proof by observing that A is row-equivalent to B .
 - (ii) Meaning, there exist elementary matrices $F_1, \dots, F_n$ such that $A = F_n \cdot \dots \cdot F_1 B$.
 - (iii) There exist elementary matrices $G_1, \dots, G_m$ such that $B = G_1 \cdot \dots \cdot G_m C$.
 - (iv) Combine the matrix equations from steps (ii) and (iii).
62. Let A be a nonsingular matrix. Prove that if B is row-equivalent to A , then B is also nonsingular.

2.5 Applications of Matrix Operations

Stochastic Matrices

Many types of applications involve a finite set of *states* $\{S_1, S_2, \dots, S_n\}$ of a given population. For instance, residents of a city may live downtown or in the suburbs. Voters may vote Democrat, Republican, or for a third party. Soft drink consumers may buy Coca-Cola, Pepsi Cola, or another brand.

The probability that a member of a population will change from the j th state to the i th state is represented by a number p_{ij} , where $0 \leq p_{ij} \leq 1$. A probability of $p_{ij} = 0$ means that the member is certain *not* to change from the j th state to the i th state, whereas a probability of $p_{ij} = 1$ means that the member is certain to change from the j th state to the i th state.

$$P = \begin{array}{c} \begin{array}{cccc} \text{From} & & & \\ \begin{array}{c} S_1 \quad S_2 \quad \cdot \quad \cdot \quad S_n \end{array} & & & \\ \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix} & \begin{array}{c} S_1 \\ S_2 \\ \vdots \\ S_n \end{array} & \text{To} \end{array} \end{array}$$

P is called the **matrix of transition probabilities** because it gives the probabilities of each possible type of transition (or change) within the population.

At each transition, each member in a given state must either stay in that state or change to another state. For probabilities, this means that the sum of the entries in any column of P is 1. For instance, in the first column we have

$$p_{11} + p_{21} + \cdots + p_{n1} = 1.$$

In general, such a matrix is called **stochastic** (the term “stochastic” means “regarding conjecture”). That is, an $n \times n$ matrix P is called a **stochastic matrix** if each entry is a number between 0 and 1 and each column of P adds up to 1.

EXAMPLE 1

Examples of Stochastic Matrices and Nonstochastic Matrices

The matrices in parts (a) and (b) are stochastic, but the matrix in part (c) is not.

$$\begin{array}{lll} \text{(a)} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \text{(b)} \begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{4} & 0 & \frac{3}{4} \\ \frac{1}{4} & \frac{2}{3} & 0 \end{bmatrix} & \text{(c)} \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.2 & 0.3 & 0.4 \\ 0.3 & 0.4 & 0.5 \end{bmatrix} \end{array}$$

Example 2 describes the use of a stochastic matrix to measure consumer preferences.

EXAMPLE 2

A Consumer Preference Model

Two competing companies offer cable television service to a city of 100,000 households. The changes in cable subscriptions each year are shown in Figure 2.1. Company A now has 15,000 subscribers and Company B has 20,000 subscribers. How many subscribers will each company have 1 year from now?

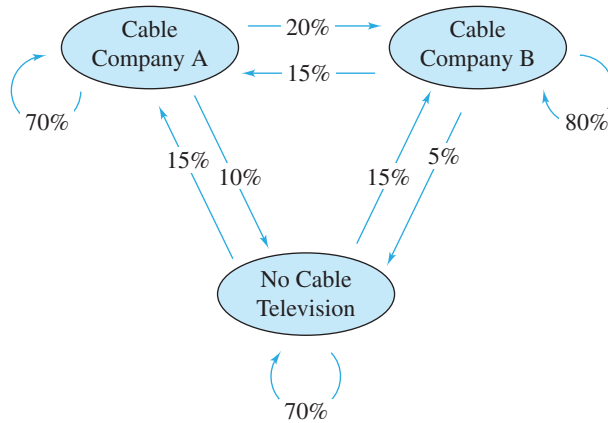


Figure 2.1

SOLUTION The matrix representing the given transition probabilities is

$$P = \begin{array}{c} \begin{array}{ccc} \text{From} & & \\ \text{A} & \text{B} & \text{None} \end{array} \\ \left[\begin{array}{ccc} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{array} \right] \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array} \end{array} \left. \vphantom{\begin{array}{ccc} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{array}} \right\} \begin{array}{l} \text{To} \end{array}$$

and the **state matrix** representing the current populations in the three states is

$$X = \begin{bmatrix} 15,000 \\ 20,000 \\ 65,000 \end{bmatrix} \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array}$$

To find the state matrix representing the populations in the three states after one year, multiply P by X to obtain

$$PX = \begin{bmatrix} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{bmatrix} \begin{bmatrix} 15,000 \\ 20,000 \\ 65,000 \end{bmatrix} = \begin{bmatrix} 23,250 \\ 28,750 \\ 48,000 \end{bmatrix}$$

After one year, Company A will have 23,250 subscribers and Company B will have 28,750 subscribers.

One of the appeals of the matrix solution in Example 2 is that once the model has been created, it becomes easy to find the state matrices representing future years by repeatedly multiplying by the matrix P . This process is demonstrated in Example 3.

EXAMPLE 3 A Consumer Preference Model

Assuming the matrix of transition probabilities from Example 2 remains the same year after year, find the number of subscribers each cable television company will have after (a) 3 years, (b) 5 years, and (c) 10 years. (The answers in this example have been rounded to the nearest person.)

SOLUTION

- (a) From Example 2 you know that the number of subscribers after 1 year is

$$PX = \begin{bmatrix} 23,250 \\ 28,750 \\ 48,000 \end{bmatrix} \quad \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array} \quad \text{After 1 year}$$

Because the matrix of transition probabilities is the same from the first to the third year, the number of subscribers after 3 years is

$$P^3X = \begin{bmatrix} 30,283 \\ 39,042 \\ 30,675 \end{bmatrix} \quad \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array} \quad \text{After 3 years}$$

After 3 years, Company A will have 30,283 subscribers and Company B will have 39,042 subscribers.

- (b) The number of subscribers after 5 years is

$$P^5X = \begin{bmatrix} 32,411 \\ 43,812 \\ 23,777 \end{bmatrix} \quad \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array} \quad \text{After 5 years}$$

After 5 years, Company A will have 32,411 subscribers and Company B will have 43,812 subscribers.

- (c) The number of subscribers after 10 years is

$$P^{10}X = \begin{bmatrix} 33,287 \\ 47,147 \\ 19,566 \end{bmatrix} \quad \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array} \quad \text{After 10 years}$$

After 10 years, Company A will have 33,287 subscribers and Company B will have 47,147 subscribers.

In Example 3, notice that there is little difference between the numbers of subscribers after 5 years and after 10 years. If the process shown in this example is continued, the numbers of subscribers eventually reach a **steady state**. That is, as long as the matrix P doesn't change, the matrix product P^nX approaches a limit $\bar{X}$. In this particular example, the limit is the steady state matrix

$$\bar{X} = \begin{bmatrix} 33,333 \\ 47,619 \\ 19,048 \end{bmatrix} \quad \begin{array}{l} \text{A} \\ \text{B} \\ \text{None} \end{array} \quad \text{Steady state}$$

You can check to see that $P\bar{X} = \bar{X}$.

Cryptography

A **cryptogram** is a message written according to a secret code (the Greek word *kryptos* means “hidden”). This section describes a method of using matrix multiplication to **encode** and **decode** messages.

Begin by assigning a number to each letter in the alphabet (with 0 assigned to a blank space), as follows.

0 =	—	14 =	N
1 =	A	15 =	O
2 =	B	16 =	P
3 =	C	17 =	Q
4 =	D	18 =	R
5 =	E	19 =	S
6 =	F	20 =	T
7 =	G	21 =	U
8 =	H	22 =	V
9 =	I	23 =	W
10 =	J	24 =	X
11 =	K	25 =	Y
12 =	L	26 =	Z
13 =	M		

Then the message is converted to numbers and partitioned into **uncoded row matrices**, each having n entries, as demonstrated in Example 4.

EXAMPLE 4

Forming Uncoded Row Matrices

Write the uncoded row matrices of size 1×3 for the message MEET ME MONDAY.

SOLUTION Partitioning the message (including blank spaces, but ignoring punctuation) into groups of three produces the following uncoded row matrices.

$$\begin{array}{ccccccc}
 [13 & 5 & 5] & [20 & 0 & 13] & [5 & 0 & 13] & [15 & 14 & 4] & [1 & 25 & 0] \\
 M & E & E & T & \text{—} & M & E & \text{—} & M & O & N & D & A & Y & \text{—}
 \end{array}$$

Note that a blank space is used to fill out the last uncoded row matrix.

To **encode** a message, choose an $n \times n$ invertible matrix A and multiply the uncoded row matrices (on the right) by A to obtain **coded row matrices**. This process is demonstrated in Example 5.

EXAMPLE 5 Encoding a Message

Use the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to encode the message MEET ME MONDAY.

SOLUTION

The coded row matrices are obtained by multiplying each of the uncoded row matrices found in Example 4 by the matrix A , as follows.

<i>Uncoded Row Matrix</i>	<i>Encoding Matrix A</i>	<i>Coded Row Matrix</i>
$[13 \quad 5 \quad 5]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [13 \quad -26 \quad 21]$
$[20 \quad 0 \quad 13]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [33 \quad -53 \quad -12]$
$[5 \quad 0 \quad 13]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [18 \quad -23 \quad -42]$
$[15 \quad 14 \quad 4]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [5 \quad -20 \quad 56]$
$[1 \quad 25 \quad 0]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [-24 \quad 23 \quad 77]$

The sequence of coded row matrices is

$$[13 \quad -26 \quad 21][33 \quad -53 \quad -12][18 \quad -23 \quad -42][5 \quad -20 \quad 56][-24 \quad 23 \quad 77].$$

Finally, removing the brackets produces the cryptogram below.

$$13 \quad -26 \quad 21 \quad 33 \quad -53 \quad -12 \quad 18 \quad -23 \quad -42 \quad 5 \quad -20 \quad 56 \quad -24 \quad 23 \quad 77$$

For those who do not know the matrix A , decoding the cryptogram found in Example 5 is difficult. But for an authorized receiver who knows the matrix A , decoding is simple. The receiver need only multiply the coded row matrices by A^{-1} to retrieve the uncoded row matrices. In other words, if

$$X = [x_1 \quad x_2 \quad \cdots \quad x_n]$$

is an uncoded $1 \times n$ matrix, then $Y = XA$ is the corresponding encoded matrix. The receiver of the encoded matrix can decode Y by multiplying on the right by A^{-1} to obtain

$$YA^{-1} = (XA)A^{-1} = X.$$

This procedure is demonstrated in Example 6.

EXAMPLE 6 Decoding a Message



Simulation

Explore this concept further with an electronic simulation available on the website college.hmco.com/pic/larsonELA6e.

Use the inverse of the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to decode the cryptogram

$$13 \quad -26 \quad 21 \quad 33 \quad -53 \quad -12 \quad 18 \quad -23 \quad -42 \quad 5 \quad -20 \quad 56 \quad -24 \quad 23 \quad 77.$$

SOLUTION Begin by using Gauss-Jordan elimination to find A^{-1} .

$$\begin{array}{c} [A : I] \\ \left[\begin{array}{ccc|ccc} 1 & -2 & 2 & 1 & 0 & 0 \\ -1 & 1 & 3 & 0 & 1 & 0 \\ 1 & -1 & -4 & 0 & 0 & 1 \end{array} \right] \end{array} \rightarrow \begin{array}{c} [I : A^{-1}] \\ \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & -10 & -8 \\ 0 & 1 & 0 & -1 & -6 & -5 \\ 0 & 0 & 1 & 0 & -1 & -1 \end{array} \right] \end{array}$$

Now, to decode the message, partition the message into groups of three to form the coded row matrices

$$[13 \quad -26 \quad 21] [33 \quad -53 \quad -12] [18 \quad -23 \quad -42] [5 \quad -20 \quad 56] [-24 \quad 23 \quad 77].$$

To obtain the decoded row matrices, multiply each coded row matrix by A^{-1} (on the right).

Coded Row Matrix	Decoding Matrix A^{-1}	Decoded Row Matrix
$[13 \quad -26 \quad 21]$	$\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}$	$= [13 \quad 5 \quad 5]$
$[33 \quad -53 \quad -12]$	$\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}$	$= [20 \quad 0 \quad 13]$
$[18 \quad -23 \quad -42]$	$\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}$	$= [5 \quad 0 \quad 13]$
$[5 \quad -20 \quad 56]$	$\begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}$	$= [15 \quad 14 \quad 4]$

$$\begin{array}{ccc}
 \text{Coded Row} & \text{Decoding} & \text{Decoded} \\
 \text{Matrix} & \text{Matrix } A^{-1} & \text{Row Matrix}
 \end{array}
 \quad
 \begin{bmatrix} -24 & 23 & 77 \end{bmatrix}
 \begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}
 =
 \begin{bmatrix} 1 & 25 & 0 \end{bmatrix}$$

The sequence of decoded row matrices is

$$\begin{bmatrix} 13 & 5 & 5 \end{bmatrix}
 \begin{bmatrix} 20 & 0 & 13 \end{bmatrix}
 \begin{bmatrix} 5 & 0 & 13 \end{bmatrix}
 \begin{bmatrix} 15 & 14 & 4 \end{bmatrix}
 \begin{bmatrix} 1 & 25 & 0 \end{bmatrix}$$

and the message is

$$\begin{array}{cccccccccccccccc}
 13 & 5 & 5 & 20 & 0 & 13 & 5 & 0 & 13 & 15 & 14 & 4 & 1 & 25 & 0. \\
 M & E & E & T & _ & M & E & _ & M & O & N & D & A & Y & _
 \end{array}$$

Leontief Input-Output Models

Matrix algebra has proved effective in analyzing problems concerning the input and output of an economic system. The model discussed here, developed by the American economist Wassily W. Leontief (1906–1999), was first published in 1936. In 1973, Leontief was awarded a Nobel prize for his work in economics.

Suppose that an economic system has n different industries $I_1, I_2, \dots, I_n$, each of which has **input** needs (raw materials, utilities, etc.) and an **output** (finished product). In producing each unit of output, an industry may use the outputs of other industries, including itself. For example, an electric utility uses outputs from other industries, such as coal and water, and even uses its own electricity.

Let d_{ij} be the amount of output the j th industry needs from the i th industry to produce one unit of output per year. The matrix of these coefficients is called the **input-output matrix**.

$$D = \begin{array}{c} \underbrace{\begin{matrix} I_1 & I_2 & \cdots & I_n \end{matrix}}_{\text{User (Output)}} \\ \left[\begin{array}{cccc} d_{11} & d_{12} & \cdots & d_{1n} \\ d_{21} & d_{22} & \cdots & d_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{n1} & d_{n2} & \cdots & d_{nn} \end{array} \right] \underbrace{\begin{matrix} I_1 \\ I_2 \\ \vdots \\ I_n \end{matrix}}_{\text{Supplier (Input)}} \end{array}$$

To understand how to use this matrix, imagine $d_{12} = 0.4$. This means that 0.4 unit of Industry 1's product must be used to produce one unit of Industry 2's product. If $d_{33} = 0.2$, then 0.2 unit of Industry 3's product is needed to produce one unit of its own product. For this model to work, the values of d_{ij} must satisfy $0 \leq d_{ij} \leq 1$ and the sum of the entries in any column must be less than or equal to 1.

EXAMPLE 7 Forming an Input-Output Matrix

Consider a simple economic system consisting of three industries: electricity, water, and coal. Production, or output, of one unit of electricity requires 0.5 unit of itself, 0.25 unit of water, and 0.25 unit of coal. Production, or output, of one unit of water requires 0.1 unit of electricity, 0.6 unit of itself, and 0 units of coal. Production, or output, of one unit of coal requires 0.2 unit of electricity, 0.15 unit of water, and 0.5 unit of itself. Find the input-output matrix for this system.

SOLUTION The column entries show the amounts each industry requires from the others, as well as from itself, in order to produce one unit of output.

$$\begin{array}{c}
 \text{User (Output)} \\
 \begin{array}{ccc}
 E & W & C
 \end{array} \\
 \left[\begin{array}{ccc}
 0.5 & 0.1 & 0.2 \\
 0.25 & 0.6 & 0.15 \\
 0.25 & 0 & 0.5
 \end{array} \right] \begin{array}{l} E \\ W \\ C \end{array} \left. \vphantom{\begin{array}{ccc} 0.5 & 0.1 & 0.2 \\ 0.25 & 0.6 & 0.15 \\ 0.25 & 0 & 0.5 \end{array}} \right\} \text{Supplier (Input)}
 \end{array}$$

The row entries show the amounts each industry supplies to the other industries, as well as to itself, in order for that particular industry to produce one unit of output. For instance, the electricity industry supplies 0.5 unit to itself, 0.1 unit to water, and 0.2 unit to coal.

To develop the Leontief input-output model further, let the total output of the i th industry be denoted by x_i . If the economic system is **closed** (meaning that it sells its products only to industries within the system, as in the example above), then the total output of the i th industry is given by the linear equation

$$x_i = d_{i1}x_1 + d_{i2}x_2 + \cdots + d_{in}x_n. \quad \text{Closed system}$$

On the other hand, if the industries within the system sell products to nonproducing groups (such as governments or charitable organizations) outside the system, then the system is called **open** and the total output of the i th industry is given by

$$x_i = d_{i1}x_1 + d_{i2}x_2 + \cdots + d_{in}x_n + e_i, \quad \text{Open system}$$

where e_i represents the external demand for the i th industry's product. The collection of total outputs for an open system is represented by the following system of n linear equations.

$$\begin{aligned}
 x_1 &= d_{11}x_1 + d_{12}x_2 + \cdots + d_{1n}x_n + e_1 \\
 x_2 &= d_{21}x_1 + d_{22}x_2 + \cdots + d_{2n}x_n + e_2 \\
 &\vdots \\
 x_n &= d_{n1}x_1 + d_{n2}x_2 + \cdots + d_{nn}x_n + e_n
 \end{aligned}$$

The matrix form of this system is

$$X = DX + E,$$

where X is called the **output matrix** and E is called the **external demand matrix**.

EXAMPLE 8 Solving for the Output Matrix of an Open System

An economic system composed of three industries has the input-output matrix shown below.

$$D = \begin{array}{c} \text{User (Output)} \\ \begin{array}{ccc} \text{A} & \text{B} & \text{C} \end{array} \\ \begin{bmatrix} 0.1 & 0.43 & 0 \\ 0.15 & 0 & 0.37 \\ 0.23 & 0.03 & 0.02 \end{bmatrix} \begin{array}{c} \text{A} \\ \text{B} \\ \text{C} \end{array} \end{array} \left. \vphantom{\begin{bmatrix} 0.1 & 0.43 & 0 \\ 0.15 & 0 & 0.37 \\ 0.23 & 0.03 & 0.02 \end{bmatrix}} \right\} \text{Supplier (Input)}$$

Find the output matrix X if the external demands are

$$E = \begin{bmatrix} 20,000 \\ 30,000 \\ 25,000 \end{bmatrix} \begin{array}{c} \text{A} \\ \text{B} \\ \text{C} \end{array}$$

(The answers in this example have been rounded to the nearest unit.)

SOLUTION Letting I be the identity matrix, write the equation $X = DX + E$ as $IX - DX = E$, which means

$$(I - D)X = E.$$

Using the matrix D above produces

$$I - D = \begin{bmatrix} 0.9 & -0.43 & 0 \\ -0.15 & 1 & -0.37 \\ -0.23 & -0.03 & 0.98 \end{bmatrix}.$$

Finally, applying Gauss-Jordan elimination to the system of linear equations represented by $(I - D)X = E$ produces

$$\begin{bmatrix} 0.9 & -0.43 & 0 & 20,000 \\ -0.15 & 1 & -0.37 & 30,000 \\ -0.23 & -0.03 & 0.98 & 25,000 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 46,616 \\ 0 & 1 & 0 & 51,058 \\ 0 & 0 & 1 & 38,014 \end{bmatrix}.$$

So, the output matrix is

$$X = \begin{bmatrix} 46,616 \\ 51,058 \\ 38,014 \end{bmatrix} \begin{array}{c} \text{A} \\ \text{B} \\ \text{C} \end{array}$$

To produce the given external demands, the outputs of the three industries must be as follows.

Output for Industry A: 46,616 units

Output for Industry B: 51,058 units

Output for Industry C: 38,014 units

The economic systems described in Examples 7 and 8 are, of course, simple ones. In the real world, an economic system would include many industries or industrial groups. For example, an economic analysis of some of the producing groups in the United States would include the products listed below (taken from the *Statistical Abstract of the United States*).

1. Farm products (grains, livestock, poultry, bulk milk)
2. Processed foods and feeds (beverages, dairy products)
3. Textile products and apparel (yarns, threads, clothing)
4. Hides, skins, and leather (shoes, upholstery)
5. Fuels and power (coal, gasoline, electricity)
6. Chemicals and allied products (drugs, plastic resins)
7. Rubber and plastic products (tires, plastic containers)
8. Lumber and wood products (plywood, pencils)
9. Pulp, paper, and allied products (cardboard, newsprint)
10. Metals and metal products (plumbing fixtures, cans)
11. Machinery and equipment (tractors, drills, computers)
12. Furniture and household durables (carpets, appliances)
13. Nonmetallic mineral products (glass, concrete, bricks)
14. Transportation equipment (automobiles, trucks, planes)
15. Miscellaneous products (toys, cameras, linear algebra texts)

A matrix of order 15×15 would be required to represent even these broad industrial groupings using the Leontief input-output model. A more detailed analysis could easily require an input-output matrix of order greater than 100×100 . Clearly, this type of analysis could be done only with the aid of a computer.

Least Squares Regression Analysis

You will now look at a procedure that is used in statistics to develop linear models. The next example demonstrates a visual method for approximating a line of best fit for a given set of data points.

EXAMPLE 9 A Visual Straight-Line Approximation

Determine the straight line that best fits the points.

$(1, 1)$, $(2, 2)$, $(3, 4)$, $(4, 4)$, and $(5, 6)$

SOLUTION Plot the points, as shown in Figure 2.2. It appears that a good choice would be the line whose slope is 1 and whose y-intercept is 0.5. The equation of this line is

$$y = 0.5 + x.$$

An examination of the line shown in Figure 2.2 reveals that you can improve the fit by rotating the line counterclockwise slightly, as shown in Figure 2.3. It seems clear that this new line, the equation of which is $y = 1.2x$, fits the given points better than the original line.

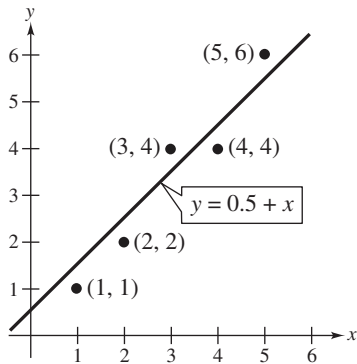


Figure 2.2

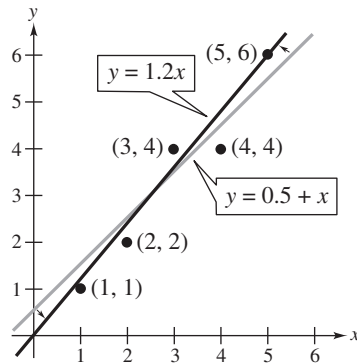


Figure 2.3

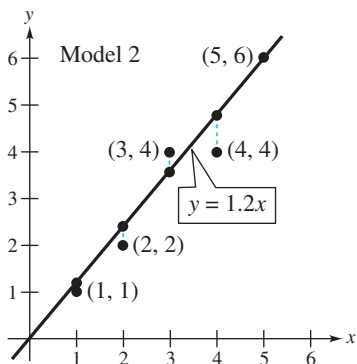
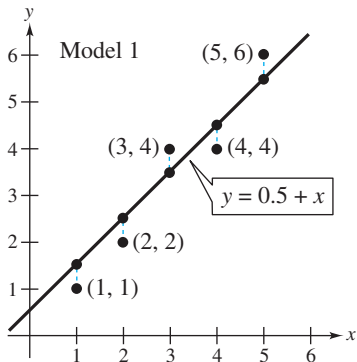


Figure 2.4

One way of measuring how well a function $y = f(x)$ fits a set of points

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

is to compute the differences between the values from the function $f(x_i)$ and the actual values y_i , as shown in Figure 2.4. By squaring these differences and summing the results, you obtain a measure of error that is called the **sum of squared error**. The sums of squared errors for our two linear models are shown in Table 2.1 below.

TABLE 2.1

Model 1: $f(x) = 0.5 + x$				Model 2: $f(x) = 1.2x$			
x_i	y_i	$f(x_i)$	$[y_i - f(x_i)]^2$	x_i	y_i	$f(x_i)$	$[y_i - f(x_i)]^2$
1	1	1.5	$(-0.5)^2$	1	1	1.2	$(-0.2)^2$
2	2	2.5	$(-0.5)^2$	2	2	2.4	$(-0.4)^2$
3	4	3.5	$(+0.5)^2$	3	4	3.6	$(+0.4)^2$
4	4	4.5	$(-0.5)^2$	4	4	4.8	$(-0.8)^2$
5	6	5.5	$(+0.5)^2$	5	6	6.0	$(+0.0)^2$
Total			1.25	Total			1.00

The sums of squared errors confirm that the second model fits the given points better than the first.

Of all possible linear models for a given set of points, the model that has the best fit is defined to be the one that minimizes the sum of squared error. This model is called the **least squares regression line**, and the procedure for finding it is called the **method of least squares**.

Definition of Least Squares Regression Line

For a set of points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the **least squares regression line** is given by the linear function

$$f(x) = a_0 + a_1x$$

that minimizes the sum of squared error

$$[y_1 - f(x_1)]^2 + [y_2 - f(x_2)]^2 + \dots + [y_n - f(x_n)]^2.$$

To find the least squares regression line for a set of points, begin by forming the system of linear equations

$$\begin{aligned} y_1 &= f(x_1) + [y_1 - f(x_1)] \\ y_2 &= f(x_2) + [y_2 - f(x_2)] \\ &\vdots \\ y_n &= f(x_n) + [y_n - f(x_n)] \end{aligned}$$

where the right-hand term, $[y_i - f(x_i)]$, of each equation is thought of as the error in the approximation of y_i by $f(x_i)$. Then write this error as

$$e_i = y_i - f(x_i)$$

so that the system of equations takes the form

$$\begin{aligned} y_1 &= (a_0 + a_1x_1) + e_1 \\ y_2 &= (a_0 + a_1x_2) + e_2 \\ &\vdots \\ y_n &= (a_0 + a_1x_n) + e_n. \end{aligned}$$

Now, if you define Y , X , A , and E as

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad A = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}, \quad E = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$$

the n linear equations may be replaced by the matrix equation

$$Y = XA + E.$$

Note that the matrix X has two columns, a column of 1's (corresponding to a_0) and a column containing the x_i 's. This matrix equation can be used to determine the coefficients of the least squares regression line, as follows.

Matrix Form for Linear Regression

For the regression model $Y = XA + E$, the coefficients of the least squares regression line are given by the matrix equation

$$A = (X^T X)^{-1} X^T Y$$

and the sum of squared error is

$$E^T E.$$

REMARK: You will learn more about this procedure in Section 5.4.

Example 10 demonstrates the use of this procedure to find the least squares regression line for the set of points from Example 9.

EXAMPLE 10

Finding the Least Squares Regression Line

Find the least squares regression line for the points $(1, 1)$, $(2, 2)$, $(3, 4)$, $(4, 4)$, and $(5, 6)$ (see Figure 2.5). Then find the sum of squared error for this regression line.

SOLUTION

Using the five points below, the matrices X and Y are

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 4 \\ 6 \end{bmatrix}.$$

This means that

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}$$

and

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 63 \end{bmatrix}.$$

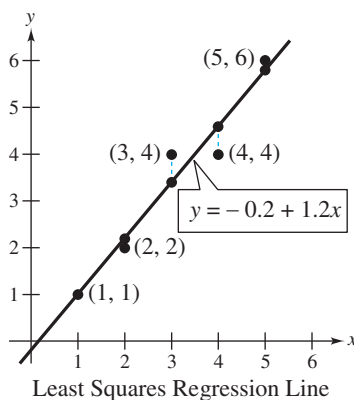


Figure 2.5

Now, using $(X^T X)^{-1}$ to find the coefficient matrix A , you have

$$A = (X^T X)^{-1} X^T Y = \frac{1}{50} \begin{bmatrix} 55 & -15 \\ -15 & 5 \end{bmatrix} \begin{bmatrix} 17 \\ 63 \end{bmatrix} = \begin{bmatrix} -0.2 \\ 1.2 \end{bmatrix}.$$

The least squares regression line is

$$y = -0.2 + 1.2x.$$

(See Figure 2.5.) The sum of squared error for this line can be shown to be 0.8, which means that this line fits the data better than either of the two experimental linear models determined earlier.

SECTION 2.5 Exercises

Stochastic Matrices

In Exercises 1–6, determine whether the matrix is stochastic.

1. $\begin{bmatrix} \frac{2}{5} & -\frac{2}{5} \\ \frac{3}{5} & \frac{7}{5} \end{bmatrix}$

2. $\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$

3. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

4. $\begin{bmatrix} 0.3 & 0.1 & 0.8 \\ 0.5 & 0.2 & 0.1 \\ 0.2 & 0.7 & 0.1 \end{bmatrix}$

5. $\begin{bmatrix} \frac{1}{3} & \frac{1}{6} & \frac{1}{4} \\ \frac{1}{3} & \frac{2}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \end{bmatrix}$

6. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

-  7. The market research department at a manufacturing plant determines that 20% of the people who purchase the plant's product during any month will not purchase it the next month. On the other hand, 30% of the people who do not purchase the product during any month will purchase it the next month. In a population of 1000 people, 100 people purchased the product this month. How many will purchase the product next month? In 2 months?
-  8. A medical researcher is studying the spread of a virus in a population of 1000 laboratory mice. During any week there is an 80% probability that an infected mouse will overcome the virus, and during the same week there is a 10% probability that a noninfected mouse will become infected. One hundred mice are currently infected with the virus. How many will be infected next week? In 2 weeks?

-  9. A population of 10,000 is grouped as follows: 5000 nonsmokers, 2500 smokers of one pack or less per day, and 2500 smokers of more than one pack per day. During any month there is a 5% probability that a nonsmoker will begin smoking a pack or less per day, and a 2% probability that a nonsmoker will begin smoking more than a pack per day. For smokers who smoke a pack or less per day, there is a 10% probability of quitting and a 10% probability of increasing to more than a pack per day. For smokers who smoke more than a pack per day, there is a 5% probability of quitting and a 10% probability of dropping to a pack or less per day. How many people will be in each of the 3 groups in 1 month? In 2 months?
-  10. A population of 100,000 consumers is grouped as follows: 20,000 users of Brand A, 30,000 users of Brand B, and 50,000 who use neither brand. During any month a Brand A user has a 20% probability of switching to Brand B and a 5% probability of not using either brand. A Brand B user has a 15% probability of switching to Brand A and a 10% probability of not using either brand. A nonuser has a 10% probability of purchasing Brand A and a 15% probability of purchasing Brand B. How many people will be in each group in 1 month? In 2 months? In 3 months?
-  11. A college dormitory houses 200 students. Those who watch an hour or more of television on any day always watch for less than an hour the next day. One-fourth of those who watch television for less than an hour one day will watch an hour or more the next day. Half of the students watched television for an hour or more today. How many will watch television for an hour or more tomorrow? In 2 days? In 30 days?

-  12. For the matrix of transition probabilities

$$P = \begin{bmatrix} 0.6 & 0.1 & 0.1 \\ 0.2 & 0.7 & 0.1 \\ 0.2 & 0.2 & 0.8 \end{bmatrix},$$

find P^2X and P^3X for the state matrix

$$X = \begin{bmatrix} 100 \\ 100 \\ 800 \end{bmatrix}.$$

Then find the steady state matrix for P .

13. Prove that the product of two 2×2 stochastic matrices is stochastic.
14. Let P be a 2×2 stochastic matrix. Prove that there exists a 2×1 state matrix X with nonnegative entries such that $PX = X$.

Cryptography

In Exercises 15–18, find the uncoded row matrices of the indicated size for the given messages. Then encode the message using the matrix A .

15. *Message:* SELL CONSOLIDATED

Row Matrix Size: 1×3

Encoding Matrix: $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{bmatrix}$

16. *Message:* PLEASE SEND MONEY

Row Matrix Size: 1×3

Encoding Matrix: $A = \begin{bmatrix} 4 & 2 & 1 \\ -3 & -3 & -1 \\ 3 & 2 & 1 \end{bmatrix}$

17. *Message:* COME HOME SOON

Row Matrix Size: 1×2

Encoding Matrix: $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$

18. *Message:* HELP IS COMING

Row Matrix Size: 1×4

Encoding Matrix: $A = \begin{bmatrix} -2 & 3 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 2 \\ 3 & 1 & -2 & -4 \end{bmatrix}$

In Exercises 19–22, find A^{-1} and use it to decode the cryptogram.

19. $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix},$

11 21 64 112 25 50 29 53 23 46 40 75 55 92

20. $A = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix},$

85 120 6 8 10 15 84 117 42 56 90 125 60 80 30
45 19 26

21. $A = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 7 & 9 \\ -1 & -4 & -7 \end{bmatrix},$

13 19 10 -1 -33 -77 3 -2 -14 4 1 -9 -5 -25
-47 4 1 -9

22. $A = \begin{bmatrix} 3 & -4 & 2 \\ 0 & 2 & 1 \\ 4 & -5 & 3 \end{bmatrix},$

112 -140 83 19 -25 13 72 -76 61 95 -118 71 20 21
38 35 -23 36 42 -48 32

23. The cryptogram below was encoded with a 2×2 matrix.

8 21 -15 -10 -13 -13 5 10 5 25 5 19 -1 6 20 40
-18 -18 1 16

The last word of the message is __RON. What is the message?

24. The cryptogram below was encoded with a 2×2 matrix.

5 2 25 11 -2 -7 -15 -15 32 14 -8 -13 38 19 -19
-19 37 16

The last word of the message is __SUE. What is the message?

-  25. Use a graphing utility or computer software program with matrix capabilities to find A^{-1} . Then decode the cryptogram.

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

38 -14 29 56 -15 62 17 3 38 18 20 76 18 -5 21 29
-7 32 32 9 77 36 -8 48 33 -5 51 41 3 79 12 1 26
58 -22 49 63 -19 69 28 8 67 31 -11 27 41 -18 28

26. A code breaker intercepted the encoded message below.

$$\begin{bmatrix} 45 & -35 & 38 & -30 & 18 & -18 & 35 & -30 & 81 & -60 & 42 & -28 & 75 \\ -55 & 2 & -2 & 22 & -21 & 15 & -10 & & & & & & \end{bmatrix}$$

Let $A^{-1} = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$.

- (a) You know that $[45 \ -35]A^{-1} = [10 \ 15]$ and $[38 \ -30]A^{-1} = [8 \ 14]$, where A^{-1} is the inverse of the encoding matrix A . Write and solve two systems of equations to find w , x , y , and z .
- (b) Decode the message.

Leontief Input-Output Models

27. A system composed of two industries, coal and steel, has the following input requirements.

- (a) To produce \$1.00 worth of output, the coal industry requires \$0.10 of its own product and \$0.80 of steel.
- (b) To produce \$1.00 worth of output, the steel industry requires \$0.10 of its own product and \$0.20 of coal.

Find D , the input-output matrix for this system. Then solve for the output matrix X in the equation $X = DX + E$, where the external demand is

$$E = \begin{bmatrix} 10,000 \\ 20,000 \end{bmatrix}.$$

28. An industrial system has two industries with the following input requirements.

- (a) To produce \$1.00 worth of output, Industry A requires \$0.30 of its own product and \$0.40 of Industry B's product.
- (b) To produce \$1.00 worth of output, Industry B requires \$0.20 of its own product and \$0.40 of Industry A's product.

Find D , the input-output matrix for this system. Then solve for the output matrix X in the equation $X = DX + E$, where the external demand is

$$E = \begin{bmatrix} 50,000 \\ 30,000 \end{bmatrix}.$$



29. A small community includes a farmer, a baker, and a grocer and has the input-output matrix D and external demand matrix E shown below.

	Farmer	Baker	Grocer		
$D =$	$\begin{bmatrix} 0.40 & 0.50 & 0.50 \\ 0.30 & 0.00 & 0.30 \\ 0.20 & 0.20 & 0.00 \end{bmatrix}$			$\begin{matrix} \text{Farmer} \\ \text{Baker} \\ \text{Grocer} \end{matrix}$	and $E = \begin{bmatrix} 1000 \\ 1000 \\ 1000 \end{bmatrix}$

Solve for the output matrix X in the equation $X = DX + E$.



30. An industrial system has three industries and the input-output matrix D and external demand matrix E shown below.

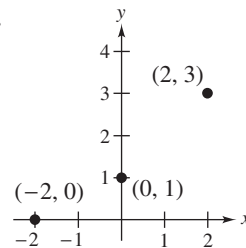
$$D = \begin{bmatrix} 0.2 & 0.4 & 0.4 \\ 0.4 & 0.2 & 0.2 \\ 0.0 & 0.2 & 0.2 \end{bmatrix} \quad \text{and} \quad E = \begin{bmatrix} 5000 \\ 2000 \\ 8000 \end{bmatrix}$$

Solve for the output matrix X in the equation $X = DX + E$.

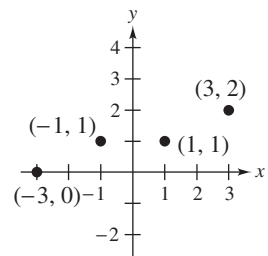
Least Squares Regression Analysis

In Exercises 31–34, (a) sketch the line that appears to be the best fit for the given points, (b) use the method of least squares to find the least squares regression line, and (c) calculate the sum of the squared error.

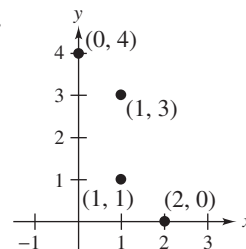
31.



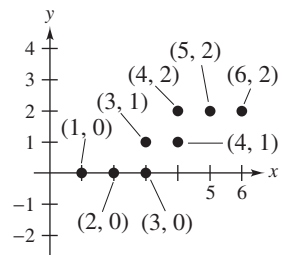
32.



33.



34.



In Exercises 35–42, find the least squares regression line.

35. $(0, 0), (1, 1), (2, 4)$
36. $(1, 0), (3, 3), (5, 6)$
37. $(-2, 0), (-1, 1), (0, 1), (1, 2)$
38. $(-4, -1), (-2, 0), (2, 4), (4, 5)$
39. $(-5, 1), (1, 3), (2, 3), (2, 5)$
40. $(-3, 4), (-1, 2), (1, 1), (3, 0)$
41. $(-5, 10), (-1, 8), (3, 6), (7, 4), (5, 5)$
42. $(0, 6), (4, 3), (5, 0), (8, -4), (10, -5)$

43. A fuel refiner wants to know the demand for a certain grade of gasoline as a function of the price. The daily sales y (in gallons) for three different prices of the product are shown in the table.

Price (x)	\$3.00	\$3.25	\$3.50
Demand (y)	4500	3750	3300

- (a) Find the least squares regression line for these data.
 (b) Estimate the demand when the price is \$3.40.

44. A hardware retailer wants to know the demand for a rechargeable power drill as a function of the price. The monthly sales for four different prices of the drill are shown in the table.

Price (x)	\$25	\$30	\$35	\$40
Demand (y)	82	75	67	55

- (a) Find the least squares regression line for these data.
 (b) Estimate the demand when the price is \$32.95.

45. The table shows the numbers y of motor vehicle registrations (in millions) in the United States for the years 2000 through 2004. (Source: U.S. Federal Highway Administration)

Year	2000	2001	2002	2003	2004
Number (y)	221.5	230.4	229.6	231.4	237.2

- (a) Use the method of least squares to find the least squares regression line for the data. Let t represent the year, with $t = 0$ corresponding to 2000.
 (b) Use the linear regression capabilities of a graphing utility to find a linear model for the data. Let t represent the year, with $t = 0$ corresponding to 2000.

46. A wildlife management team studied the reproduction rates of deer in 3 tracts of a wildlife preserve. Each tract contained 5 acres. In each tract the number of females x and the percent of females y that had offspring the following year were recorded. The results are shown in the table.

Number (x)	100	120	140
Percent (y)	75	68	55

- (a) Use the method of least squares to find the least squares regression line that models the data.
 (b) Use a graphing utility to graph the model and the data in the same viewing window.
 (c) Use the model to create a table of estimated values of y . Compare the estimated values with the actual data.
 (d) Use the model to estimate the percent of females that had offspring when there were 170 females.
 (e) Use the model to estimate the number of females when 40% of the females had offspring.

CHAPTER 2 Review Exercises

In Exercises 1–6, perform the indicated matrix operations.

1. $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 5 & -4 \end{bmatrix} - 3 \begin{bmatrix} 5 & 3 & -6 \\ 0 & -2 & 5 \end{bmatrix}$

2. $-2 \begin{bmatrix} 1 & 2 \\ 5 & -4 \\ 6 & 0 \end{bmatrix} + 8 \begin{bmatrix} 7 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 2 \\ 5 & -4 \\ 6 & 0 \end{bmatrix} \begin{bmatrix} 6 & -2 & 8 \\ 4 & 0 & 0 \end{bmatrix}$

4. $\begin{bmatrix} 1 & 5 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} 6 & -2 & 8 \\ 4 & 0 & 0 \end{bmatrix}$

5. $\begin{bmatrix} 1 & 3 & 2 \\ 0 & 2 & -4 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & 2 \end{bmatrix}$

6. $\begin{bmatrix} 2 & 1 \\ 6 & 0 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ -3 & 1 \end{bmatrix} + \begin{bmatrix} -2 & 4 \\ 0 & 4 \end{bmatrix}$

In Exercises 7–10, write out the system of linear equations represented by the matrix equation.

7. $\begin{bmatrix} 5 & 4 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -22 \end{bmatrix}$

8. $\begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$

$$9. \begin{bmatrix} 0 & 1 & -2 \\ -1 & 3 & 1 \\ 2 & -2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

$$10. \begin{bmatrix} 0 & 1 & 2 \\ 3 & 2 & 1 \\ 4 & -3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -7 \end{bmatrix}$$

In Exercises 11–14, write the system of linear equations in matrix form.

$$11. \begin{aligned} 2x - y &= 5 \\ 3x + 2y &= -4 \end{aligned}$$

$$12. \begin{aligned} 2x + y &= -8 \\ x + 4y &= -4 \end{aligned}$$

$$13. \begin{aligned} 2x_1 + 3x_2 + x_3 &= 10 \\ 2x_1 - 3x_2 - 3x_3 &= 22 \\ 4x_1 - 2x_2 + 3x_3 &= -2 \end{aligned}$$

$$14. \begin{aligned} -3x_1 - x_2 + x_3 &= 0 \\ 2x_1 + 4x_2 - 5x_3 &= -3 \\ x_1 - 2x_2 + 3x_3 &= 1 \end{aligned}$$

In Exercises 15–18, find A^T , $A^T A$, and AA^T .

$$15. \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & 2 \end{bmatrix}$$

$$16. \begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix}$$

$$17. \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$$

$$18. \begin{bmatrix} 1 & -2 & -3 \end{bmatrix}$$

In Exercises 19–22, find the inverse of the matrix (if it exists).

$$19. \begin{bmatrix} 3 & -1 \\ 2 & -1 \end{bmatrix}$$

$$20. \begin{bmatrix} 4 & -1 \\ -8 & 2 \end{bmatrix}$$

$$21. \begin{bmatrix} 2 & 3 & 1 \\ 2 & -3 & -3 \\ 4 & 0 & 3 \end{bmatrix}$$

$$22. \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

In Exercises 23–26, write the system of linear equations in the form $A\mathbf{x} = \mathbf{b}$. Then find A^{-1} and use it to solve for $\mathbf{x}$.

$$23. \begin{aligned} 5x_1 + 4x_2 &= 2 \\ -x_1 + x_2 &= -22 \end{aligned}$$

$$24. \begin{aligned} 3x_1 + 2x_2 &= 1 \\ x_1 + 4x_2 &= -3 \end{aligned}$$

$$25. \begin{aligned} -x_1 + x_2 + 2x_3 &= 1 \\ 2x_1 + 3x_2 + x_3 &= -2 \\ 5x_1 + 4x_2 + 2x_3 &= 4 \end{aligned}$$

$$26. \begin{aligned} x_1 + x_2 + 2x_3 &= 0 \\ x_1 - x_2 + x_3 &= -1 \\ 2x_1 + x_2 + x_3 &= 2 \end{aligned}$$

In Exercises 27 and 28, find A .

$$27. (3A)^{-1} = \begin{bmatrix} 4 & -1 \\ 2 & 3 \end{bmatrix}$$

$$28. (2A)^{-1} = \begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix}$$

In Exercises 29 and 30, find x such that the matrix A is nonsingular.

$$29. A = \begin{bmatrix} 3 & 1 \\ x & -1 \end{bmatrix}$$

$$30. A = \begin{bmatrix} 2 & x \\ 1 & 4 \end{bmatrix}$$

In Exercises 31 and 32, find the inverse of the elementary matrix.

$$31. \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$32. \begin{bmatrix} 1 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

In Exercises 33–36, factor A into a product of elementary matrices.

$$33. A = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$$

$$34. A = \begin{bmatrix} -3 & 13 \\ 1 & -4 \end{bmatrix}$$

$$35. A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 4 \end{bmatrix}$$

$$36. A = \begin{bmatrix} 3 & 0 & 6 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

37. Find two 2×2 matrices A such that $A^2 = I$.

38. Find two 2×2 matrices A such that $A^2 = O$.

39. Find three 2×2 matrices A such that $A^2 = A$.

40. Find 2×2 matrices A and B such that $AB = O$ but $BA \neq O$.

In Exercises 41 and 42, let the matrices X , Y , Z , and W be

$$X = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \quad Y = \begin{bmatrix} -1 \\ 0 \\ 3 \\ 2 \end{bmatrix}, \quad Z = \begin{bmatrix} 3 \\ 4 \\ -1 \\ 2 \end{bmatrix}, \quad W = \begin{bmatrix} 3 \\ 2 \\ -4 \\ -1 \end{bmatrix}.$$

41. (a) Find scalars a , b , and c such that $W = aX + bY + cZ$.

(b) Show that there do not exist scalars a and b such that $Z = aX + bY$.

42. Show that if $aX + bY + cZ = O$, then $a = b = c = 0$.

43. Let A , B , and $A + B$ be nonsingular matrices. Prove that $A^{-1} + B^{-1}$ is nonsingular by showing that

$$(A^{-1} + B^{-1})^{-1} = A(A + B)^{-1}B.$$

44. **Writing** Let A , B , and C be $n \times n$ matrices and let C be nonsingular. If $AC = CB$, is it true that $A = B$? If so, prove it. If not, explain why and find an example for which the hypothesis is false.

In Exercises 45 and 46, find the LU -factorization of the matrix.

$$45. \begin{bmatrix} 2 & 5 \\ 6 & 14 \end{bmatrix}$$

$$46. \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

In Exercises 47 and 48, use the LU -factorization of the coefficient matrix to solve the linear system.

$$\begin{aligned} 47. \quad & x + z = 3 \\ & 2x + y + 2z = 7 \\ & 3x + 2y + 6z = 8 \end{aligned}$$

$$\begin{aligned} 48. \quad & 2x_1 + x_2 + x_3 - x_4 = 7 \\ & 3x_2 + x_3 - x_4 = -3 \\ & -2x_3 = 2 \\ & 2x_1 + x_2 + x_3 - 2x_4 = 8 \end{aligned}$$

True or False? In Exercises 49–52, determine whether each statement is true or false. If a statement is true, give a reason or cite an appropriate statement from the text. If a statement is false, provide an example that shows the statement is not true in all cases or cite an appropriate statement from the text.

49. (a) Addition of matrices is not commutative.
 (b) The transpose of the sum of matrices is equal to the sum of the transposes of the matrices.
50. (a) The product of a 2×3 matrix and a 3×5 matrix is a matrix that is 5×2 .
 (b) The transpose of a product is equal to the product of transposes in reverse order.
51. (a) All $n \times n$ matrices are invertible.
 (b) If an $n \times n$ matrix A is not symmetric, then $A^T A$ is not symmetric.
52. (a) If A and B are $n \times n$ matrices and A is invertible, then $(ABA^{-1})^2 = AB^2A^{-1}$.
 (b) If A and B are nonsingular $n \times n$ matrices, then $A + B$ is a nonsingular matrix.
53. At a convenience store, the numbers of gallons of 87 octane, 89 octane, and 93 octane gasoline sold on Friday, Saturday, and Sunday of a particular holiday weekend are shown by the matrix.

$$\begin{array}{l} \text{Friday} \\ \text{Saturday} \\ \text{Sunday} \end{array} \begin{array}{ccc} 87 & 89 & 93 \\ \left[\begin{array}{ccc} 580 & 840 & 320 \\ 560 & 420 & 160 \\ 860 & 1020 & 540 \end{array} \right] = A \end{array}$$

A second matrix gives the selling prices per gallon and the profits per gallon for the three grades of gasoline sold by the convenience store.

$$\begin{array}{l} \text{Selling Price} \\ \text{(per gallon)} \end{array} \begin{array}{l} \text{Profit} \\ \text{(per gallon)} \end{array} \begin{array}{l} 87 \\ 89 \\ 93 \end{array} \begin{array}{cc} \left[\begin{array}{cc} 3.05 & 0.05 \\ 3.15 & 0.08 \\ 3.25 & 0.10 \end{array} \right] = B$$

- (a) Find AB . What is the meaning of AB in the context of the situation?
 (b) Find the convenience store's total gasoline sales profit for Friday through Sunday.

54. At a certain dairy mart, the numbers of gallons of skim, 2%, and whole milk that are sold on Friday, Saturday, and Sunday of a particular week are shown by the matrix.

$$\begin{array}{l} \text{Friday} \\ \text{Saturday} \\ \text{Sunday} \end{array} \begin{array}{ccc} \text{Skim} & \text{2\%} & \text{Whole} \\ \left[\begin{array}{ccc} 40 & 64 & 52 \\ 60 & 82 & 76 \\ 76 & 96 & 84 \end{array} \right] = A \end{array}$$

A second matrix gives the selling prices per gallon and the profits per gallon for the three types of milk sold by the dairy mart.

$$\begin{array}{l} \text{Selling Price} \\ \text{(per gallon)} \end{array} \begin{array}{l} \text{Profit} \\ \text{(per gallon)} \end{array} \begin{array}{l} \text{Skim} \\ \text{2\%} \\ \text{Whole} \end{array} \begin{array}{cc} \left[\begin{array}{cc} 3.32 & 1.32 \\ 3.22 & 1.07 \\ 3.12 & 0.92 \end{array} \right] = B$$

- (a) Find AB . What is the meaning of AB in the context of the situation?
 (b) Find the dairy mart's total profit for Friday through Sunday.
55. The numbers of calories burned by individuals of different body weights performing different types of aerobic exercises for a 20-minute time period are shown in the matrix.

$$\begin{array}{l} \text{120-lb Person} \\ \text{150-lb Person} \end{array} \begin{array}{l} \text{Bicycling} \\ \text{Jogging} \\ \text{Walking} \end{array} \begin{array}{cc} \left[\begin{array}{cc} 109 & 136 \\ 127 & 159 \\ 64 & 79 \end{array} \right] = A$$

A 120-pound person and a 150-pound person bicycled for 40 minutes, jogged for 10 minutes, and walked for 60 minutes.

- (a) Organize the amounts of time spent exercising in a matrix B .
 (b) Find the product BA .
 (c) Explain the meaning of the matrix product BA as it applies to this situation.

56. The final grades in a particular linear algebra course at a liberal arts college are determined by grades on two midterms and a final exam. The grades for six students and two possible grading systems are shown in the matrices below.

	Midterm 1	Midterm 2	Final Exam
Student 1	78	82	80
Student 2	84	88	85
Student 3	92	93	90
Student 4	88	86	90
Student 5	74	78	80
Student 6	96	95	98

$$= A$$

	Grading System 1	Grading System 2
Midterm 1	0.25	0.20
Midterm 2	0.25	0.20
Final Exam	0.50	0.60

$$= B$$

- (a) Describe each grading system in matrix B and how it is to be applied.
- (b) Compute the numerical grades for the six students using the two grading systems.
- (c) Assign each student a letter grade for each grading system using the letter grade scale shown in the table below.

Numerical Grade Range	Letter Grade
90–100	A
80–89	B
70–79	C
60–69	D
0–59	F

Note: If necessary, round up to the nearest integer.

Stochastic Matrices

In Exercises 57 and 58, determine whether the matrix is stochastic.

57.
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.1 \\ 0 & 0.1 & 0.5 \end{bmatrix}$$

58.
$$\begin{bmatrix} 0.3 & 0.4 & 0.1 \\ 0.2 & 0.4 & 0.5 \\ 0.5 & 0.2 & 0.4 \end{bmatrix}$$

In Exercises 59 and 60, use the given matrix of transition probabilities P and state matrix X to find the state matrices PX , P^2X , and P^3X .

59.
$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{3}{4} \end{bmatrix}, \quad X = \begin{bmatrix} 128 \\ 64 \end{bmatrix}$$

60.
$$P = \begin{bmatrix} 0.6 & 0.2 & 0.0 \\ 0.2 & 0.7 & 0.1 \\ 0.2 & 0.1 & 0.9 \end{bmatrix}, \quad X = \begin{bmatrix} 1000 \\ 1000 \\ 1000 \end{bmatrix}$$

61. A country is divided into 3 regions. Each year, 10% of the residents of Region 1 move to Region 2 and 5% move to Region 3; 15% of the residents of Region 2 move to Region 1 and 5% move to Region 3; and 10% of the residents of Region 3 move to Region 1 and 10% move to Region 2. This year each region has a population of 100,000. Find the population of each region (a) in 1 year and (b) in 3 years.
62. Find the steady state matrix for the populations described in Exercise 61.

Cryptography

In Exercises 63 and 64, find the uncoded row matrices of the indicated size for the given message. Then encode the message using the matrix A .

63. Message: ONE IF BY LAND

Row Matrix Size: 1×2

Encoding Matrix:
$$A = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$$

64. Message: BEAM ME UP SCOTTY

Row Matrix Size: 1×3

Encoding Matrix:
$$A = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 1 & 3 \\ -2 & -1 & -3 \end{bmatrix}$$

In Exercises 65–68, find A^{-1} to decode the cryptogram. Then decode the message.

65.
$$A = \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix},$$

–45 34 36 –24 –43 37 –23 22 –37 29 57 –38 –39 31

66.
$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix},$$

11 52 –8 –9 –13 –39 5 20 12 56 5 20 –2 7 9 41 25 100

$$67. A = \begin{bmatrix} 2 & -1 & -1 \\ -5 & 2 & 2 \\ 5 & -1 & -2 \end{bmatrix},$$

$$\begin{matrix} 58 & -3 & -25 & -48 & 28 & 19 & -40 & 13 & 13 & -98 & 39 & 39 & 118 & -25 & -48 \\ 28 & -14 & -14 & & & & & & & & & & & & \end{matrix}$$

$$68. A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix},$$

$$\begin{matrix} 23 & 20 & 132 & 54 & 128 & 102 & 32 & 21 & 203 & 6 & 10 & 23 & 21 & 15 & 129 & 36 & 46 & 173 \\ 29 & 72 & 45 & & & & & & & & & & & & & & & \end{matrix}$$

 In Exercises 69 and 70, use a graphing utility or computer software program with matrix capabilities to find A^{-1} , then decode the cryptogram.

$$69. \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

$$\begin{matrix} -2 & 2 & 5 & 39 & -53 & -72 & -6 & -9 & 93 & 4 & -12 & 27 & 31 & -49 & -16 & 19 \\ -24 & -46 & -8 & -7 & 99 & & & & & & & & & & & \end{matrix}$$

$$70. \begin{bmatrix} 2 & 0 & 1 \\ 2 & -1 & 0 \\ 1 & 2 & -4 \end{bmatrix}$$

$$\begin{matrix} 66 & 27 & -31 & 37 & 5 & -9 & 61 & 46 & -73 & 46 & -14 & 9 & 94 & 21 & -49 & 32 & -4 & 12 \\ 66 & 31 & -53 & 47 & 33 & -67 & & & & & & & & & & & & \end{matrix}$$

Leontief Input-Output Models

71. An industrial system has two industries with the input requirements shown below.

(a) To produce \$1.00 worth of output, Industry A requires \$0.20 of its own product and \$0.30 of Industry B's product.

(b) To produce \$1.00 worth of output, Industry B requires \$0.10 of its own product and \$0.50 of Industry A's product.

Find D , the input-output matrix for this system. Then solve for the output matrix X in the equation $X = DX + E$, where the external demand is

$$E = \begin{bmatrix} 40,000 \\ 80,000 \end{bmatrix}.$$



72. An industrial system with three industries has the input-output matrix D and the external demand matrix E shown below.

$$D = \begin{bmatrix} 0.1 & 0.3 & 0.2 \\ 0.0 & 0.2 & 0.3 \\ 0.4 & 0.1 & 0.1 \end{bmatrix} \quad \text{and} \quad E = \begin{bmatrix} 3000 \\ 3500 \\ 8500 \end{bmatrix}$$

Solve for the output matrix X in the equation $X = DX + E$.

Least Squares Regression Analysis



In Exercises 73–76, find the least squares regression line for the points.

73. (1, 5), (2, 4), (3, 2)

74. (2, 1), (3, 3), (4, 2), (5, 4), (6, 4)

75. (-2, 4), (-1, 2), (0, 1), (1, -2), (2, -3)

76. (1, 1), (1, 3), (1, 2), (1, 4), (2, 5)

77. A farmer used four test plots to determine the relationship between wheat yield (in kilograms) per square kilometer and the amount of fertilizer (in hundreds of kilograms) per square kilometer. The results are shown in the table.

Fertilizer, x	1.0	1.5	2.0	2.5
Yield, y	32	41	48	53

(a) Find the least squares regression line for these data.

(b) Estimate the yield for a fertilizer application of 160 kilograms per square kilometer.

78. The Consumer Price Index (CPI) for all items for the years 2001 to 2005 is shown in the table. (Source: Bureau of Labor Statistics)

Year	2001	2002	2003	2004	2005
CPI, y	177.1	179.9	184.0	188.9	195.3

(a) Find the least squares regression line for these data. Let x represent the year, with $x = 0$ corresponding to 2000.

(b) Estimate the CPI for the years 2010 and 2015.

79. The table shows the average monthly cable television rates y in the United States (in dollars) for the years 2000 through 2005. (Source: *Broadband Cable Databook*)

Year	2000	2001	2002	2003	2004	2005
Rate, y	30.37	32.87	34.71	36.59	38.14	39.63

- (a) Use the method of least squares to find the least squares regression line for the data. Let x represent the year, with $x = 0$ corresponding to 2000.
- (b) Use the linear regression capabilities of a graphing utility to find a linear model for the data. How does this model compare with the model obtained in part (a)?
- (c) Use the linear model to create a table of estimated values for y . Compare the estimated values with the actual data.
- (d) Use the linear model to predict the average monthly rate in 2010.
- (e) Use the linear model to predict when the average monthly rate will be \$51.00.

-  **80.** The table shows the numbers of cellular phone subscribers y (in millions) in the United States for the years 2000 through 2005. (Source: Cellular Telecommunications and Internet Association)

<i>Year</i>	2000	2001	2002	2003	2004	2005
<i>Subscribers, y</i>	109.5	128.3	140.8	158.7	182.1	207.9

- (a) Use the method of least squares to find the least squares regression line for the data. Let x represent the year, with $x = 0$ corresponding to 2000.
- (b) Use the linear regression capabilities of a graphing utility to find a linear model for the data. How does this model compare with the model obtained in part (a)?

- (c) Use the linear model to create a table of estimated values for y . Compare the estimated values with the actual data.
- (d) Use the linear model to predict the number of subscribers in 2010.
- (e) Use the linear model to predict when the number of subscribers will be 260 million.

- 81.** The table shows the average salaries y (in millions of dollars) of Major League baseball players in the United States for the years 2000 through 2005. (Source: Major League Baseball)

<i>Year</i>	2000	2001	2002	2003	2004	2005
<i>Salary, y</i>	1.8	2.1	2.3	2.4	2.3	2.5

- (a) Use the method of least squares to find the least squares regression line for the data. Let x represent the year, with $x = 0$ corresponding to 2000.
- (b) Use the linear regression capabilities of a graphing utility or computer software program to find a linear model for the data. How does this model compare with the model obtained in part (a)?
- (c) Use the linear model to create a table of estimated values for y . Compare the estimated values with the actual data.
- (d) Use the linear model to predict the average salary in 2010.
- (e) Use the linear model to predict when the average salary will be 3.7 million.

CHAPTER 2 Projects

1 Exploring Matrix Multiplication

The first two test scores for Anna, Bruce, Chris, and David are shown in the table. Use the table to create a matrix M to represent the data. Input matrix M into a graphing utility or computer software program and use it to answer the following questions.

	<i>Test 1</i>	<i>Test 2</i>
<i>Anna</i>	84	96
<i>Bruce</i>	56	72
<i>Chris</i>	78	83
<i>David</i>	82	91

1. Which test was more difficult? Which was easier? Explain.
2. How would you rank the performances of the four students?
3. Describe the meanings of the matrix products $M\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $M\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.
4. Describe the meanings of the matrix products $\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}M$ and $\begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}M$.
5. Describe the meanings of the matrix products $M\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\frac{1}{2}M\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
6. Describe the meanings of the matrix products $\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}M$ and $\frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}M$.
7. Describe the meaning of the matrix product $\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}M\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
8. Use matrix multiplication to express the combined overall average score on both tests.
9. How could you use matrix multiplication to scale the scores on test 1 by a factor of 1.1?

2 Nilpotent Matrices

Let A be a nonzero square $n \times n$ matrix. Is it possible that a positive integer k exists such that $A^k = 0$? For example, find A^3 for the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

A square matrix A is said to be **nilpotent of index k** if $A \neq 0$, $A^2 \neq 0, \dots$, $A^{k-1} \neq 0$, but $A^k = 0$. In this project you will explore the world of nilpotent matrices.

1. What is the index of the nilpotent matrix A ?
2. Use a graphing utility or computer software program to determine which of the matrices below are nilpotent and to find their indices.

(a) $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

(e) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

(f) $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

3. Find 3×3 nilpotent matrices of indices 2 and 3.
4. Find 4×4 nilpotent matrices of indices 2, 3, and 4.
5. Find a nilpotent matrix of index 5.
6. Are nilpotent matrices invertible? Prove your answer.
7. If A is nilpotent, what can you say about A^T ? Prove your answer.
8. If A is nilpotent, show that $I - A$ is invertible.